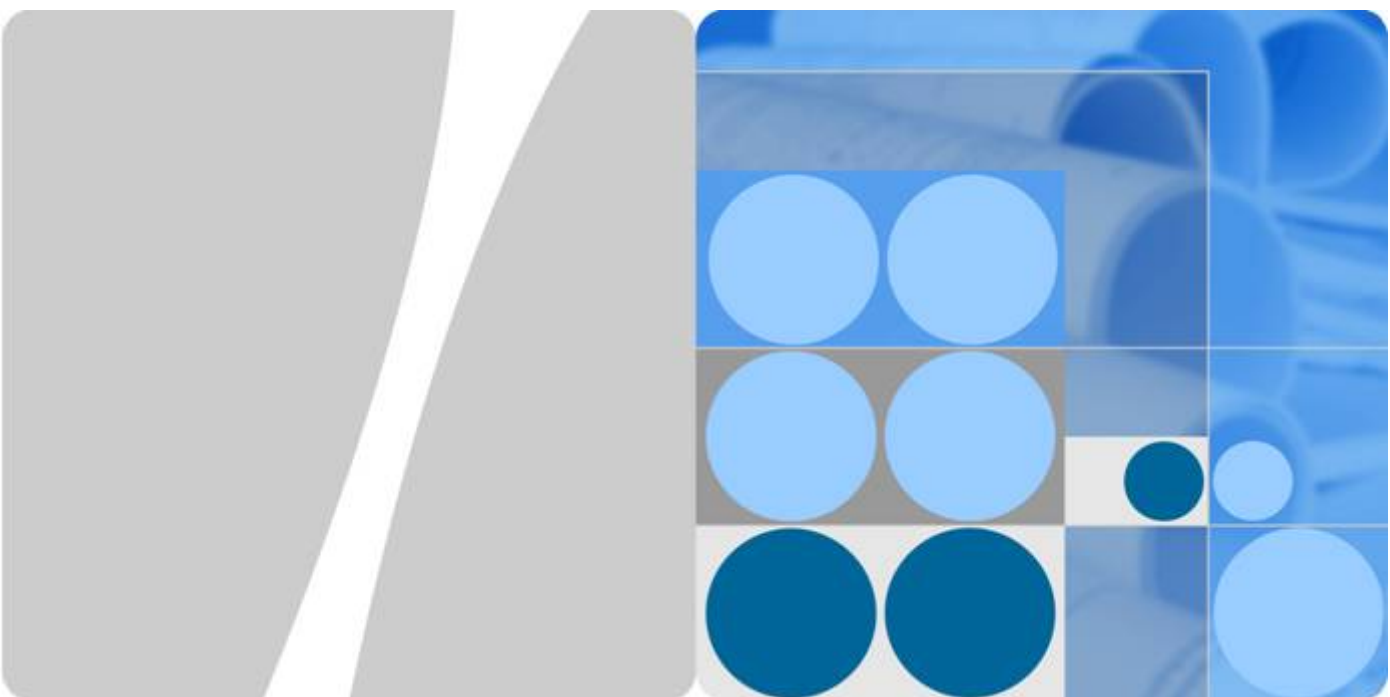




CloudEngine 7800&6800&5800 Series Switches

Hardware Description

Issue 16
Date 2015-08-26

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About This Document

Intended Audience

This document describes hardware components of the CE7800&6800&5800 series switches, including the chassis, power modules, fan modules, cables, and optical modules. You can find useful information about CE7800&6800&5800 series switches hardware components from this document.

This document is intended for:

- Network planning engineers
- Hardware installation engineers
- Commissioning engineers
- On-site maintenance engineers
- System maintenance engineers

Symbol Conventions

The symbols that may be found in this document are defined as follows.

Symbol	Description
 DANGER	Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
 WARNING	Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
 CAUTION	Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury.

Symbol	Description
 NOTICE	Indicates a potentially hazardous situation which, if not avoided, could result in equipment damage, data loss, performance deterioration, or unanticipated results. NOTICE is used to address practices not related to personal injury.
 NOTE	Calls attention to important information, best practices and tips. NOTE is used to address information not related to personal injury, equipment damage, and environment deterioration.

Command Conventions

The command conventions that may be found in this document are defined as follows.

Convention	Description
Boldface	The keywords of a command line are in boldface .
<i>Italic</i>	Command arguments are in <i>italics</i> .
[]	Items (keywords or arguments) in brackets [] are optional.
{ x y ... }	Optional items are grouped in braces and separated by vertical bars. One item is selected.
[x y ...]	Optional items are grouped in brackets and separated by vertical bars. One item is selected or no item is selected.
{ x y ... }*	Optional items are grouped in braces and separated by vertical bars. A minimum of one item or a maximum of all items can be selected.
[x y ...]*	Optional items are grouped in brackets and separated by vertical bars. Several items or no item can be selected.
&<1-n>	The parameter before the & sign can be repeated 1 to n times.
#	A line starting with the # sign is comments.

Declaration

This manual is only a reference for you to configure your devices. The contents in the manual, such as command line syntax, and command outputs, are based on the device conditions in the

lab. The manual provides instructions for general scenarios, but do not cover all usage scenarios of all product models. The contents in the manual may be different from your actual device situations due to the differences in software versions, models, and configuration files. The manual will not list every possible difference. You should configure your devices according to actual situations.

The specifications provided in this manual are tested in lab environment (for example, the tested device has been installed with a certain type of boards or only one protocol is run on the device). Results may differ from the listed specifications when you attempt to obtain the maximum values with multiple functions enabled on the device.

Change History

Changes between document issues are cumulative. The latest document issue contains all the changes made in earlier issues.

Issue 16 (2015-08-26)

This version has the following updates:

The following information is modified:

- **2 Chassis**
- **3 Power Module**
- **4 Fan Module**

Issue 15 (2015-06-10)

This version has the following updates:

The following information is modified:

- **2.3 Appearance and Structure**
- **6.2 SFP/SFP+ Modules**

Issue 14 (2015-05-30)

This version has the following updates:

The following information is added:

- CE6810-32T16S4Q-LI
- CE6810-24S2Q-LI
- CE6850-48T6Q-HI
- CE6851-48S6Q-HI
- CE6850U-48S6Q-HI
- CE6850U-24S2Q-HI
- CE5855-48T4S2Q-EI
- CE5855-24T4S2Q-EI
- 150 W AC Power Module (ES0W2PSA0150)

- FAN-040A Series Fan Modules

Issue 13 (2015-04-26)

This version has the following updates:

The following information is modified:

- [2.3 Appearance and Structure](#)
- [2.5 Ports](#)

Issue 12 (2015-01-20)

This version has the following updates:

The following information is added:

- CE6850-48S6Q-HI
- 600 W AC&240 V DC Power Module
- 600 W High-Voltage DC Power Module
- FAN-060A Series Fan Modules

Issue 11 (2014-12-01)

This version has the following updates:

The following information is modified:

- [2.3 Appearance and Structure](#)
- [5.8 AOC Cable](#)

Issue 10 (2014-09-20)

This version has the following updates:

The following information is added:

- CE6810-48S4Q-LI
- CE6810-48S-LI

Issue 09 (2014-08-01)

This version has the following updates:

The following information is modified:

- [1 Version Support for Components](#)
- [3 Power Module](#)
- [4 Fan Module](#)

Issue 08 (2014-04-21)

This version has the following updates:

The following information is added:

- CE7850-32Q-EI
- CE6810-48S4Q-EI

Issue 07 (2013-12-31)

This version has the following updates:

The following information is added:

- CE5850-48T4S2Q-HI

Issue 06 (2013-12-01)

This version has the following updates:

The following information is modified:

- [2.6 Specifications](#)

Issue 05 (2013-11-20)

This version has the following updates:

The following information is modified:

- [2.6 Specifications](#)

Issue 04 (2013-10-15)

This version has the following updates:

The following information is modified:

- [3 Power Module](#)

Issue 03 (2013-08-01)

This version has the following updates:

The following information is added:

- [3.4 350 W DC Power Module](#)
- [3.5 600 W AC Power Module](#)
- [4.2 FAN-40SB Series Fan Modules](#)
- [5.2 DC Power Cable](#)

The following information is modified:

- [2 Chassis](#)
- [6 Optical Module](#)

Issue 02 (2013-04-20)

This version has the following updates:

The following information is modified:

- **2.4 Indicators**

Issue 01 (2013-03-15)

Initial commercial release.

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1 Version Support for Components

About This Chapter

NOTE

The availability of device models and modules described in this document will be specified in product change notices (PCNs). For details, contact the product manager of Huawei local office.

The versions mentioned in this document refer to the software versions released for the CE7800&6800&5800 series switches.

[1.1 Components Available in V100R001C00](#)

[1.2 Components Available in V100R002C00](#)

[1.3 Components Available in V100R003C00](#)

[1.4 Components Available in V100R003C10](#)

[1.5 Components Available in V100R005C00](#)

[1.6 Components Available in V100R005C10](#)

1.1 Components Available in V100R001C00

Available Models

Table 1-1 lists the switch models available in V100R001C00.

Table 1-1 Models available in V100R001C00

Series	Model	Maximum Number of Ports Supported and Port Description
CE6800	CE6850-48S4Q-EI	52 total: 48 10GE SFP+ optical ports, 4 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE6850-48T4Q-EI	52 total: 48 10GBASE-T Ethernet electrical ports, 4 40GE QSFP+ optical ports (can be split into four 10GE ports)
CE5800	CE5850-48T4S2Q-EI	54 total: 48 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports, 2 40GE QSFP+ optical ports (cannot be split into four 10GE ports)

Available Power Modules

Table 1-2 lists the power modules available in V100R001C00.

Table 1-2 Power modules available in V100R001C00

Power Module Type	Power Module	Power Description
AC power module	PAC-150WA	150 W AC power module (natural heat dissipation)
	PAC-350WA-B	350 W AC power module (air exhaust on front panel)
	PAC-350WA-F	350 W AC power module (air intake on front panel)

Available Fan Modules

Table 1-3 lists the fan modules available in V100R001C00.

Table 1-3 Fan modules available in V100R001C00

Series	Model	Description
FAN-40EA	FAN-40EA-B	FAN-40EA (air exhaust on front panel)
	FAN-40EA-F	FAN-40EA (air intake on front panel)

1.2 Components Available in V100R002C00



The components marked * are the new components added to V100R002C00.

Available Models

Table 1-4 lists the switch models available in V100R002C00.

Table 1-4 Models available in V100R002C00

Series	Model	Maximum Number of Ports Supported and Port Description
CE6800	CE6850-48S4Q-EI	52 total: 48 10GE SFP+ optical ports, 4 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE6850-48T4Q-EI	52 total: 48 10GBASE-T Ethernet electrical ports, 4 40GE QSFP+ optical ports (can be split into four 10GE ports)
CE5800	CE5810-24T4S-EI*	28 total: 24 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports
	CE5810-48T4S-EI*	52 total: 48 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports
	CE5850-48T4S2Q-EI	54 total: 48 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports, 2 40GE QSFP+ optical ports (cannot be split into four 10GE ports)

Available Power Modules

Table 1-5 lists the power modules available in V100R002C00.

Table 1-5 Power modules available in V100R002C00

Power Module Type	Power Module	Power Description
AC power module	PAC-150WA	150 W AC power module (natural heat dissipation)
	PAC-350WA-B	350 W AC power module (air exhaust on front panel)
	PAC-350WA-F	350 W AC power module (air intake on front panel)
	PAC-600WA-B*	600 W AC power module (air exhaust on front panel)
	PAC-600WA-F*	600 W AC power module (air intake on front panel)
DC power module	PDC-350WA-B*	350 W DC power module (air exhaust on front panel)
	PDC-350WA-F*	350 W DC power module (air intake on front panel)

Available Fan Modules

Table 1-6 lists the fan modules available in V100R002C00.

Table 1-6 Fan modules available in V100R002C00

Series	Model	Description
FAN-40EA	FAN-40EA-B	FAN-40EA (air exhaust on front panel)
	FAN-40EA-F	FAN-40EA (air intake on front panel)
FAN-40SB	FAN-40SB-B*	FAN-40SB (air exhaust on front panel)
	FAN-40SB-F*	FAN-40SB (air intake on front panel)

1.3 Components Available in V100R003C00

NOTE

The components marked * are the new components added to V100R003C00.

Available Models

Table 1-7 lists the switch models available in V100R003C00.

Table 1-7 Models available in V100R003C00

Series	Model	Maximum Number of Ports Supported and Port Description
CE7800	CE7850-32Q-EI*	32 total: 32 40GE QSFP+ optical ports (can be split into four 10GE ports)
CE6800	CE6810-48S4Q-EI*	52 total: 48 10GE SFP+ optical ports, 4 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE6850-48S4Q-EI	52 total: 48 10GE SFP+ optical ports, 4 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE6850-48T4Q-EI	52 total: 48 10GBASE-T Ethernet electrical ports, 4 40GE QSFP+ optical ports (can be split into four 10GE ports)
CE5800	CE5810-24T4S-EI	28 total: 24 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports
	CE5810-48T4S-EI	52 total: 48 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports
	CE5850-48T4S2Q-EI	54 total: 48 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports, 2 40GE QSFP+ optical ports (cannot be split into four 10GE ports)
	CE5850-48T4S2Q-HI*	54 total: 48 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports, 2 40GE QSFP+ optical ports (can be split into four 10GE ports)

Available Power Modules

Table 1-8 lists the power modules available in V100R003C00.

Table 1-8 Power modules available in V100R003C00

Power Module Type	Power Module	Power Description
AC power module	PAC-150WA	150 W AC power module (natural heat dissipation)
	PAC-350WA-B	350 W AC power module (air exhaust on front panel)
	PAC-350WA-F	350 W AC power module (air intake on front panel)
	PAC-600WA-B	600 W AC power module (air exhaust on front panel)
	PAC-600WA-F	600 W AC power module (air intake on front panel)
DC power module	PDC-350WA-B	350 W DC power module (air exhaust on front panel)
	PDC-350WA-F	350 W DC power module (air intake on front panel)

Available Fan Modules

Table 1-9 lists the fan modules available in V100R003C00.

Table 1-9 Fan modules available in V100R003C00

Series	Model	Description
FAN-40EA	FAN-40EA-B	FAN-40EA (air exhaust on front panel)
	FAN-40EA-F	FAN-40EA (air intake on front panel)
FAN-40SB	FAN-40SB-B	FAN-40SB (air exhaust on front panel)
	FAN-40SB-F	FAN-40SB (air intake on front panel)
FAN-40HA	FAN-40HA-B*	FAN-40HA (air exhaust on front panel)

Series	Model	Description
	FAN-40HA-F*	FAN-40HA (air intake on front panel)

1.4 Components Available in V100R003C10

NOTE

The components marked * are the new components added to V100R003C10.

Available Models

Table 1-10 lists the switch models available in V100R003C10.

Table 1-10 Models available in V100R003C10

Series	Model	Maximum Number of Ports Supported and Port Description
CE7800	CE7850-32Q-EI	32 total: 32 40GE QSFP+ optical ports (can be split into four 10GE ports)
CE6800	CE6810-48S4Q-EI	52 total: 48 10GE SFP+ optical ports, 4 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE6850-48S4Q-EI	52 total: 48 10GE SFP+ optical ports, 4 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE6850-48T4Q-EI	52 total: 48 10GBASE-T Ethernet electrical ports, 4 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE6810-48S4Q-LI*	52 total: 48 10GE SFP+ optical ports, 4 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE6810-48S-LI*	48 total: 48 10GE SFP+ optical ports
CE5800	CE5810-24T4S-EI	28 total: 24 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports

Series	Model	Maximum Number of Ports Supported and Port Description
	CE5810-48T4S-EI	52 total: 48 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports
	CE5850-48T4S2Q-EI	54 total: 48 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports, 2 40GE QSFP+ optical ports (cannot be split into four 10GE ports)
	CE5850-48T4S2Q-HI	54 total: 48 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports, 2 40GE QSFP+ optical ports (can be split into four 10GE ports)

Available Power Modules

Table 1-11 lists the power modules available in V100R003C10.

Table 1-11 Power modules available in V100R003C10

Power Module Type	Power Module	Power Description
AC power module	PAC-150WA	150 W AC power module (natural heat dissipation)
	PAC-350WA-B	350 W AC power module (air exhaust on front panel)
	PAC-350WA-F	350 W AC power module (air intake on front panel)
	PAC-600WA-B	600 W AC power module (air exhaust on front panel)
	PAC-600WA-F	600 W AC power module (air intake on front panel)
DC power module	PDC-350WA-B	350 W DC power module (air exhaust on front panel)
	PDC-350WA-F	350 W DC power module (air intake on front panel)

Available Fan Modules

Table 1-12 lists the fan modules available in V100R003C10.

Table 1-12 Fan modules available in V100R003C10

Series	Model	Description
FAN-40EA	FAN-40EA-B	FAN-40EA (air exhaust on front panel)
	FAN-40EA-F	FAN-40EA (air intake on front panel)
FAN-40SB	FAN-40SB-B	FAN-40SB (air exhaust on front panel)
	FAN-40SB-F	FAN-40SB (air intake on front panel)
FAN-40HA	FAN-40HA-B	FAN-40HA (air exhaust on front panel)
	FAN-40HA-F	FAN-40HA (air intake on front panel)

1.5 Components Available in V100R005C00



NOTE

The components marked * are the new components added to V100R005C00.

Available Models

Table 1-13 lists the switch models available in V100R005C00.

Table 1-13 Models available in V100R005C00

Series	Model	Maximum Number of Ports Supported and Port Description
CE7800	CE7850-32Q-EI	32 total: 32 40GE QSFP+ optical ports (can be split into four 10GE ports)
CE6800	CE6810-48S4Q-EI	52 total: 48 10GE SFP+ optical ports, 4 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE6850-48S4Q-EI	52 total: 48 10GE SFP+ optical ports, 4 40GE QSFP+ optical ports (can be split into four 10GE ports)

Series	Model	Maximum Number of Ports Supported and Port Description
	CE6850-48T4Q-EI	52 total: 48 10GBASE-T Ethernet electrical ports, 4 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE6810-48S4Q-LI	52 total: 48 10GE SFP+ optical ports, 4 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE6810-48S-LI	48 total: 48 10GE SFP+ optical ports
	CE6850-48S6Q-HI*	54 total: 48 10GE SFP+ optical ports, 6 40GE QSFP+ optical ports (can be split into four 10GE ports)
CE5800	CE5810-24T4S-EI	28 total: 24 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports
	CE5810-48T4S-EI	52 total: 48 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports
	CE5850-48T4S2Q-EI	54 total: 48 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports, 2 40GE QSFP+ optical ports (cannot be split into four 10GE ports)
	CE5850-48T4S2Q-HI	54 total: 48 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports, 2 40GE QSFP+ optical ports (can be split into four 10GE ports)

Available Power Modules

Table 1-14 lists the power modules available in V100R005C00.

Table 1-14 Power modules available in V100R005C00

Power Module Type	Power Module	Power Description
AC power module	PAC-150WA	150 W AC power module (natural heat dissipation)

Power Module Type	Power Module	Power Description
	PAC-350WA-B	350 W AC power module (air exhaust on front panel)
	PAC-350WA-F	350 W AC power module (air intake on front panel)
	PAC-600WA-B	600 W AC power module (air exhaust on front panel)
	PAC-600WA-F	600 W AC power module (air intake on front panel)
	PAC-600WB-B*	600 W AC&240 V DC power module (air exhaust on front panel)
	PAC-600WB-F*	600 W AC&240 V DC power module (air intake on front panel)
DC power module	PDC-350WA-B	350 W DC power module (air exhaust on front panel)
	PDC-350WA-F	350 W DC power module (air intake on front panel)
High-voltage DC power module	PHD-600WA-B*	600 W high-voltage DC power module (air exhaust on front panel)
	PHD-600WA-F*	600 W high-voltage DC power module (air intake on front panel)

Available Fan Modules

Table 1-15 lists the fan modules available in V100R005C00.

Table 1-15 Fan modules available in V100R005C00

Series	Model	Description
FAN-40EA	FAN-40EA-B	FAN-40EA series fan modules (air exhaust on front panel)
	FAN-40EA-F	FAN-40EA series fan modules (air intake on front panel)
FAN-40SB	FAN-40SB-B	FAN-40SB series fan modules (air exhaust on front panel)

Series	Model	Description
	FAN-40SB-F	FAN-40SB series fan modules (air intake on front panel)
FAN-40HA	FAN-40HA-B	FAN-40HA series fan modules (air exhaust on front panel)
	FAN-40HA-F	FAN-40HA series fan modules (air intake on front panel)
FAN-060A	FAN-060A-B*	FAN-060A series fan modules (air exhaust on front panel)
	FAN-060A-F*	FAN-060A series fan modules (air intake on front panel)

1.6 Components Available in V100R005C10

NOTE

The components marked * are the new components added to V100R005C10.

Available Models

Table 1-16 lists the switch models available in V100R005C10.

Table 1-16 Models available in V100R005C10

Series	Model	Maximum Number of Ports Supported and Port Description
CE7800	CE7850-32Q-EI	32 total: 32 40GE QSFP+ optical ports (can be split into four 10GE ports)
CE6800	CE6810-48S4Q-EI	52 total: 48 10GE SFP+ optical ports, 4 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE6850-48S4Q-EI	52 total: 48 10GE SFP+ optical ports, 4 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE6850-48T4Q-EI	52 total: 48 10GBASE-T Ethernet electrical ports, 4 40GE QSFP+ optical ports (can be split into four 10GE ports)

Series	Model	Maximum Number of Ports Supported and Port Description
	CE6810-48S4Q-LI	52 total: 48 10GE SFP+ optical ports, 4 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE6810-48S-LI	48 total: 48 10GE SFP+ optical ports
	CE6850-48S6Q-HI	54 total: 48 10GE SFP+ optical ports, 6 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE6851-48S6Q-HI*	54 total: 48 10GE SFP+ optical ports, 6 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE6850U-48S6Q-HI*	54 total: 48 10GE SFP+/FC optical ports, 6 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE6850-48T6Q-HI*	54 total: 48 10GBASE-T Ethernet electrical ports, 6 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE6850U-24S2Q-HI*	26 total: 24 10GE SFP+/FC optical ports, 2 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE6810-32T16S4Q-LI*	52 total: 32 10GBASE-T Ethernet electrical ports, 16 10GE SFP+ optical ports, 4 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE6810-24S2Q-LI*	26 total: 24 10GE SFP+ optical ports, 2 40GE QSFP+ optical ports (can be split into four 10GE ports)
CE5800	CE5810-24T4S-EI	28 total: 24 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports
	CE5810-48T4S-EI	52 total: 48 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports

Series	Model	Maximum Number of Ports Supported and Port Description
	CE5850-48T4S2Q-EI	54 total: 48 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports, 2 40GE QSFP+ optical ports (cannot be split into four 10GE ports)
	CE5850-48T4S2Q-HI	54 total: 48 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports, 2 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE5855-48T4S2Q-EI*	54 total: 48 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports, 2 40GE QSFP+ optical ports (can be split into four 10GE ports)
	CE5855-24T4S2Q-EI*	30 total: 24 10/100/1000BASE-T Ethernet electrical ports, 4 10GE SFP+ optical ports, 2 40GE QSFP+ optical ports (can be split into four 10GE ports)

Available Power Modules

Table 1-17 lists the power modules available in V100R005C10.

Table 1-17 Power modules available in V100R005C10

Power Module Type	Power Module	Power Description
AC power module	PAC-150WA	150 W AC power module (natural heat dissipation)
	ES0W2PSA0150*	150 W AC power module (natural heat dissipation)
	PAC-350WA-B	350 W AC power module (air exhaust on front panel)
	PAC-350WA-F	350 W AC power module (air intake on front panel)
	PAC-600WA-B	600 W AC power module (air exhaust on front panel)
	PAC-600WA-F	600 W AC power module (air intake on front panel)

Power Module Type	Power Module	Power Description
	PAC-600WB-B	600 W AC&240 V DC power module (air exhaust on front panel)
	PAC-600WB-F	600 W AC&240 V DC power module (air intake on front panel)
DC power module	PDC-350WA-B	350 W DC power module (air exhaust on front panel)
	PDC-350WA-F	350 W DC power module (air intake on front panel)
High-voltage DC power module	PHD-600WA-B	600 W high-voltage DC power module (air exhaust on front panel)
	PHD-600WA-F	600 W high-voltage DC power module (air intake on front panel)

Available Fan Modules

Table 1-18 lists the fan modules available in V100R005C10.

Table 1-18 Fan modules available in V100R005C10

Series	Model	Description
FAN-40EA	FAN-40EA-B	FAN-40EA series fan modules (air exhaust on front panel)
	FAN-40EA-F	FAN-40EA series fan modules (air intake on front panel)
FAN-40SB	FAN-40SB-B	FAN-40SB series fan modules (air exhaust on front panel)
	FAN-40SB-F	FAN-40SB series fan modules (air intake on front panel)
FAN-40HA	FAN-40HA-B	FAN-40HA series fan modules (air exhaust on front panel)
	FAN-40HA-F	FAN-40HA series fan modules (air intake on front panel)
FAN-060A	FAN-060A-B	FAN-060A series fan modules (air exhaust on front panel)

Series	Model	Description
	FAN-060A-F	FAN-060A series fan modules (air intake on front panel)
FAN-040A	FAN-040A-B*	FAN-040A series fan modules (air exhaust on front panel)
	FAN-040A-F*	FAN-040A series fan modules (air intake on front panel)

2 Chassis

About This Chapter

- [2.1 Naming Conventions](#)
- [2.2 Version Mapping](#)
- [2.3 Appearance and Structure](#)
- [2.4 Indicators](#)
- [2.5 Ports](#)
- [2.6 Specifications](#)
- [2.7 Ordering Information](#)

2.1 Naming Conventions

Figure 2-1 shows the CE7800&6800&5800 series switches naming conventions.

Figure 2-1 CE7800&6800&5800 series switches naming conventions

CE6850U-48S6Q-EI

A B C D E F

Table 2-1 describes the CE7800&6800&5800 series switches naming conventions.

Table 2-1 CE7800&6800&5800 series switches naming conventions

Field	Meaning
A	CloudEngine series data center switches ● CE78: CE7800 series ● CE68: CE6800 series ● CE58: CE5800 series
B	Product model category: ● 10: basic model ● 50: advanced model
C	Special function flag. This flag is not present if the product does not provide special functions. U : The product supports the unified port function.
D	Number and type of downlink interfaces: ● T: GE/10GBase-T electrical interfaces ● S: GE/10GE SFP+ optical interfaces ● Q: quad small form-factor pluggable plus (QSFP+) optical interfaces
E	Number and types of uplink interfaces: ● T: GE/10GBase-T electrical interfaces ● S: GE/10GE SFP+ optical interfaces ● Q: QSFP+ optical interfaces NOTE This field is not present in the product name if the product has only fixed interfaces and the uplink and downlink interfaces are the same type or if the product supports flexible service units.

Field	Meaning
F	Product model type: ● LI: model providing basic functions ● EI: model providing enhanced functions ● HI: model providing advanced functions

2.2 Version Mapping

Table 2-2 lists the mapping between the CE7800&6800&5800 series switches and software versions.

Table 2-2 Mapping between the CE7800&6800&5800 series switches and software versions

Series		Model	Available Version
CE7800	CE7850	CE7850-32Q-EI	V100R003C00 and later versions
CE6800	CE6810	CE6810-48S4Q-EI	V100R003C00 and later versions
		CE6810-48S4Q-LI	V100R003C10 and later versions
		CE6810-48S-LI	V100R003C10 and later versions
		CE6810-32T16S4Q-LI	V100R005C10 and later versions
		CE6810-24S2Q-LI	V100R005C10 and later versions
	CE6850	CE6850-48S4Q-EI	V100R001C00 and later versions
		CE6850-48T4Q-EI	V100R001C00 and later versions
		CE6850-48S6Q-HI	V100R005C00 and later versions
		CE6850-48T6Q-HI	V100R005C10 and later versions
		CE6851-48S6Q-HI	V100R005C10 and later versions
		CE6850U-48S6Q-HI	V100R005C10 and later versions
		CE6850U-24S2Q-HI	V100R005C10 and later versions
CE5800	CE5810	CE5810-24T4S-EI	V100R002C00 and later versions
		CE5810-48T4S-EI	V100R002C00 and later versions
	CE5850	CE5850-48T4S2Q-EI	V100R001C00 and later versions
		CE5850-48T4S2Q-HI	V100R003C00 and later versions
	CE5855	CE5855-48T4S2Q-EI	V100R005C10 and later versions

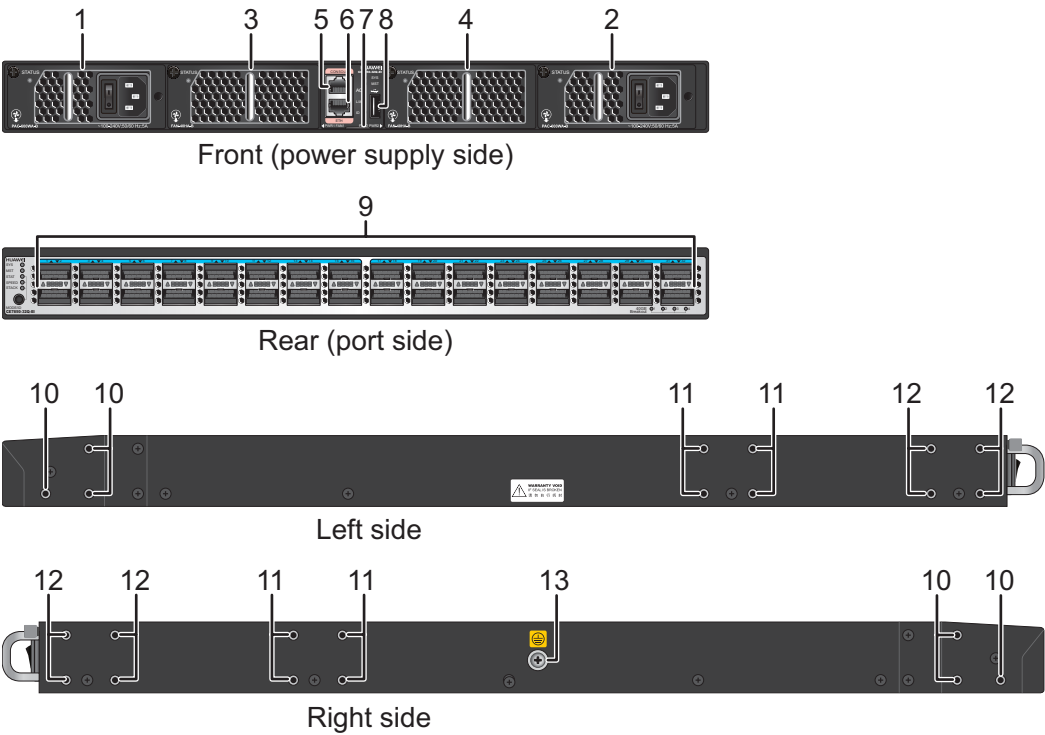
Series		Model	Available Version
		CE5855-24T4S2Q-EI	V100R005C10 and later versions

2.3 Appearance and Structure

 **NOTE**
The figures in this document are for reference only.

CE7850-32Q-EI

Figure 2-2 Appearance of the CE7850-32Q-EI

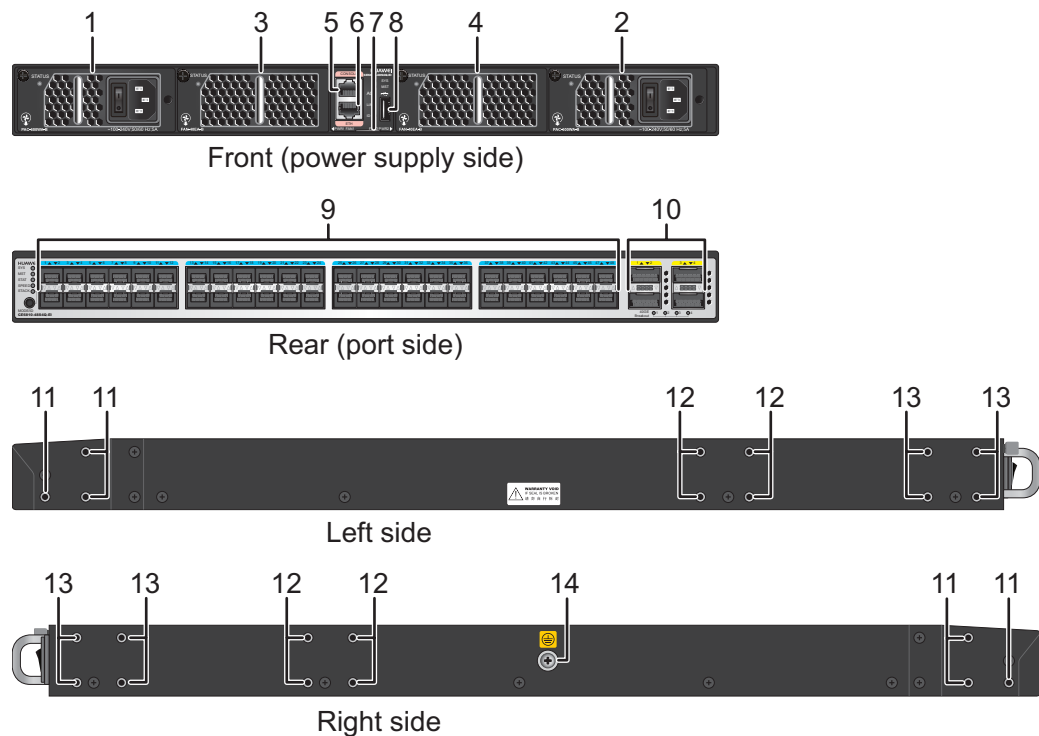


1	Power supply slot 1 Applicable power modules: ● 600 W AC power module	2	Power supply slot 2 Applicable power modules: ● 600 W AC power module
3	Fan slot 1 Applicable fan modules: ● FAN-40HA series fan modules	4	Fan slot 2 Applicable fan modules: ● FAN-40HA series fan modules
5	Console port	6	ETH management port (RJ45)

7	Barcode label NOTE This label is drawable, and you can pull it outward to view the barcode and MAC address of the switch.	8	USB port
9	Thirty-two 40GE QSFP+ Ethernet optical ports NOTE A 40GE QSFP+ port can be split into four 10GE ports. Applicable modules and cables: ● 40GE optical module ● 10 m QSFP+ AOC cable (QSFP+ to QSFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to 4*SFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to QSFP+)	10	Three port-side mounting holes for mounting brackets
11	Four middle mounting holes for mounting brackets	12	Four power-supply-side mounting holes for mounting brackets
13	Ground screw	-	-

CE6810-48S4Q-EI

Figure 2-3 Appearance of the CE6810-48S4Q-EI

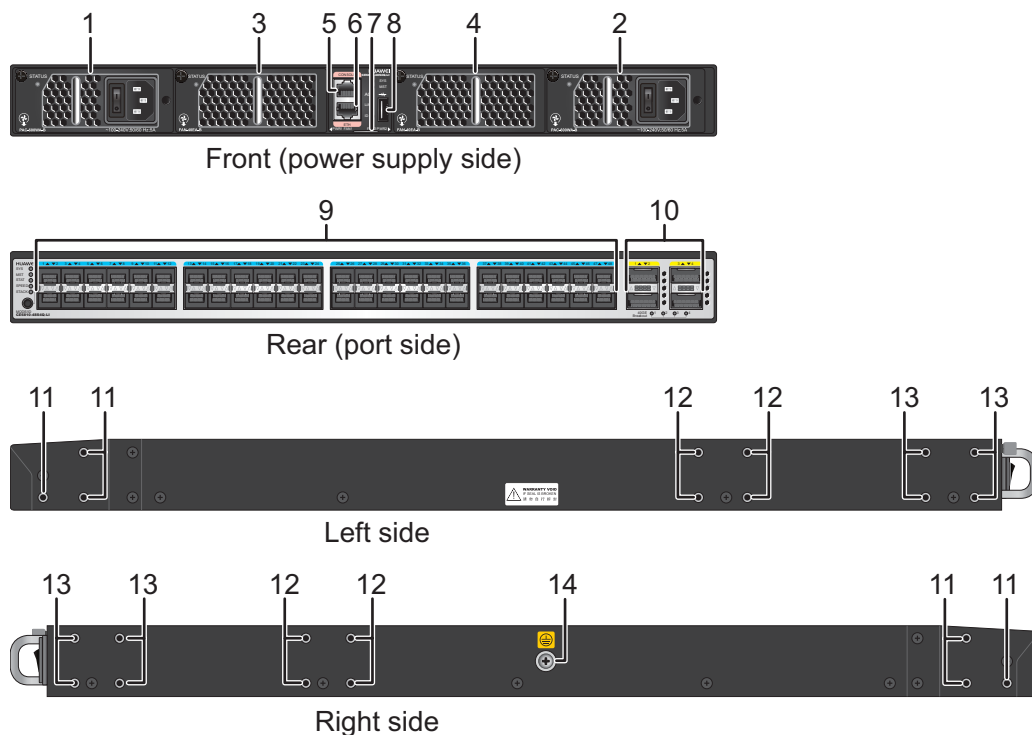


1	Power supply slot 1 Applicable power modules: ● 350 W DC power module ● 600 W AC power module	2	Power supply slot 2 Applicable power modules: ● 350 W DC power module ● 600 W AC power module
3	Fan slot 1 Applicable fan modules: ● FAN-40EA series fan modules	4	Fan slot 2 Applicable fan modules: ● FAN-40EA series fan modules
5	Console port	6	ETH management port (RJ45)
7	Barcode label NOTE This label is drawable, and you can pull it outward to view the barcode and MAC address of the switch.	8	USB port

9	Forty-eight 10GE SFP+ Ethernet optical ports Applicable modules and cables: ● 10GE optical module (OSXD22N00 and LE2MXSC80FF0 not supported) ● 10GE-CWDM optical module ● GE optical module ● GE-CWDM optical module ● GE copper module (works at 10 Mbit/s, 100 Mbit/s, or 1000 Mbit/s) ● 3 m, 10 m, 20 m SFP+ AOC cables ● 1 m, 3 m, 5 m, 7 m (active), 10 m (active) SFP+ copper cables	10	Four 40GE QSFP+ Ethernet optical ports NOTE A 40GE QSFP+ port can be split into four 10GE ports. Applicable modules and cables: ● 40GE optical module ● 10 m QSFP+ AOC cable (QSFP+ to QSFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to 4*SFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to QSFP+)
11	Three port-side mounting holes for mounting brackets	12	Four middle mounting holes for mounting brackets
13	Four power-supply-side mounting holes for mounting brackets	14	Ground screw

CE6810-48S4Q-LI

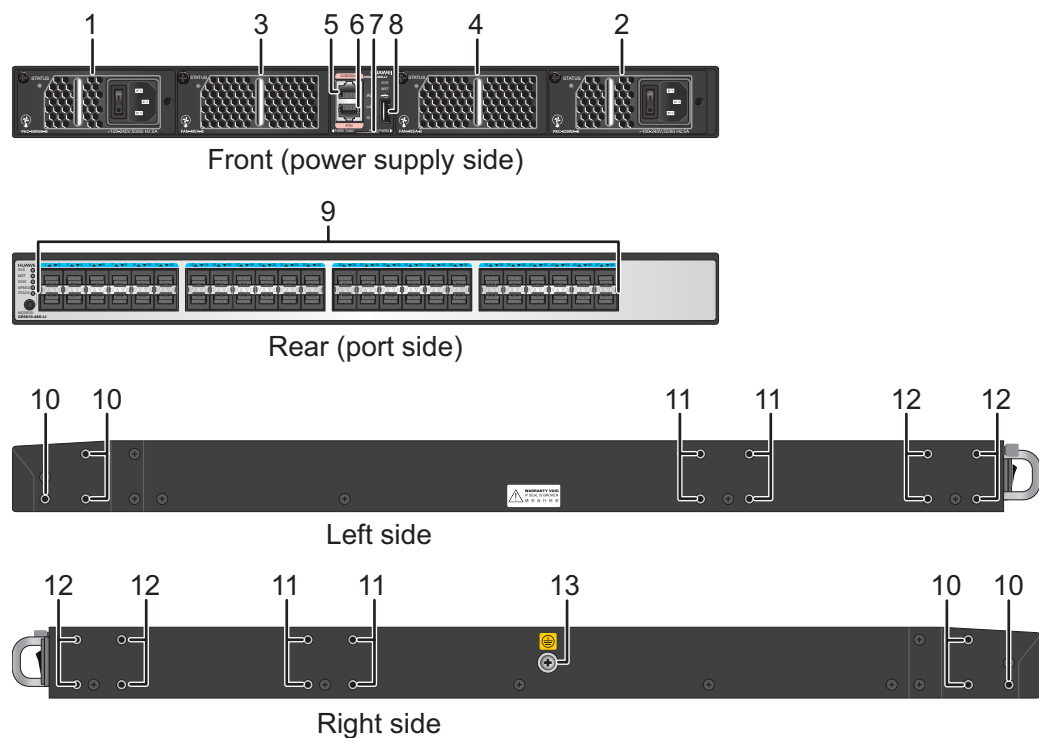
Figure 2-4 Appearance of the CE6810-48S4Q-LI



1	Power supply slot 1 Applicable power modules: ● 350 W DC power module ● 600 W AC power module	2	Power supply slot 2 Applicable power modules: ● 350 W DC power module ● 600 W AC power module
3	Fan slot 1 Applicable fan modules: ● FAN-40EA series fan modules	4	Fan slot 2 Applicable fan modules: ● FAN-40EA series fan modules
5	Console port	6	ETH management port (RJ45)
7	Barcode label NOTE This label is drawable, and you can pull it outward to view the barcode and MAC address of the switch.	8	USB port
9	Forty-eight 10GE SFP+ Ethernet optical ports Applicable modules and cables: ● 10GE optical module (OSXD22N00 and LE2MXSC80FF0 not supported) ● 10GE-CWDM optical module ● GE optical module ● GE-CWDM optical module ● GE copper module (works at 10 Mbit/s, 100 Mbit/s, or 1000 Mbit/s) ● 3 m, 10 m, 20 m SFP+ AOC cables ● 1 m, 3 m, 5 m, 7 m (active), 10 m (active) SFP+ copper cables	10	Four 40GE QSFP+ Ethernet optical ports NOTE A 40GE QSFP+ port can be split into four 10GE ports. Applicable modules and cables: ● 40GE optical module ● 10 m QSFP+ AOC cable (QSFP+ to QSFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to 4*SFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to QSFP+)
11	Three port-side mounting holes for mounting brackets	12	Four middle mounting holes for mounting brackets
13	Four power-supply-side mounting holes for mounting brackets	14	Ground screw

CE6810-48S-LI

Figure 2-5 Appearance of the CE6810-48S-LI

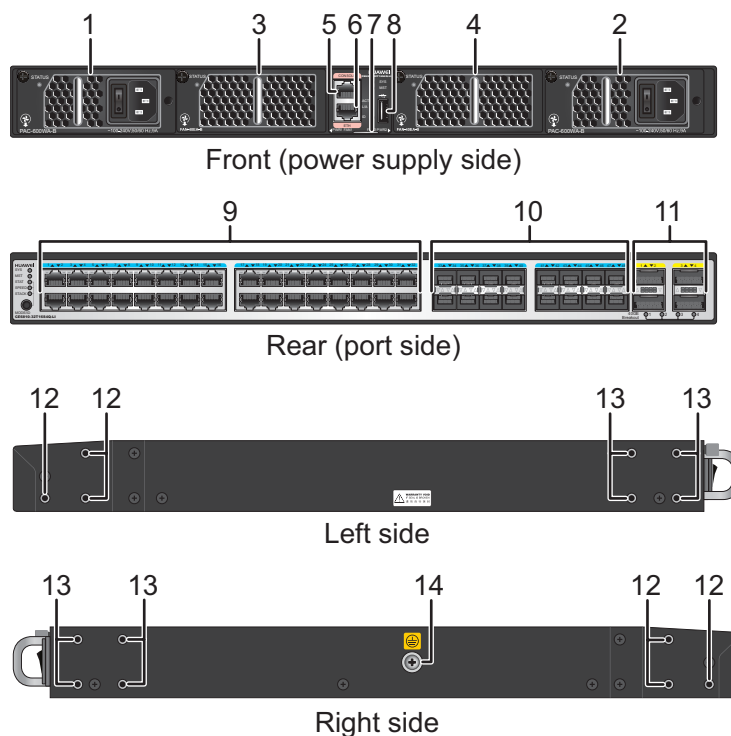


1	Power supply slot 1 Applicable power modules: ● 350 W DC power module ● 600 W AC power module	2	Power supply slot 2 Applicable power modules: ● 350 W DC power module ● 600 W AC power module
3	Fan slot 1 Applicable fan modules: ● FAN-40EA series fan modules	4	Fan slot 2 Applicable fan modules: ● FAN-40EA series fan modules
5	Console port	6	ETH management port (RJ45)
7	Barcode label NOTE This label is drawable, and you can pull it outward to view the barcode and MAC address of the switch.	8	USB port

9	Forty-eight 10GE SFP+ Ethernet optical ports Applicable modules and cables: ● 10GE optical module (OSXD22N00 and LE2MXSC80FF0 not supported) ● 10GE-CWDM optical module ● GE optical module ● GE-CWDM optical module ● GE copper module (works at 10 Mbit/s, 100 Mbit/s, or 1000 Mbit/s) ● 3 m, 10 m, 20 m SFP+ AOC cables ● 1 m, 3 m, 5 m, 7 m (active), 10 m (active) SFP+ copper cables	10	Three port-side mounting holes for mounting brackets
11	Four middle mounting holes for mounting brackets	12	Four power-supply-side mounting holes for mounting brackets
13	Ground screw	-	-

CE6810-32T16S4Q-LI

Figure 2-6 Appearance of the CE6810-32T16S4Q-LI

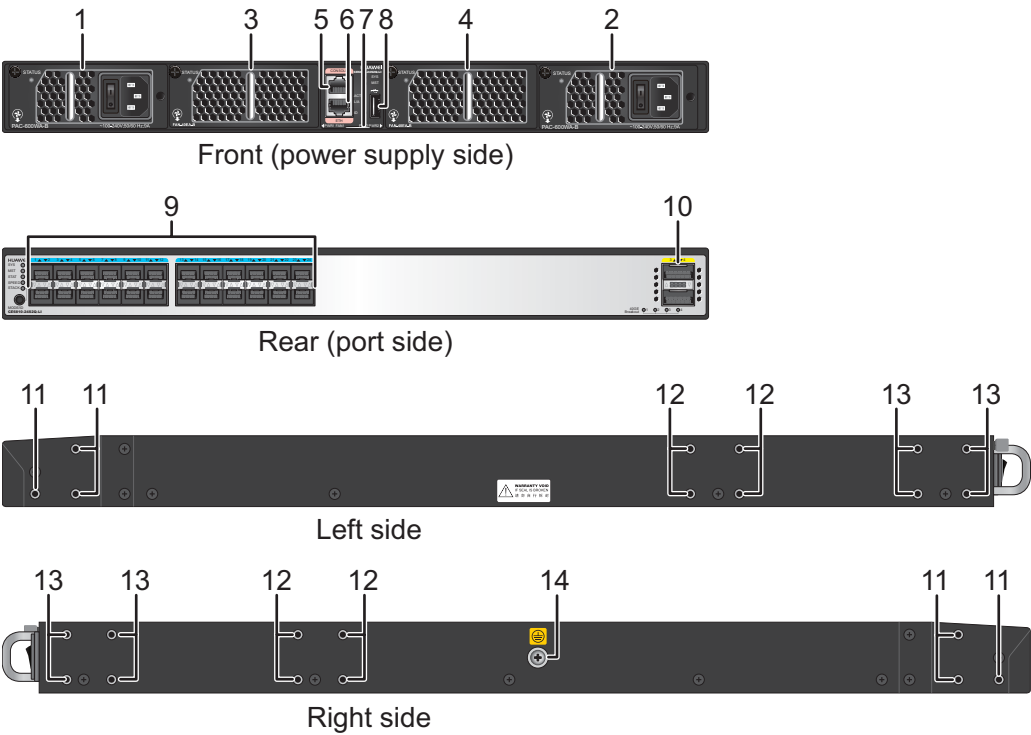


1	Power supply slot 1 Applicable power modules: ● 350 W DC power module ● 600 W AC power module	2	Power supply slot 2 Applicable power modules: ● 350 W DC power module ● 600 W AC power module
3	Fan slot 1 Applicable fan modules: ● FAN-40EA series fan modules	4	Fan slot 2 Applicable fan modules: ● FAN-40EA series fan modules
5	Console port	6	ETH management port (RJ45)
7	Barcode label NOTE This label is drawable, and you can pull it outward to view the barcode and MAC address of the switch.	8	USB port
9	Thirty-two 10GBASE-T Ethernet electrical ports	10	Sixteen 10GE SFP+ Ethernet optical ports Applicable modules and cables: ● 10GE optical module (OSXD22N00 and LE2MXSC80FF0 not supported) ● 10GE-CWDM optical module ● GE optical module ● GE-CWDM optical module ● GE copper module (works at 10 Mbit/s, 100 Mbit/s, or 1000 Mbit/s) ● 3 m, 10 m, 20 m SFP+ AOC cables ● 1 m, 3 m, 5 m, 7 m (active), 10 m (active) SFP+ copper cables
11	Four 40GE QSFP+ Ethernet optical ports NOTE A 40GE QSFP+ port can be split into four 10GE ports. Applicable modules and cables: ● 40GE optical module ● 10 m QSFP+ AOC cable (QSFP+ to QSFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to 4*SFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to QSFP+)	12	Three port-side mounting holes for mounting brackets

13	Four power-supply-side mounting holes for mounting brackets	14	Ground screw
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CE6810-24S2Q-LI

Figure 2-7 Appearance of the CE6810-24S2Q-LI

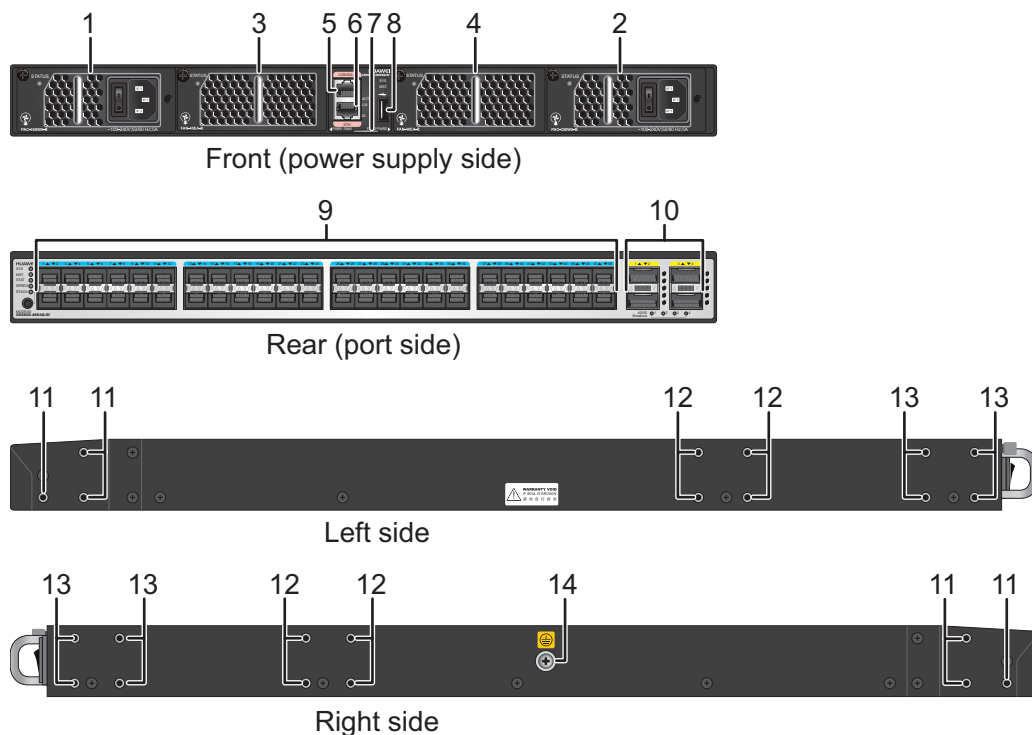


1	Power supply slot 1 Applicable power modules: ● 350 W DC power module ● 600 W AC power module	2	Power supply slot 2 Applicable power modules: ● 350 W DC power module ● 600 W AC power module
3	Fan slot 1 Applicable fan modules: ● FAN-40EA series fan modules	4	Fan slot 2 Applicable fan modules: ● FAN-40EA series fan modules
5	Console port	6	ETH management port (RJ45)
7	Barcode label NOTE This label is drawable, and you can pull it outward to view the barcode and MAC address of the switch.	8	USB port

9	Twenty-four 10GE SFP+ Ethernet optical ports Applicable modules and cables: ● 10GE optical module (OSXD22N00 and LE2MXSC80FF0 not supported) ● 10GE-CWDM optical module ● GE optical module ● GE-CWDM optical module ● GE copper module (works at 10 Mbit/s, 100 Mbit/s, or 1000 Mbit/s) ● 3 m, 10 m, 20 m SFP+ AOC cables ● 1 m, 3 m, 5 m, 7 m (active), 10 m (active) SFP+ copper cables	10	Two 40GE QSFP+ Ethernet optical ports NOTE A 40GE QSFP+ port can be split into four 10GE ports. Applicable modules and cables: ● 40GE optical module ● 10 m QSFP+ AOC cable (QSFP+ to QSFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to 4*SFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to QSFP+)
11	Three port-side mounting holes for mounting brackets	12	Four middle mounting holes for mounting brackets
13	Four power-supply-side mounting holes for mounting brackets	14	Ground screw

CE6850-48S4Q-EI

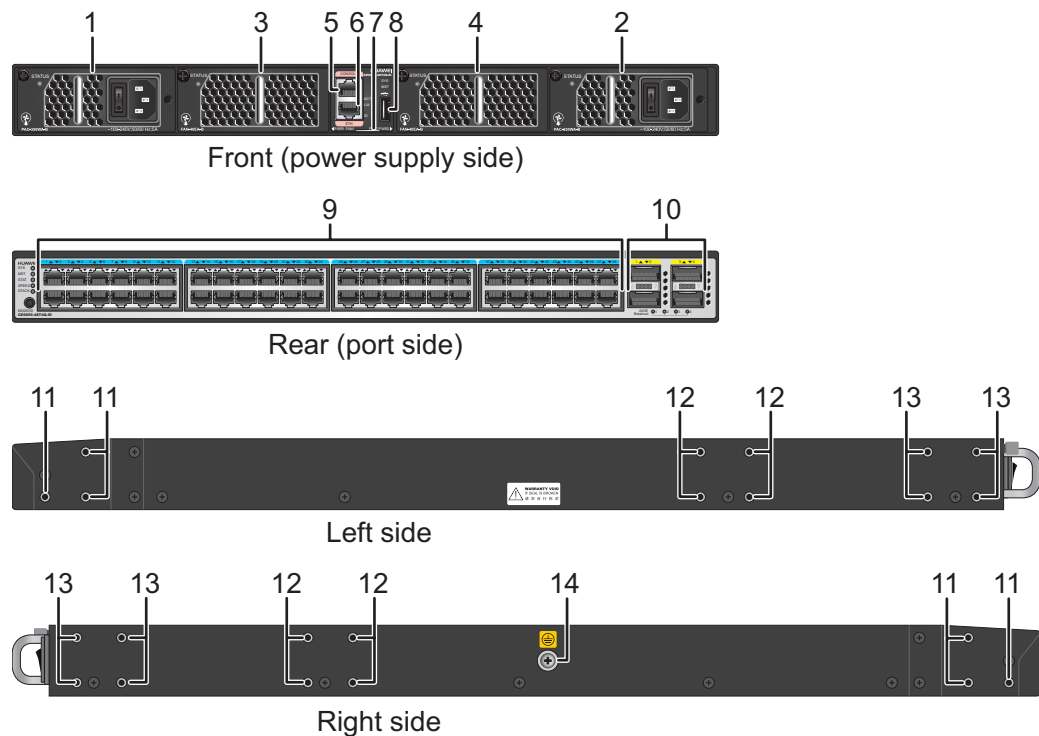
Figure 2-8 Appearance of the CE6850-48S4Q-EI



1	Power supply slot 1 Applicable power modules: ● 350 W AC power module ● 350 W DC power module	2	Power supply slot 2 Applicable power modules: ● 350 W AC power module ● 350 W DC power module
3	Fan slot 1 Applicable fan modules: ● FAN-40EA series fan modules	4	Fan slot 2 Applicable fan modules: ● FAN-40EA series fan modules
5	Console port	6	ETH management port (RJ45)
7	Barcode label NOTE This label is drawable, and you can pull it outward to view the barcode and MAC address of the switch.	8	USB port
9	Forty-eight 10GE SFP+ Ethernet optical ports Applicable modules and cables: ● 10GE optical module ● 10GE-CWDM optical module ● GE optical module ● GE-CWDM optical module ● GE copper module (works at 10 Mbit/s, 100 Mbit/s, or 1000 Mbit/s) ● 3 m, 10 m, 20 m SFP+ AOC cables ● 1 m, 3 m, 5 m, 7 m (active), 10 m (active) SFP+ copper cables	10	Four 40GE QSFP+ Ethernet optical ports NOTE A 40GE QSFP+ port can be split into four 10GE ports. Applicable modules and cables: ● 40GE optical module ● 10 m QSFP+ AOC cable (QSFP+ to QSFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to 4*SFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to QSFP+)
11	Three port-side mounting holes for mounting brackets	12	Four middle mounting holes for mounting brackets
13	Four power-supply-side mounting holes for mounting brackets	14	Ground screw

CE6850-48T4Q-EI

Figure 2-9 Appearance of the CE6850-48T4Q-EI

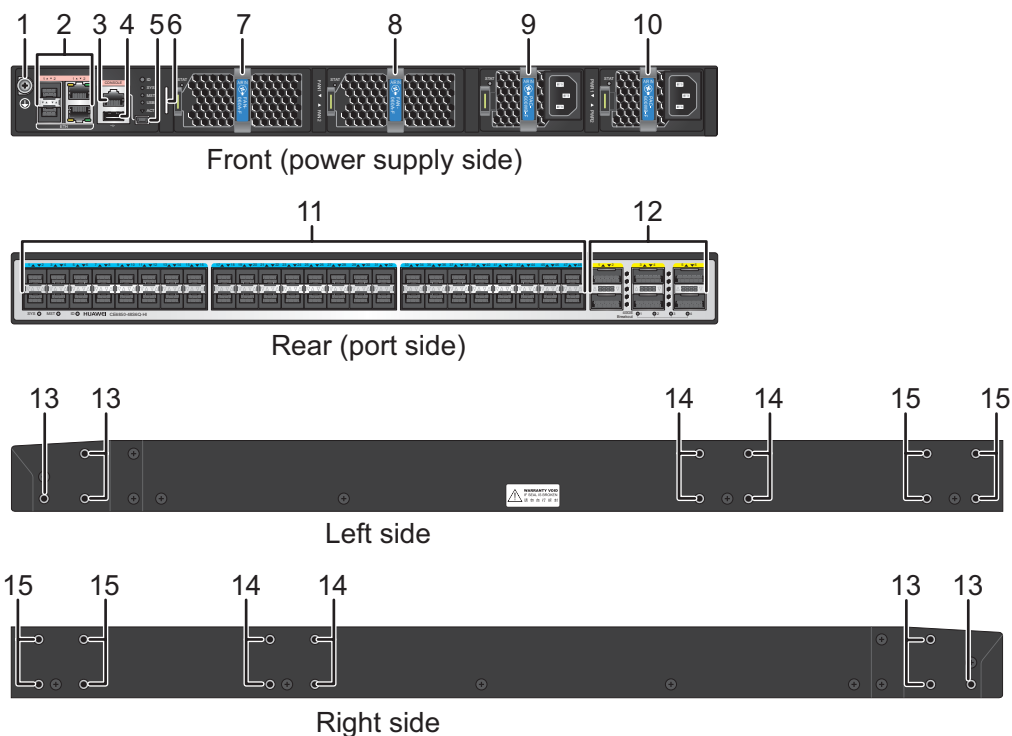


1	Power supply slot 1 Applicable power modules: ● 350 W AC power module ● 600 W AC power module	2	Power supply slot 2 Applicable power modules: ● 350 W AC power module ● 600 W AC power module
3	Fan slot 1 Applicable fan modules: ● FAN-40EA series fan modules	4	Fan slot 2 Applicable fan modules: ● FAN-40EA series fan modules
5	Console port	6	ETH management port (RJ45)
7	Barcode label NOTE This label is drawable, and you can pull it outward to view the barcode and MAC address of the switch.	8	USB port

9	Forty-eight 10GBASE-T Ethernet electrical ports NOTE When a CE6850-48T4Q-EI switch uses 350 W AC power modules and all its ports are in use, the length of each network cable used on the switch cannot exceed 30 m.	10	Four 40GE QSFP+ Ethernet optical ports NOTE A 40GE QSFP+ port can be split into four 10GE ports. Applicable modules and cables: ● 40GE optical module ● 10 m QSFP+ AOC cable (QSFP+ to QSFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to 4*SFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to QSFP+)
11	Three port-side mounting holes for mounting brackets	12	Four middle mounting holes for mounting brackets
13	Four power-supply-side mounting holes for mounting brackets	14	Ground screw

CE6850-48S6Q-HI

Figure 2-10 Appearance of the CE6850-48S6Q-HI

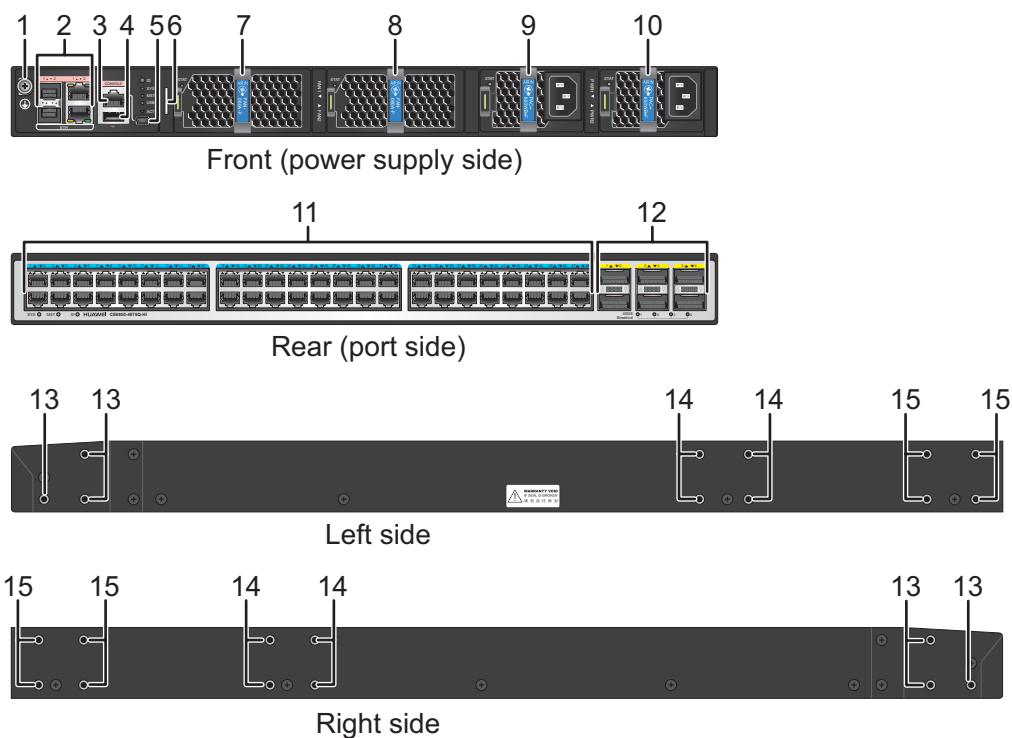


1	Ground screw	2	Two ETH management ports (combo) Applicable transceiver modules for the GE optical port of the combo port: ● FE optical module ● GE optical module ● GE-CWDM optical module NOTE The combo optical port uses a 100M or GE optical module and matching fibers. A 100M optical module can be used only after the switch starts successfully. If a 10GE optical module is installed, the interface can go Up, but the system displays an alarm message, indicating that the interface does not support the optical module. If a GE copper module is installed and the remote interface has a GE copper module installed, the local interface can go Up but does not support rate configuration.
3	Console port	4	USB port
5	Mini USB port	6	Barcode label NOTE This label is drawable, and you can pull it outward to view the barcode and MAC address of the switch.
7	Fan slot 1 Applicable fan modules: ● FAN-060A series fan modules	8	Fan slot 2 Applicable fan modules: ● FAN-060A series fan modules
9	Power supply slot 1 Applicable power modules: ● 600 W AC&240 V DC power module ● 600 W high-voltage DC power module	10	Power supply slot 2 Applicable power modules: ● 600 W AC&240 V DC power module ● 600 W high-voltage DC power module

11	Forty-eight 10GE SFP+ Ethernet optical ports Applicable transceiver modules and cables: ● 10GE optical module (OSXD22N00 and LE2MXSC80FF0 not supported) ● 10GE-CWDM optical module ● GE optical module ● GE-CWDM optical module ● GE copper module (only works at 1000 Mbit/s) ● 3 m, 10 m, 20 m SFP+ AOC cables ● 1 m, 3 m, 5 m, 7 m (active), 10 m (active) SFP+ copper cables	1 2	Six 40GE QSFP+ Ethernet optical ports NOTE A 40GE QSFP+ port can be split into four 10GE ports. Applicable modules and cables: ● 40GE optical module ● 10 m QSFP+ AOC cable (QSFP+ to QSFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to 4*SFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to QSFP+)
13	Three port-side mounting holes for mounting brackets	1 4	Four middle mounting holes for mounting brackets
15	Four power-supply-side mounting holes for mounting brackets	-	-

CE6850-48T6Q-HI

Figure 2-11 Appearance of the CE6850-48T6Q-HI

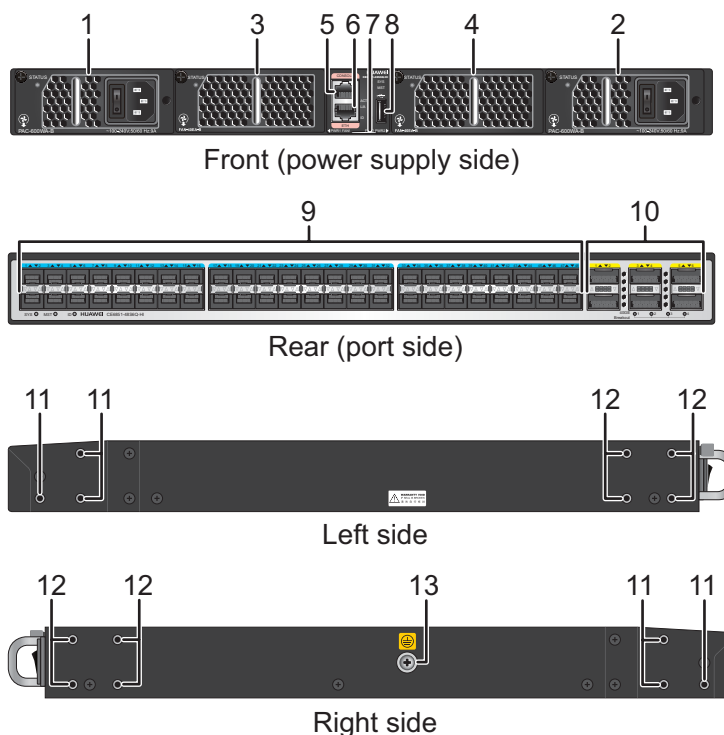


1	Ground screw	2	Two ETH management ports (combo) Applicable transceiver modules for the GE optical port of the combo port: ● FE optical module ● GE optical module ● GE-CWDM optical module NOTE The combo optical port uses a 100M or GE optical module and matching fibers. A 100M optical module can be used only after the switch starts successfully. If a 10GE optical module is installed, the interface can go Up, but the system displays an alarm message, indicating that the interface does not support the optical module. If a GE copper module is installed and the remote interface has a GE copper module installed, the local interface can go Up but does not support rate configuration.
3	Console port	4	USB port
5	Mini USB port	6	Barcode label NOTE This label is drawable, and you can pull it outward to view the barcode and MAC address of the switch.
7	Fan slot 1 Applicable fan modules: ● FAN-060A series fan modules	8	Fan slot 2 Applicable fan modules: ● FAN-060A series fan modules
9	Power supply slot 1 Applicable power modules: ● 600 W AC&240 V DC power module ● 600 W high-voltage DC power module	10	Power supply slot 2 Applicable power modules: ● 600 W AC&240 V DC power module ● 600 W high-voltage DC power module

11	Forty-eight 10GBASE-T Ethernet electrical ports	1 2	Six 40GE QSFP+ Ethernet optical ports NOTE A 40GE QSFP+ port can be split into four 10GE ports. Applicable modules and cables: ● 40GE optical module ● 10 m QSFP+ AOC cable (QSFP+ to QSFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to 4*SFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to QSFP+)
13	Three port-side mounting holes for mounting brackets	1 4	Four middle mounting holes for mounting brackets
15	Four power-supply-side mounting holes for mounting brackets	-	-

CE6851-48S6Q-HI

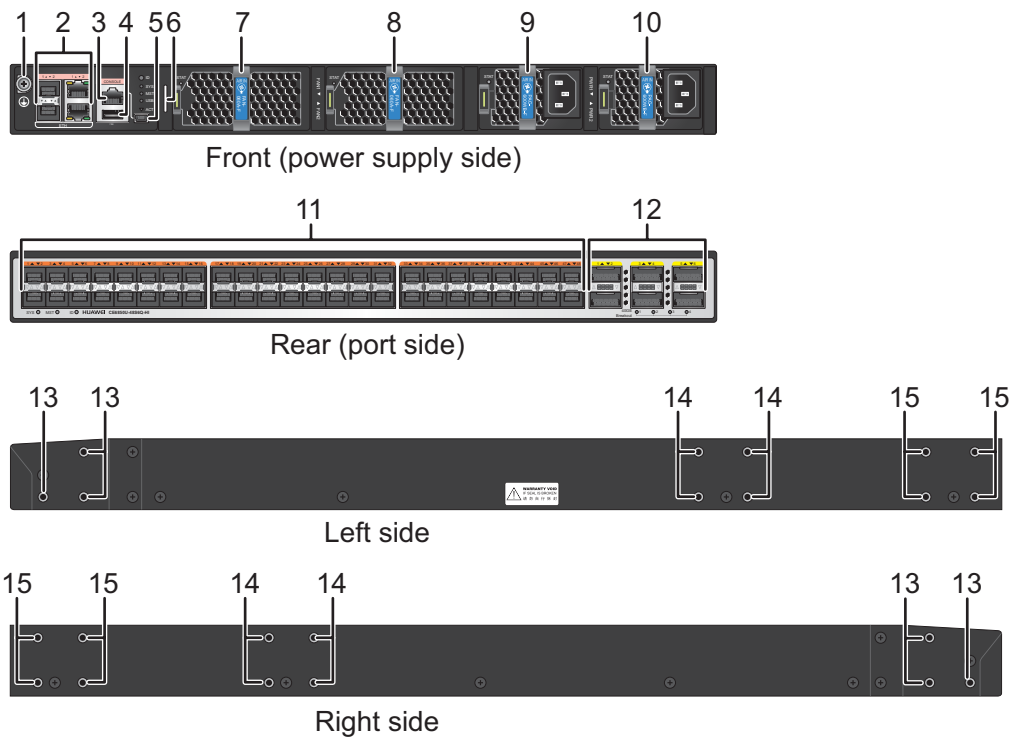
Figure 2-12 Appearance of the CE6851-48S6Q-HI



1	Power supply slot 1 Applicable power modules: ● 350 W DC power module ● 600 W AC power module	2	Power supply slot 2 Applicable power modules: ● 350 W DC power module ● 600 W AC power module
3	Fan slot 1 Applicable fan modules: ● FAN-40EA series fan modules	4	Fan slot 2 Applicable fan modules: ● FAN-40EA series fan modules
5	Console port	6	ETH management port (RJ45)
7	Barcode label NOTE This label is drawable, and you can pull it outward to view the barcode and MAC address of the switch.	8	USB port
9	Forty-eight 10GE SFP+ Ethernet optical ports Applicable transceiver modules and cables: ● 10GE optical module (OSXD22N00 and LE2MXSC80FF0 not supported) ● 10GE-CWDM optical module ● GE optical module ● GE-CWDM optical module ● GE copper module (only works at 1000 Mbit/s) ● 3 m, 10 m, 20 m SFP+ AOC cables ● 1 m, 3 m, 5 m, 7 m (active), 10 m (active) SFP+ copper cables	10	Six 40GE QSFP+ Ethernet optical ports NOTE A 40GE QSFP+ port can be split into four 10GE ports. Applicable modules and cables: ● 40GE optical module ● 10 m QSFP+ AOC cable (QSFP+ to QSFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to 4*SFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to QSFP+)
11	Three port-side mounting holes for mounting brackets	12	Four power-supply-side mounting holes for mounting brackets
13	Ground screw	-	-

CE6850U-48S6Q-HI

Figure 2-13 Appearance of the CE6850U-48S6Q-HI

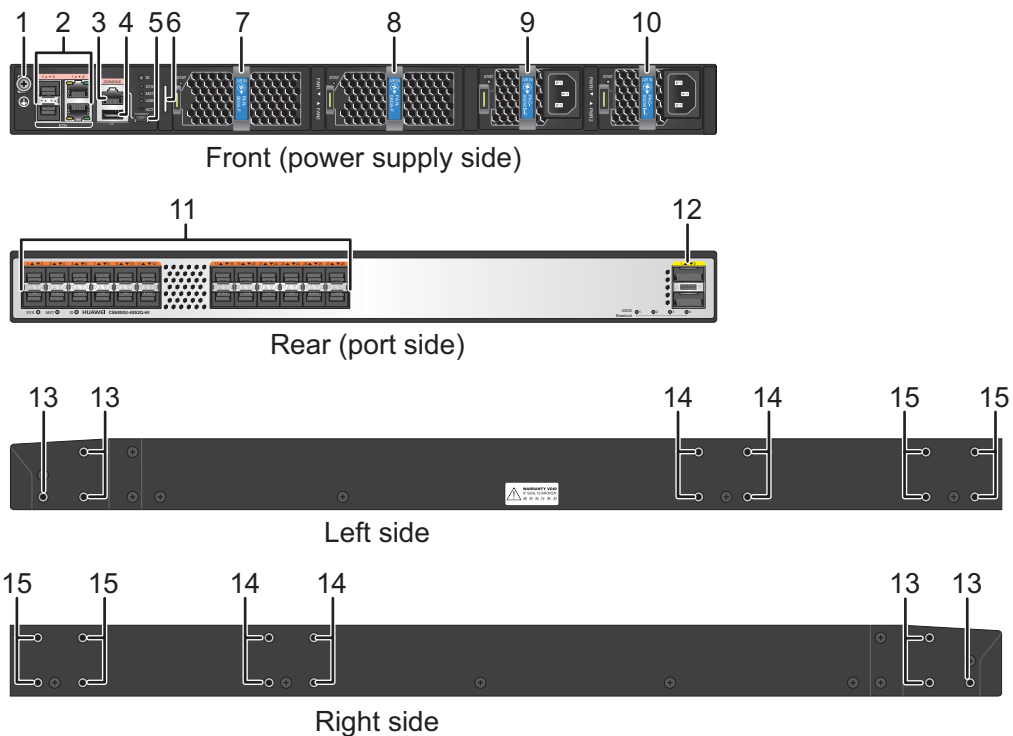


1	Ground screw	2	Two ETH management ports (combo) Applicable transceiver modules for the GE optical port of the combo port: ● FE optical module ● GE optical module ● GE-CWDM optical module NOTE The combo optical port uses a 100M or GE optical module and matching fibers. A 100M optical module can be used only after the switch starts successfully. If a 10GE optical module is installed, the interface can go Up, but the system displays an alarm message, indicating that the interface does not support the optical module. If a GE copper module is installed and the remote interface has a GE copper module installed, the local interface can go Up but does not support rate configuration.
3	Console port	4	USB port

5	Mini USB port	6	Barcode label NOTE This label is drawable, and you can pull it outward to view the barcode and MAC address of the switch.
7	Fan slot 1 Applicable fan modules: ● FAN-060A series fan modules	8	Fan slot 2 Applicable fan modules: ● FAN-060A series fan modules
9	Power supply slot 1 Applicable power modules: ● 600 W AC&240 V DC power module ● 600 W high-voltage DC power module	10	Power supply slot 2 Applicable power modules: ● 600 W AC&240 V DC power module ● 600 W high-voltage DC power module
11	Forty-eight 10GE SFP+ Ethernet optical ports Applicable transceiver modules and cables: ● FC optical module ● 10GE optical module ● 10GE-CWDM optical module ● GE optical module ● GE-CWDM optical module ● GE copper module (only works at 1000 Mbit/s) ● 3 m, 10 m, 20 m SFP+ AOC cables ● 1 m, 3 m, 5 m, 7 m (active), 10 m (active) SFP+ copper cables	12	Six 40GE QSFP+ Ethernet optical ports NOTE A 40GE QSFP+ port can be split into four 10GE ports. Applicable modules and cables: ● 40GE optical module ● 10 m QSFP+ AOC cable (QSFP+ to QSFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to 4*SFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to QSFP+)
13	Three port-side mounting holes for mounting brackets	14	Four middle mounting holes for mounting brackets
15	Four power-supply-side mounting holes for mounting brackets	-	-

CE6850U-24S2Q-HI

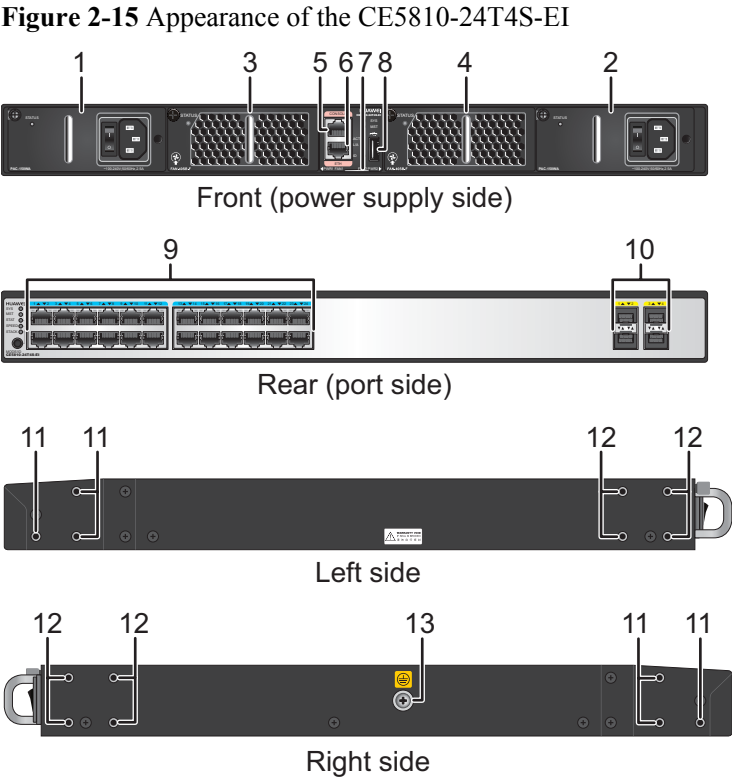
Figure 2-14 Appearance of the CE6850U-24S2Q-HI



1	Ground screw	2	Two ETH management ports (combo) Applicable transceiver modules for the GE optical port of the combo port: ● FE optical module ● GE optical module ● GE-CWDM optical module NOTE The combo optical port uses a 100M or GE optical module and matching fibers. A 100M optical module can be used only after the switch starts successfully. If a 10GE optical module is installed, the interface can go Up, but the system displays an alarm message, indicating that the interface does not support the optical module. If a GE copper module is installed and the remote interface has a GE copper module installed, the local interface can go Up but does not support rate configuration.
3	Console port	4	USB port

5	Mini USB port	6	Barcode label NOTE This label is drawable, and you can pull it outward to view the barcode and MAC address of the switch.
7	Fan slot 1 Applicable fan modules: ● FAN-060A series fan modules	8	Fan slot 2 Applicable fan modules: ● FAN-060A series fan modules
9	Power supply slot 1 Applicable power modules: ● 600 W AC&240 V DC power module ● 600 W high-voltage DC power module	10	Power supply slot 2 Applicable power modules: ● 600 W AC&240 V DC power module ● 600 W high-voltage DC power module
11	Twenty-four 10GE SFP+ Ethernet optical ports Applicable transceiver modules and cables: ● FC optical module ● 10GE optical module ● 10GE-CWDM optical module ● GE optical module ● GE-CWDM optical module ● GE copper module (only works at 1000 Mbit/s) ● 3 m, 10 m, 20 m SFP+ AOC cables ● 1 m, 3 m, 5 m, 7 m (active), 10 m (active) SFP+ copper cables	12	Two 40GE QSFP+ Ethernet optical ports NOTE A 40GE QSFP+ port can be split into four 10GE ports. Applicable modules and cables: ● 40GE optical module ● 10 m QSFP+ AOC cable (QSFP+ to QSFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to 4*SFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to QSFP+)
13	Three port-side mounting holes for mounting brackets	14	Four middle mounting holes for mounting brackets
15	Four power-supply-side mounting holes for mounting brackets	-	-

CE5810-24T4S-EI

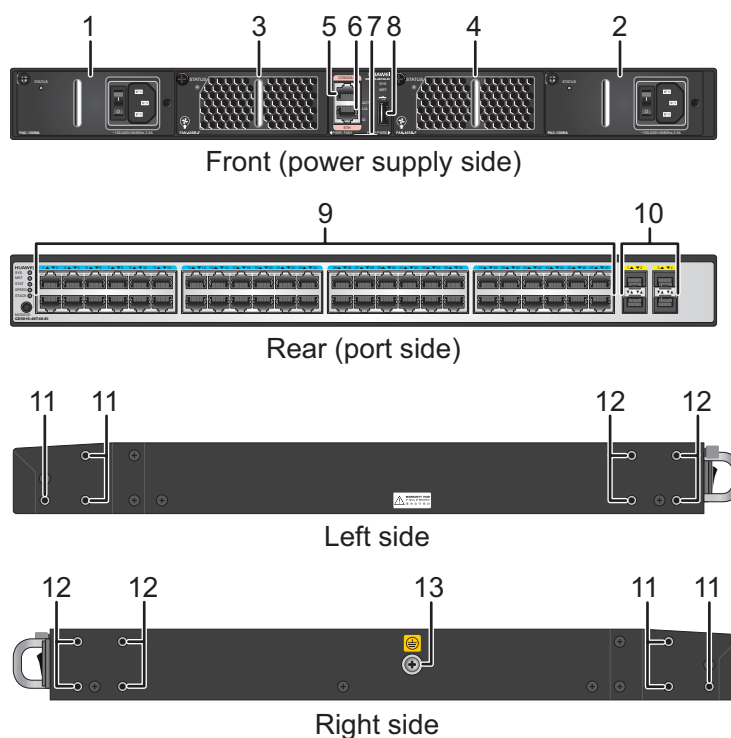


1	Power supply slot 1 Applicable power modules: ● 150 W AC power module (PAC-150WA) ● 350 W DC power module	2	Power supply slot 2 Applicable power modules: ● 150 W AC power module (PAC-150WA) ● 350 W DC power module
3	Fan slot 1 Applicable fan modules: ● FAN-40SB series fan modules	4	Fan slot 2 Applicable fan modules: ● FAN-40SB series fan modules
5	Console port	6	ETH management port (RJ45)
7	Barcode label NOTE This label is drawable, and you can pull it outward to view the barcode and MAC address of the switch.	8	USB port

9	Twenty-four 10/100/1000BASE-T Ethernet electrical ports	10	Four 10GE SFP+ Ethernet optical ports Applicable transceiver modules and cables: ● 10GE optical module ● 10GE-CWDM optical module ● GE optical module ● GE-CWDM optical module ● GE copper module (only works at 1000 Mbit/s) ● 3 m, 10 m, 20 m SFP+ AOC cables ● 1 m, 3 m, 5 m, 7 m (active), 10 m (active) SFP+ copper cables
11	Three port-side mounting holes for mounting brackets	12	Four power-supply-side mounting holes for mounting brackets
13	Ground screw	-	-

CE5810-48T4S-EI

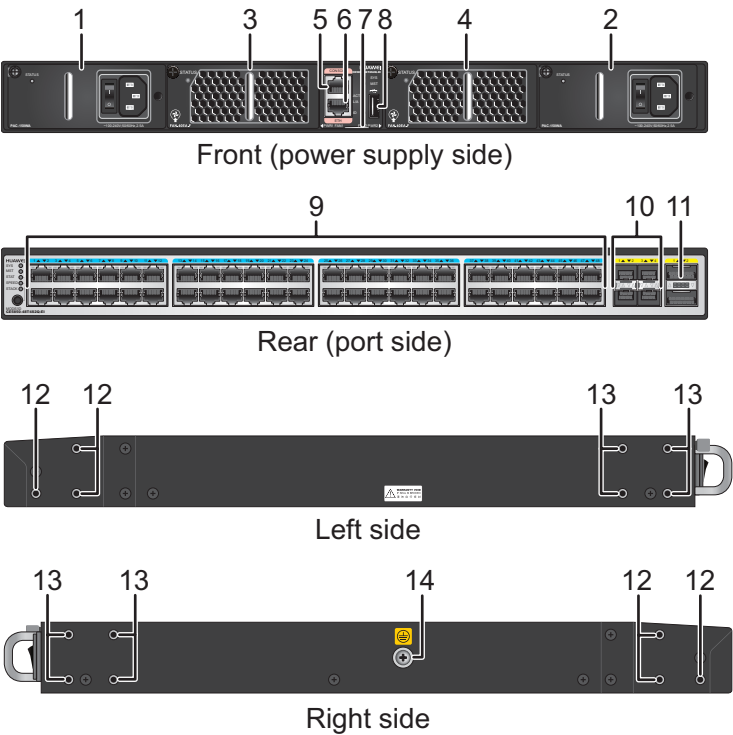
Figure 2-16 Appearance of the CE5810-48T4S-EI



1	Power supply slot 1 Applicable power modules: ● 150 W AC power module (PAC-150WA) ● 350 W DC power module	2	Power supply slot 2 Applicable power modules: ● 150 W AC power module (PAC-150WA) ● 350 W DC power module
3	Fan slot 1 Applicable fan modules: ● FAN-40SB series fan modules	4	Fan slot 2 Applicable fan modules: ● FAN-40SB series fan modules
5	Console port	6	ETH management port (RJ45)
7	Barcode label NOTE This label is drawable, and you can pull it outward to view the barcode and MAC address of the switch.	8	USB port
9	Forty-eight 10/100/1000BASE-T Ethernet electrical ports	10	Four 10GE SFP+ Ethernet optical ports Applicable transceiver modules and cables: ● 10GE optical module ● 10GE-CWDM optical module ● GE optical module ● GE-CWDM optical module ● GE copper module (only works at 1000 Mbit/s) ● 3 m, 10 m, 20 m SFP+ AOC cables ● 1 m, 3 m, 5 m, 7 m (active), 10 m (active) SFP+ copper cables
11	Three port-side mounting holes for mounting brackets	12	Four power-supply-side mounting holes for mounting brackets
13	Ground screw	-	-

CE5850-48T4S2Q-EI

Figure 2-17 Appearance of the CE5850-48T4S2Q-EI

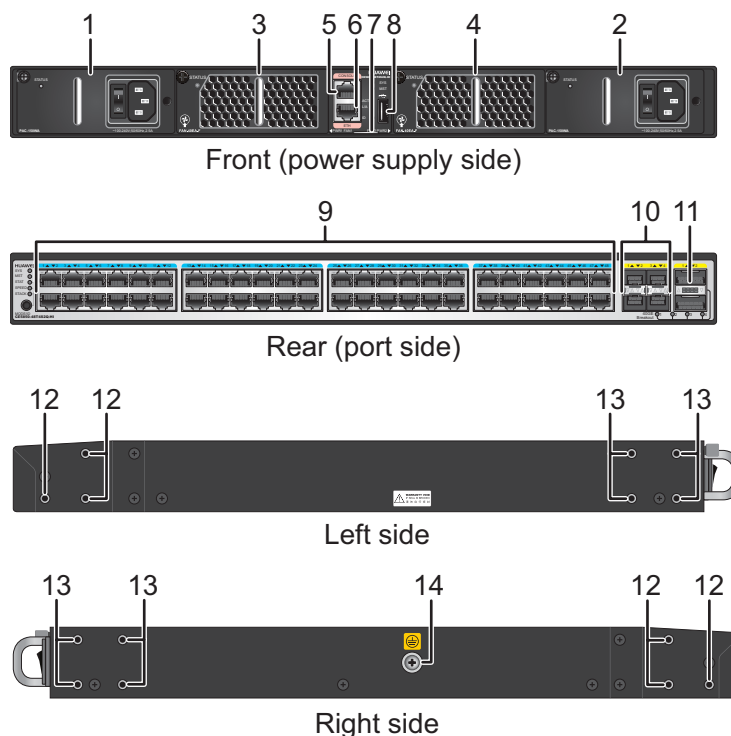


1	Power supply slot 1 Applicable power modules: ● 150 W AC power module (PAC-150WA) ● 350 W DC power module	2	Power supply slot 2 Applicable power modules: ● 150 W AC power module (PAC-150WA) ● 350 W DC power module
3	Fan slot 1 Applicable fan modules: ● FAN-40EA series fan modules	4	Fan slot 2 Applicable fan modules: ● FAN-40EA series fan modules
5	Console port	6	ETH management port (RJ45)
7	Barcode label NOTE This label is drawable, and you can pull it outward to view the barcode and MAC address of the switch.	8	USB port

9	Forty-eight 10/100/1000BASE-T Ethernet electrical ports	10	Four 10GE SFP+ Ethernet optical ports Applicable modules and cables: ● 10GE optical module ● 10GE-CWDM optical module ● GE optical module ● GE-CWDM optical module ● GE copper module (works at 10 Mbit/s, 100 Mbit/s, or 1000 Mbit/s) ● 3 m, 10 m, 20 m SFP+ AOC cables ● 1 m, 3 m, 5 m, 7 m (active), 10 m (active) SFP+ copper cables
11	Two 40GE QSFP+ Ethernet optical ports NOTE A 40GE QSFP+ port cannot be split into four 10GE ports. Applicable transceiver modules and cables: ● 40GE optical module ● 10 m QSFP+ AOC cable (QSFP+ to QSFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to QSFP+)	12	Three port-side mounting holes for mounting brackets
13	Four power-supply-side mounting holes for mounting brackets	14	Ground screw

CE5850-48T4S2Q-HI

Figure 2-18 Appearance of the CE5850-48T4S2Q-HI

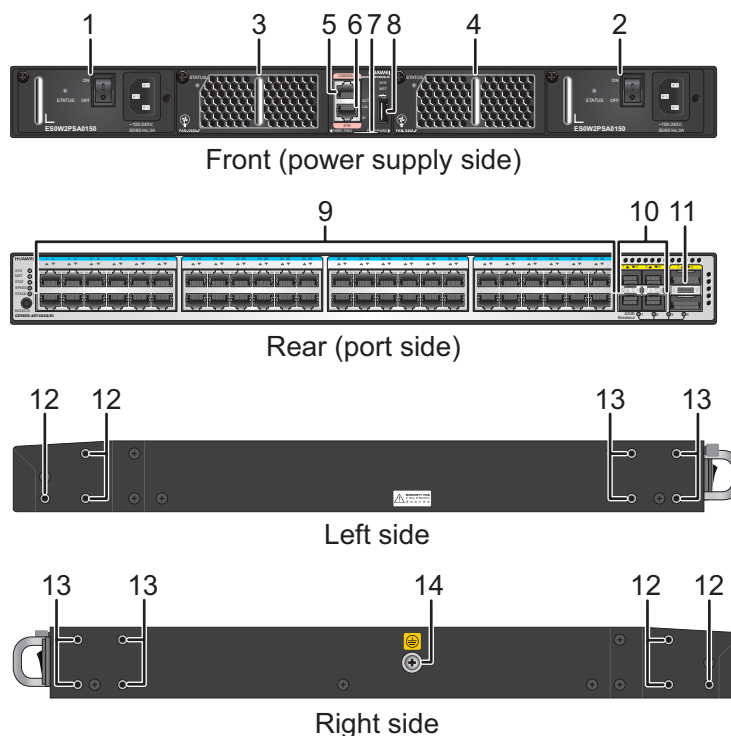


1	Power supply slot 1 Applicable power modules: ● 150 W AC power module (PAC-150WA) ● 350 W DC power module	2	Power supply slot 2 Applicable power modules: ● 150 W AC power module (PAC-150WA) ● 350 W DC power module
3	Fan slot 1 Applicable fan modules: ● FAN-40EA series fan modules	4	Fan slot 2 Applicable fan modules: ● FAN-40EA series fan modules
5	Console port	6	ETH management port (RJ45)
7	Barcode label NOTE This label is drawable, and you can pull it outward to view the barcode and MAC address of the switch.	8	USB port

9	Forty-eight 10/100/1000BASE-T Ethernet electrical ports	10	Four 10GE SFP+ Ethernet optical ports Applicable modules and cables: ● 10GE optical module ● 10GE-CWDM optical module ● GE optical module ● GE-CWDM optical module ● GE copper module (works at 10 Mbit/s, 100 Mbit/s, or 1000 Mbit/s) ● 3 m, 10 m, 20 m SFP+ AOC cables ● 1 m, 3 m, 5 m, 7 m (active), 10 m (active) SFP+ copper cables
11	Two 40GE QSFP+ Ethernet optical ports NOTE A 40GE QSFP+ port can be split into four 10GE ports. Applicable modules and cables: ● 40GE optical module ● 10 m QSFP+ AOC cable (QSFP+ to QSFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to 4*SFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to QSFP+)	12	Three port-side mounting holes for mounting brackets
13	Four power-supply-side mounting holes for mounting brackets	14	Ground screw

CE5855-48T4S2Q-EI

Figure 2-19 Appearance of the CE5855-48T4S2Q-EI

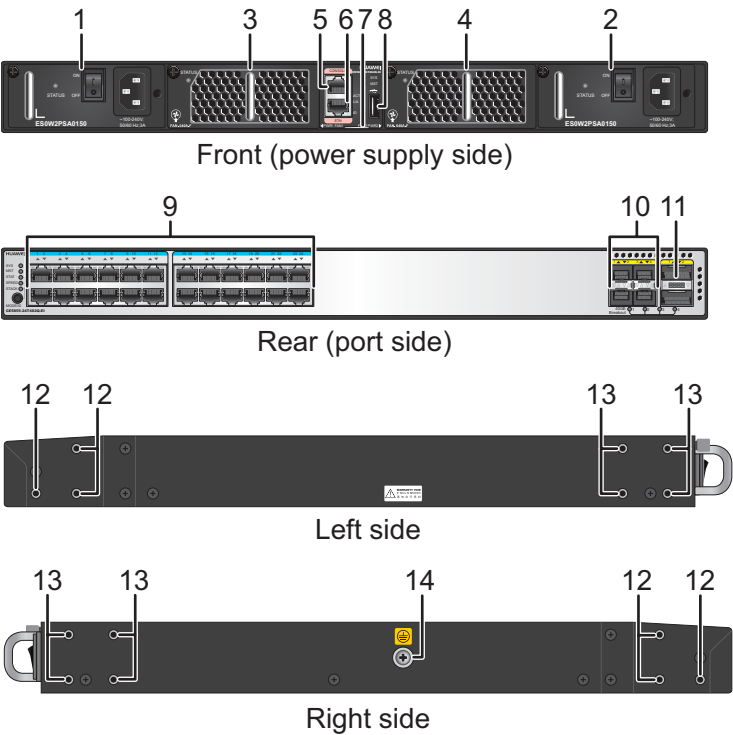


1	Power supply slot 1 Applicable power modules: ● 150 W AC power module (ES0W2PSA0150) ● 350 W DC power module	2	Power supply slot 2 Applicable power modules: ● 150 W AC power module (ES0W2PSA0150) ● 350 W DC power module
3	Fan slot 1 Applicable fan modules: ● FAN-040A series fan modules	4	Fan slot 2 Applicable fan modules: ● FAN-040A series fan modules
5	Console port	6	ETH management port (RJ45)
7	Barcode label NOTE This label is drawable, and you can pull it outward to view the barcode and MAC address of the switch.	8	USB port

9	Forty-eight 10/100/1000BASE-T Ethernet electrical ports	10	Four 10GE SFP+ Ethernet optical ports Applicable modules and cables: ● 10GE optical module (OSXD22N00 and LE2MXSC80FF0 not supported) ● 10GE-CWDM optical module ● GE optical module ● GE-CWDM optical module ● GE copper module (works at 10 Mbit/s, 100 Mbit/s, or 1000 Mbit/s) ● 3 m, 10 m, 20 m SFP+ AOC cables ● 1 m, 3 m, 5 m, 7 m (active), 10 m (active) SFP+ copper cables
11	Two 40GE QSFP+ Ethernet optical ports Applicable modules and cables: ● 40GE optical module ● 10 m QSFP+ AOC cable (QSFP+ to QSFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to 4*SFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to QSFP+) NOTE A 40GE QSFP+ port can be split into four 10GE ports.	12	Three port-side mounting holes for mounting brackets
13	Four power-supply-side mounting holes for mounting brackets	14	Ground screw

CE5855-24T4S2Q-EI

Figure 2-20 Appearance of the CE5855-24T4S2Q-EI



1	Power supply slot 1 Applicable power modules: ● 150 W AC power module (ES0W2PSA0150) ● 350 W DC power module	2	Power supply slot 2 Applicable power modules: ● 150 W AC power module (ES0W2PSA0150) ● 350 W DC power module
3	Fan slot 1 Applicable fan modules: ● FAN-040A series fan modules	4	Fan slot 2 Applicable fan modules: ● FAN-040A series fan modules
5	Console port	6	ETH management port (RJ45)
7	Barcode label NOTE This label is drawable, and you can pull it outward to view the barcode and MAC address of the switch.	8	USB port

9	Twenty-four 10/100/1000BASE-T Ethernet electrical ports	10	Four 10GE SFP+ Ethernet optical ports Applicable modules and cables: ● 10GE optical module (OSXD22N00 and LE2MXSC80FF0 not supported) ● 10GE-CWDM optical module ● GE optical module ● GE-CWDM optical module ● GE copper module (works at 10 Mbit/s, 100 Mbit/s, or 1000 Mbit/s) ● 3 m, 10 m, 20 m SFP+ AOC cables ● 1 m, 3 m, 5 m, 7 m (active), 10 m (active) SFP+ copper cables
11	Two 40GE QSFP+ Ethernet optical ports Applicable modules and cables: ● 40GE optical module ● 10 m QSFP+ AOC cable (QSFP+ to QSFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to 4*SFP+) ● 1 m, 3 m, 5 m QSFP+ copper cables (QSFP+ to QSFP+) NOTE A 40GE QSFP+ port can be split into four 10GE ports.	12	Three port-side mounting holes for mounting brackets
13	Four power-supply-side mounting holes for mounting brackets	14	Ground screw

Slot

- **Power supply slot**
The CE7800&6800&5800 series switches have two power supply slots, in which power modules can be installed to provide power to the chassis. A chassis can have one or two power modules. Double power modules can provide a higher reliability.
The CE7800&6800&5800 series switches support double power modules (1+1 backup).
 - When both power modules are working properly, they equally provide power for a chassis.
 - When one power module fails, the other one provides all power required for a chassis.
All power modules are hot swappable.
- **Fan slot**
The CE7800&6800&5800 series switches have two fan slots, in which fan modules can be installed to cool the chassis, ensuring efficient heat dissipation and system stability. A chassis must have two working fan modules to ensure normal operating.

All fan modules are hot swappable.

Airflow

The cooling systems of the CE7800&6800&5800 series switches have front-to-back or back-to-front airflow depending on the airflow direction of the power modules and fan modules used.

The airflow direction of the power modules and fan modules required on the CE7800&6800&5800 series switches depends on how the switches are installed in cabinets. Typically, cabinets in a data center have cold air flowing in from the front and hot air exhausted from the back. If CE7800&6800&5800 series switches are installed with the power supply side facing the front, you are advised to use fan modules and power modules with front-to-back airflow in the switches.

NOTE

- Front-to-back airflow: The power modules and fan modules using front-to-back airflow are marked



Air flows into the chassis from the power supply side and flows out from the port side, as shown in [Figure 2-21](#) (CE5800 as an example).

- Back-to-front airflow: The power modules and fan modules using back-to-front airflow are marked



Air flows into the chassis from the port side and flows out from the power supply side, as shown in [Figure 2-22](#) (CE5800 as an example).

- When the power module and fan module use forcible heat dissipation, they must use the same airflow method. For example, if the power module with back-to-front airflow is used, the fan module with back-to-front airflow must be used.
- When the fanless 150 W AC power module is used, the fan module with either of the airflow methods can be used.

Figure 2-21 Front-to-back (Air flows out from the port side) airflow

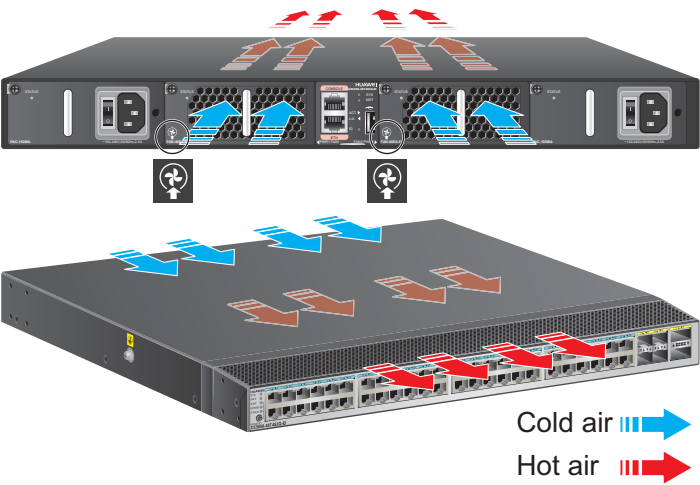
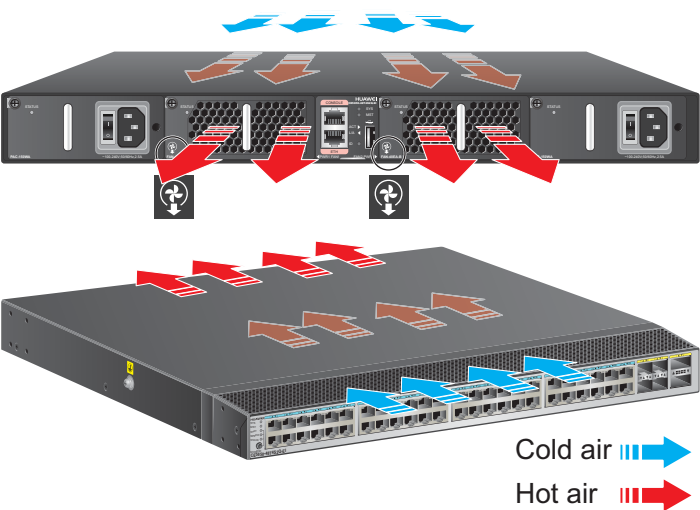


Figure 2-22 Back-to-front (air flows into from the port side) airflow



2.4 Indicators

Table 2-3 is a quick reference table for indicators of the CE7800&6800&5800 series switches.

Table 2-3 Indicator quick reference table

Switch Model	Indicator Description
CE7850-32Q-EI	The indicator symbols and meanings are described in CE7850-32Q-EI .
CE6810-48S4Q-EI	The downlink service port indicators of the CE6810-48S4Q-EI, CE6810-48S4Q-LI, CE6810-24S2Q-LI, and CE6850-48S4Q-EI are 10GE optical port indicators, and other indicators on these models are the same as those on the CE6850-48T4Q-EI. The CE6850-48T4Q-EI is used as an example here to describe the indicators.
CE6850-48S4Q-EI	

Switch Model	Indicator Description
CE6850-48T4Q-EI	The CE6851-48S6Q-HI does not have a mode switch button and the STAT/SPEED/STACK mode indicator. The downlink service port indicators of the CE6851-48S6Q-HI are 10GE optical port indicators, and other indicators on the CE6851-48S6Q-HI are the same as those on the CE6850-48T4Q-EI.
CE6810-48S4Q-LI	
CE6810-24S2Q-LI	
CE6851-48S6Q-HI	
CE6810-32T16S4Q-LI	The downlink service port indicators of the CE6810-32T16S4Q-LI are 10GE electrical port indicators and 10GE optical indicators, and other indicators are the same as those on the CE6850-48T4Q-EI. The CE6850-48T4Q-EI is used as an example here to describe the indicators.
CE6810-48S-LI	Service port indicators of the CE6810-48S-LI are 10GE optical port indicators, and the CE6810-48S-LI has no 40GE port indicators or 40GE Breakout indicators 1/2/3/4. Other indicators on the CE6810-48S-LI are the same as those on the CE6850-48T4Q-EI. The CE6850-48T4Q-EI is used as an example here to describe the indicators.
CE6850-48S6Q-HI	Indicators on the CE6850U-48S6Q-HI and CE6850U-24S2Q-HI are the same as those on the CE6850-48S6Q-HI. The CE6850-48S6Q-HI is used as an example here to describe the indicators.
CE6850U-48S6Q-HI	
CE6850U-24S2Q-HI	
CE6850-48T6Q-HI	The downlink service port indicators of the CE6850-48T6Q-HI are 10GE electrical port indicators, and other indicators are the same as those on the CE6850-48S6Q-HI. The CE6850-48S6Q-HI is used as an example here to describe the indicators.
CE5810-24T4S-EI	The CE5810-24T4S-EI and CE5810-48T4S-EI have no 40GE port indicators or 40GE Breakout indicators 1/2/3/4. Other indicators on the three models are the same as those on the CE5850-48T4S2Q-HI. The CE5850-48T4S2Q-HI is used as an example here to describe the indicators.
CE5810-48T4S-EI	
CE5850-48T4S2Q-EI	The CE5850-48T4S2Q-EI has no 40GE Breakout indicators 1/2/3/4, and other indicators are the same as those on the CE5850-48T4S2Q-HI. The CE5850-48T4S2Q-HI is used as an example here to describe the indicators.
CE5850-48T4S2Q-HI	
CE5855-48T4S2Q-EI	Indicators on the CE5855-48T4S2Q-EI and CE5855-24T4S2Q-EI are the same as those on the CE5850-48T4S2Q-HI. The CE5850-48T4S2Q-HI is used as an example here to describe the indicators.

Switch Model	Indicator Description
CE5855-24T4S2 Q-EI	

CE7850-32Q-EI

Figure 2-23 Indicators on the CE7850-32Q-EI rear panel

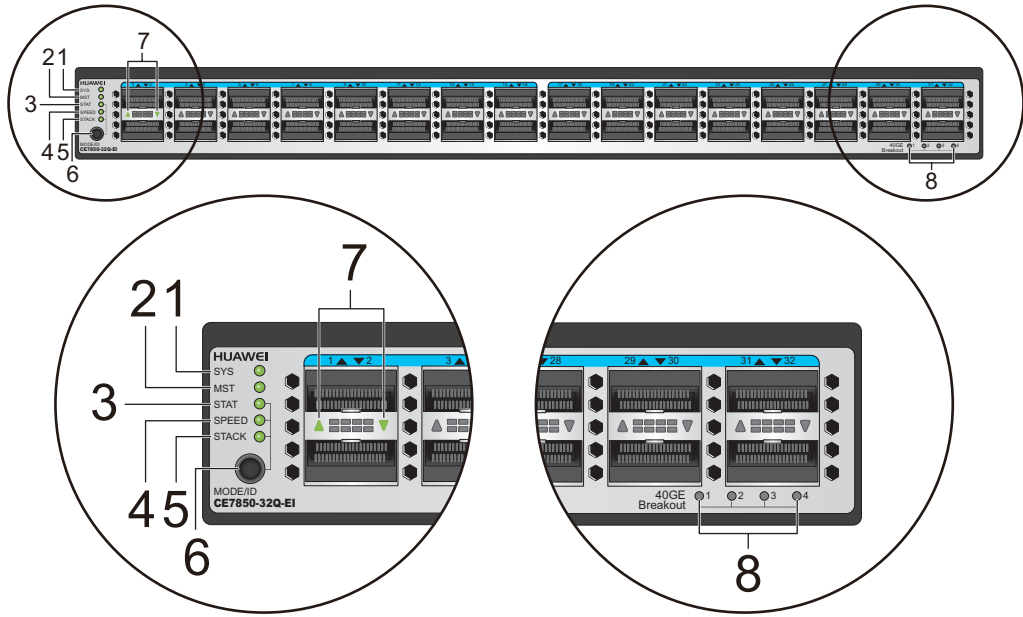


Figure 2-24 Indicators on the CE7850-32Q-EI front panel

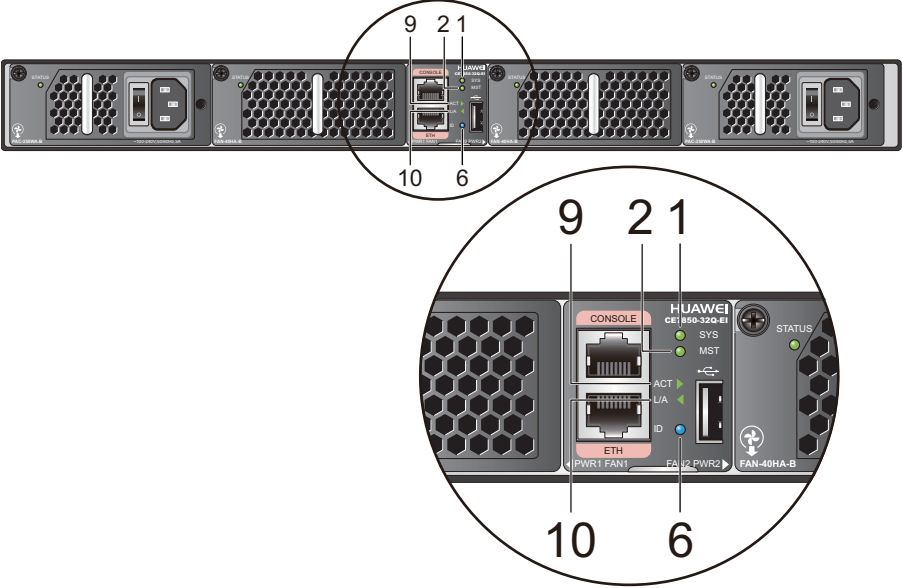


Table 2-4 Indicators on the CE7850-32Q-EI panels

No.	Indicator/Button	Color	Description
1	SYS: system status indicator	-	Off: The system is not running.
		Green	● Fast blinking: The system is starting. ● Slow blinking: The system is running normally.
		Red	Steady on: ● The system fails to start. ● At least one power module does not work normally. ● At least one fan module does not work normally.
2	MST: stack master/slave indicator	-	Off: The switch is not a stack master.
		Green	Steady on: The switch is a stack master or standalone switch.
		Amber	Steady on: A master election error or another type of error has occurred in the stack.
3	STAT: STAT mode indicator	-	Off: The STAT mode is not selected.
		Green	Steady on: The STAT mode (default mode) is selected, and service port indicators show the link connection states and link activity on ports.
4	SPEED: SPEED mode indicator	-	Off: The SPEED mode is not selected.
		Green	Steady on: The SPEED mode is selected, and service port indicators show the speed of each port.
5	STACK: STACK mode indicator	-	Off: The STACK mode is not selected.
		Green	Steady on: The STACK mode is selected, and service port indicators show the stack member ID of the local switch.

No.	Indicator/Button	Color	Description
6	MODE/ID: mode switch button and ID indicator NOTE The mode switch button on the rear panel is integrated with the ID indicator. There is only an ID indicator and no mode switch button on the front panel.	Mode switch button: -	- When you press the MODE button once, the SPEED indicator turns green and service port indicators show the speed of each port. - When you press the MODE button a second time, the STACK indicator turns green and service port indicators show the stack member ID of the local switch. - When you press the button a third time, the STAT indicator turns green (default mode) and service port indicators show the link connection states and link activity on ports. If you do not press the MODE button within 45 seconds, only the STAT indicator blinks green and the SPEED and STACK indicators are off.
		ID indicator: -	Off: The ID indicator is not used (default state).
		ID indicator: blue	Steady on: The indicator identifies the device for maintenance. The ID indicator can be turned on or off remotely to help field engineers find the device to maintain.
7	Service port indicator (40GE optical port) NOTE Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top.	Meanings of service port indicators vary in different modes. For details, see Table 2-8 . When a 40GE port is configured as four 10GE ports, this indicator shows the status of a 10GE port. The sequence number of the indicated 10GE port is identified by indicators 40GE Breakout 1/2/3/4 on the lower right corner of the panel. NOTE Each 40GE port has a single-color indicator, which shows the status of the 40GE port by default. If a 40GE port is not split and is connected to four 10GE ports on a remote device using a one-to-four high-speed cable, the 40GE port cannot go Up and its indicator is off.	
8	40GE Breakout 1/2/3/4 (sequence number indicators of 10GE ports converted from a 40GE port) NOTE Indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s.	-	Off: 40GE ports are not split into four 10GE ports.

No.	Indicator/Button	Color	Description
		Green	Steady on: At least one 40GE port has been split into four 10GE ports. When one or more 40GE ports are configured as four 10GE ports, these indicators identify the sequence numbers of the 10GE ports. A 40GE port indicator (7 in Figure 2-23) shows the status of a 10GE port converted from the 40GE port: ● When Breakout indicator 1 is on, each 40GE port indicator shows the status of the first 10GE port converted from the corresponding 40GE port. ● When Breakout indicator 2 is on, each 40GE port indicator shows the status of the second 10GE port converted from the corresponding 40GE port. ● When Breakout indicator 3 is on, each 40GE port indicator shows the status of the third 10GE port converted from the corresponding 40GE port. ● When Breakout indicator 4 is on, each 40GE port indicator shows the status of the fourth 10GE port converted from the corresponding 40GE port. The following is an example: The first 40GE port shown in Figure 2-23 is split into four 10GE ports, and the second 40GE port is not split. ● When Breakout indicator 1 is on, the indicator of 40GE port 1 shows the status of the first 10GE port converted from 40GE port 1, and the indicator of 40GE port 2 still shows the status of 40GE port 2. ● When Breakout indicator 2 is on, the indicator of 40GE port 1 shows the status of the second 10GE port converted from 40GE port 1, and the indicator of 40GE port 2 still shows the status of 40GE port 2.
9	ACT: USB-based deployment indicator	-	Off: USB-based deployment is disabled (default state).
		Green	● Steady on: A USB-based deployment has been completed. ● Blinking: The system is reading data from a USB flash drive.
		Red	Steady on: USB-based deployment has failed.

No.	Indicator/Button	Color	Description
10	L/A: ETH management port indicator	-	Off: No link is established on the port.
		Green	- Steady on: A link is established on the port. - Blinking: The port is sending or receiving data.

CE6850-48T4Q-EI

Figure 2-25 Indicators on the CE6850-48T4Q-EI rear panel

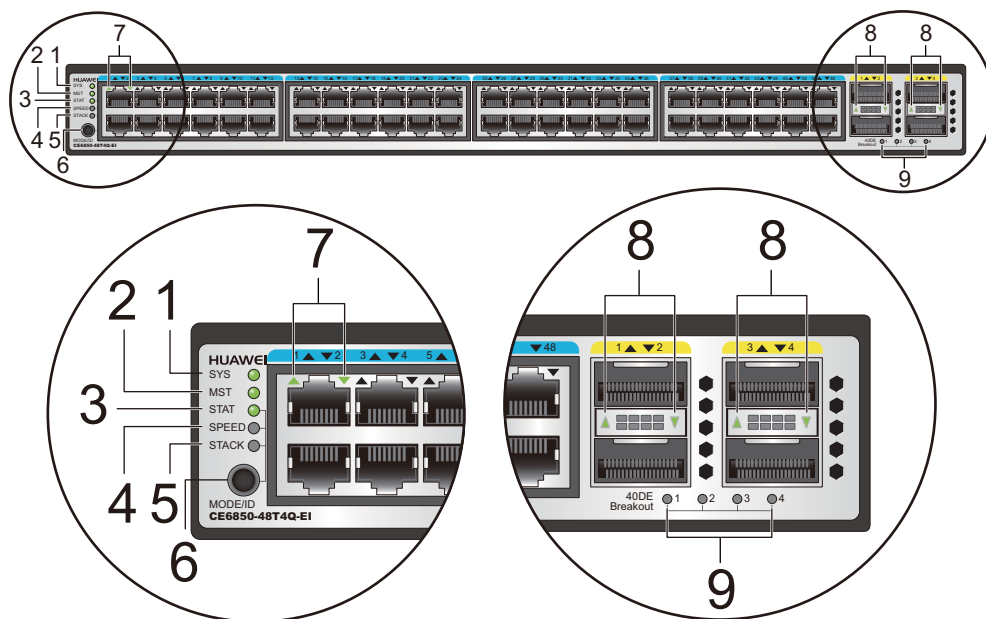


Figure 2-26 Indicators on the CE6850-48T4Q-EI front panel

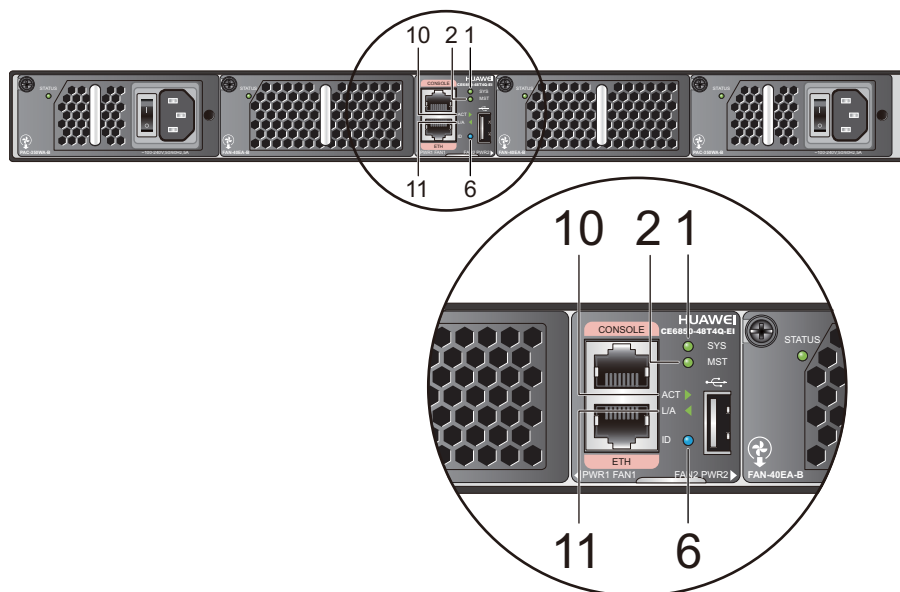


Table 2-5 Indicators on the CE6850-48T4Q-EI panels

No.	Indicator/Button	Color	Description
1	SYS: system status indicator	-	Off: The system is not running.
		Green	● Fast blinking: The system is starting. ● Slow blinking: The system is running properly.
		Red	Steady on: ● The system fails to start. ● At least one power module does not work normally. ● At least one fan module does not work normally. ● The card power consumption exceeds the rated power of the power modules.
2	MST: stack master/slave indicator	-	Off: The switch is not a stack master.
		Green	● Steady on: The switch is a stack master or standalone switch. ● Blinking: The switch is working in SVF mode. (Versions earlier than V100R005C00: Only the CE6810-48S4Q-EI supports this indicator state. V100R005C00: Only the CE6810-48S4Q-EI and CE6850-48T4Q-EI support this indicator state. V100R005C10 and later versions: Only the CE6810-48S4Q-EI, CE6810-48S4Q-LI, CE6810-48S-LI, CE6810-32T16S4Q-LI, CE6810-24S2Q-LI, and CE6850-48T4Q-EI support this indicator state.)
		Amber	Steady on: A master election error or another type of error has occurred in the stack. (The CE6810-24S2Q-LI, CE6851-48S6Q-HI, and CE6810-32T16S4Q-LI do not support this indicator state.)
3	STAT: STAT mode indicator	-	Off: The STAT mode is not selected.
		Green	Steady on: The STAT mode (default mode) is selected, and service port indicators show the link connection states and link activity on ports.
4	SPEED: SPEED mode indicator	-	Off: The SPEED mode is not selected.

No.	Indicator/Button	Color	Description
		Green	Steady on: The SPEED mode is selected, and service port indicators show the speed of each port.
5	STACK: STACK mode indicator	-	Off: The STACK mode is not selected.
		Green	Steady on: The STACK mode is selected, and service port indicators show the stack member ID or leaf ID of the local switch. NOTE In V100R002C00 and later versions, if the indicator mode on any member switch of a stack or SVF system is changed to STACK by pressing the MODE button, all the other member switches in the stack or SVF system change the stack mode to STACK. In this case, service port indicators on the member switches show stack member IDs or leaf IDs of these switches.
6	MODE/ID: mode switch button and ID indicator NOTE The mode switch button on the rear panel is integrated with the ID indicator. There is only an ID indicator and no mode switch button on the front panel.	Mode switch button: -	- When you press the MODE button once, the SPEED indicator turns green and service port indicators show the speed of each port. - When you press the MODE button a second time, the STACK indicator turns green and service port indicators show the stack member ID of the local switch. - When you press the button a third time, the STAT indicator turns green (default mode) and service port indicators show the link connection states and link activity on ports. If you do not press the MODE button within 45 seconds, only the STAT indicator blinks green and the SPEED and STACK indicators are off.
		ID indicator: or: -	Off: The ID indicator is not used (default state).
		ID indicator: or: blue	Steady on: The indicator identifies the device for maintenance. The ID indicator can be turned on or off remotely to help field engineers find the device to maintain.

No.	Indicator/Button	Color	Description
7	Service port indicator (10GE electrical port) NOTE Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top.		Meanings of service port indicators vary in different modes. For details, see Table 2-8 .
8	Service port indicator (40GE optical port) NOTE Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top.		Meanings of service port indicators vary in different modes. For details, see Table 2-8 . When a 40GE port is configured as four 10GE ports, this indicator shows the status of a 10GE port. The sequence number of the indicated 10GE port is identified by indicators 40GE Breakout 1/2/3/4 on the lower right corner of the panel. NOTE Each 40GE port has a single-color indicator, which shows the status of the 40GE port by default. If a 40GE port is not split and is connected to four 10GE ports on a remote device using a one-to-four high-speed cable, the 40GE port cannot go Up and its indicator is off.
9	40GE Breakout 1/2/3/4 (sequence number indicators of 10GE ports converted from a 40GE port) NOTE Indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s.	-	Off: 40GE ports are not split into four 10GE ports.

No.	Indicator/Button	Color	Description
		Green	Steady on: At least one 40GE port has been split into four 10GE ports. When one or more 40GE ports are configured as four 10GE ports, these indicators identify the sequence numbers of the 10GE ports. A 40GE port indicator (8 in Figure 2-25) shows the status of a 10GE port converted from the 40GE port: ● When Breakout indicator 1 is on, each 40GE port indicator shows the status of the first 10GE port converted from the corresponding 40GE port. ● When Breakout indicator 2 is on, each 40GE port indicator shows the status of the second 10GE port converted from the corresponding 40GE port. ● When Breakout indicator 3 is on, each 40GE port indicator shows the status of the third 10GE port converted from the corresponding 40GE port. ● When Breakout indicator 4 is on, each 40GE port indicator shows the status of the fourth 10GE port converted from the corresponding 40GE port. The following is an example: The first 40GE port shown in Figure 2-25 is split into four 10GE ports, and the second 40GE port is not split. ● When Breakout indicator 1 is on, the indicator of 40GE port 1 shows the status of the first 10GE port converted from 40GE port 1, and the indicator of 40GE port 2 still shows the status of 40GE port 2. ● When Breakout indicator 2 is on, the indicator of 40GE port 1 shows the status of the second 10GE port converted from 40GE port 1, and the indicator of 40GE port 2 still shows the status of 40GE port 2.
10	ACT: USB-based deployment indicator	-	Off: USB-based deployment is disabled (default state).
		Green	● Steady on: A USB-based deployment has been completed. ● Blinking: The system is reading data from a USB flash drive.
		Red	Steady on: USB-based deployment has failed.

No.	Indicator/Button	Color	Description
11	L/A: ETH management port indicator	-	Off: No link is established on the port.
		Green	● Steady on: A link is established on the port. ● Blinking: The port is sending or receiving data.

CE6850-48S6Q-HI

Figure 2-27 Indicators on the CE6850-48S6Q-HI rear panel

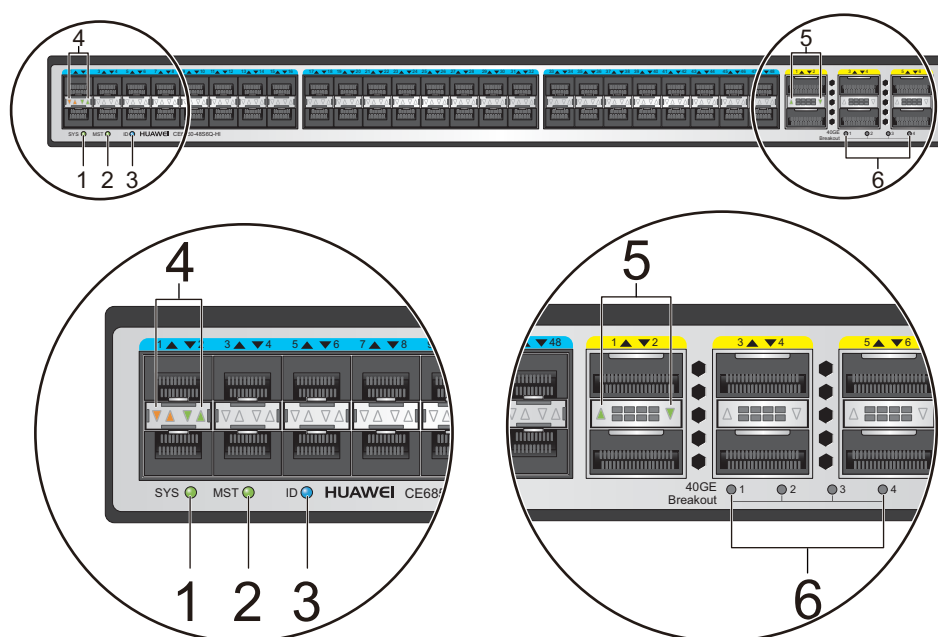


Figure 2-28 Indicators on the CE6850-48S6Q-HI front panel

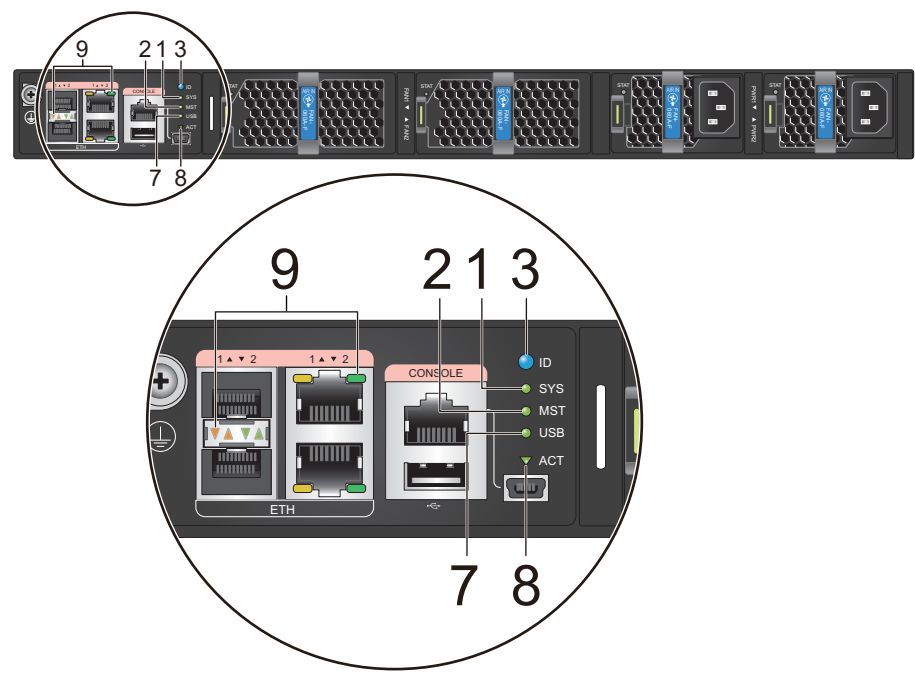


Table 2-6 Indicators on the CE6850-48S6Q-HI panels

No.	Indicator/Button	Color	Description
1	SYS: system status indicator	-	Off: The system is not running.
		Green	● Fast blinking: The system is starting. ● Slow blinking: The system is running normally.
		Red	Steady on: ● The system fails to start. ● A power module does not work normally. ● A fan module does not work normally.
2	MST: stack master/slave indicator	-	Off: The switch is not a stack master.
		Green	Steady on: The switch is a stack master or standalone switch.
3	ID: ID indicator	-	Off: The ID indicator is not used (default state).
		Blue	Steady on: The indicator identifies the device for maintenance. The ID indicator can be turned on or off remotely to help field engineers find the device to maintain.

No.	Indicator/Button	Color	Description
4	Service port indicator (10GE optical port) NOTE Each 10GE optical port has two single-color indicators. The one on the left is the ACT indicator (yellow), and the one on the right is the LINK indicator (green). Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top.	Green	● Off: The port is not connected or has been shut down. ● Steady on: A link is established on the port.
		Yellow	● Off: The port is not sending or receiving data. ● Blinking: The port is sending or receiving data.
5	Service port indicator (40GE optical port) NOTE Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top.	-	Off: The port is not connected or has been shut down.
		Green	● Steady on: A link is established on the port. ● Blinking: The port is sending or receiving data.
		When a 40GE port is configured as four 10GE ports, this indicator shows the status of a 10GE port. The sequence number of the indicated 10GE port is identified by indicators 40GE Breakout 1/2/3/4 on the lower right corner of the panel. NOTE Each 40GE port has a single-color indicator, which shows the status of the 40GE port by default. If a 40GE port is not split and is connected to four 10GE ports on a remote device using a one-to-four high-speed cable, the 40GE port cannot go Up and its indicator is off.	
6	40GE Breakout 1/2/3/4 (sequence number indicators of 10GE ports converted from a 40GE port) NOTE Indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s.	-	Off: 40GE ports are not split into four 10GE ports.

No.	Indicator/Button	Color	Description
		Green	Steady on: At least one 40GE port has been split into four 10GE ports. When one or more 40GE ports are configured as four 10GE ports, these indicators identify the sequence numbers of the 10GE ports. A port indicator (5 in Figure 2-27) shows the status of a 10GE port converted from the corresponding 40GE port: ● When Breakout indicator 1 is on, each 40GE port indicator shows the status of the first 10GE port converted from the corresponding 40GE port. ● When Breakout indicator 2 is on, each 40GE port indicator shows the status of the second 10GE port converted from the corresponding 40GE port. ● When Breakout indicator 3 is on, each 40GE port indicator shows the status of the third 10GE port converted from the corresponding 40GE port. ● When Breakout indicator 4 is on, each 40GE port indicator shows the status of the fourth 10GE port converted from the corresponding 40GE port. The following is an example: The first 40GE port shown in Figure 2-27 is split into four 10GE ports, and the second 40GE port is not split. ● When Breakout indicator 1 is on, the indicator of 40GE port 1 shows the status of the first 10GE port converted from 40GE port 1, and the indicator of 40GE port 2 still shows the status of 40GE port 2. ● When Breakout indicator 2 is on, the indicator of 40GE port 1 shows the status of the second 10GE port converted from 40GE port 1, and the indicator of 40GE port 2 still shows the status of 40GE port 2.
7	USB: USB-based deployment indicator	-	Off: USB-based deployment is disabled (default state).
		Green	● Steady on: A USB-based deployment has been completed. ● Blinking: The system is reading data from a USB flash drive.

No.	Indicator/Button	Color	Description
		Red	Steady on: USB-based deployment has failed.
8	ACT: Mini USB port indicator	-	Off: The Mini USB port is inactive, and the console port can be used.
		Green	Steady on: The Mini USB port is active, and the console port cannot be used.
9	ETH management port indicator	-	Off: No link is established on the port.
		Green	Steady on: A link is established on the port.
		Yellow	Blinking: The port is sending or receiving data.

CE5850-48T4S2Q-HI

Figure 2-29 Indicators on the CE5850-48T4S2Q-HI rear panel

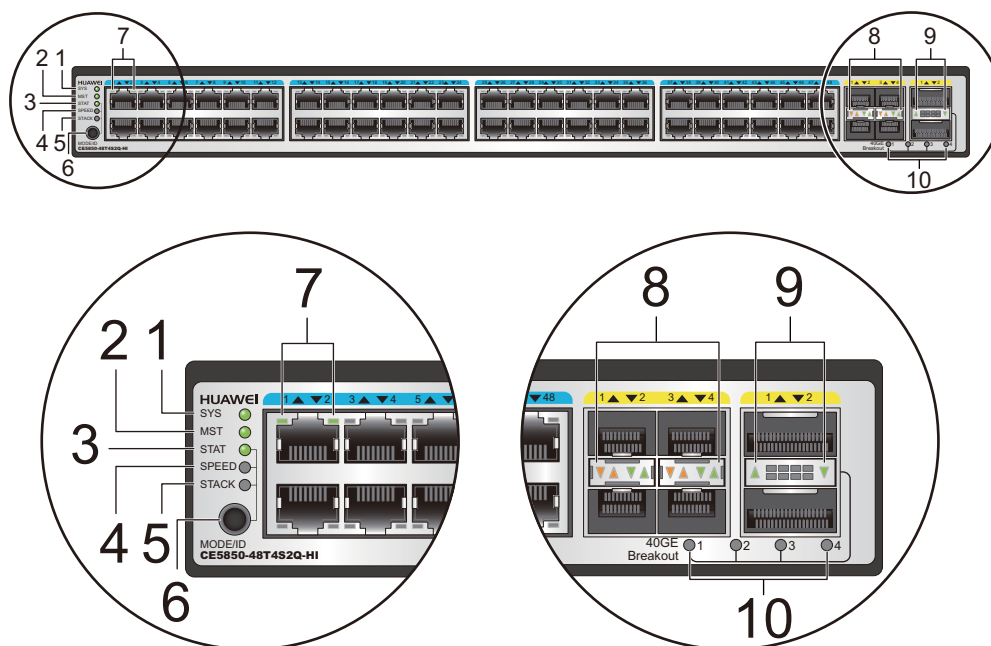


Figure 2-30 Indicators on the CE5850-48T4S2Q-HI front panel

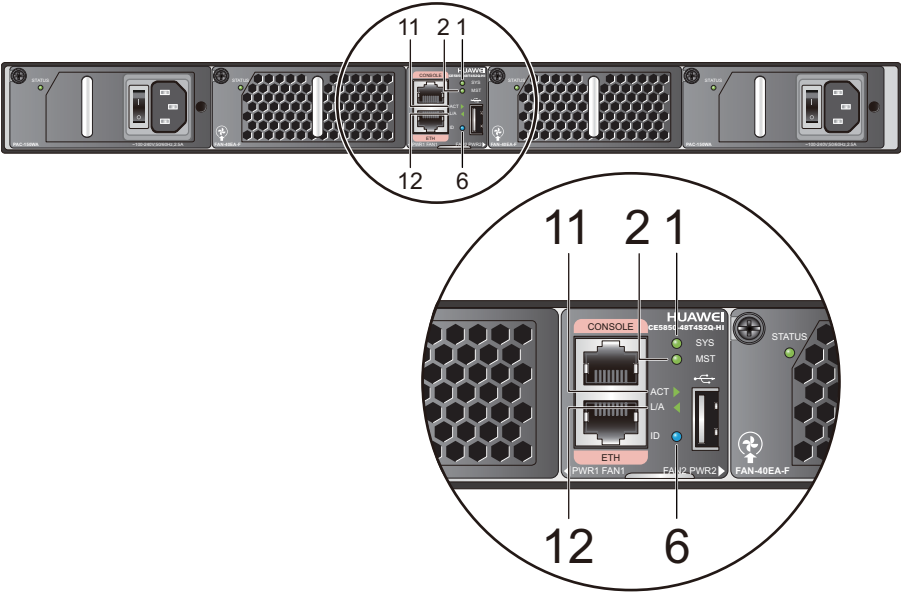


Table 2-7 Indicators on the CE5850-48T4S2Q-HI panels

No.	Indicator/Button	Color	Description
1	SYS: system status indicator	-	Off: The system is not running.
		Green	- Fast blinking: The system is starting. - Slow blinking: The system is running properly.
		Red	Steady on: - The system fails to start. - At least one power module does not work normally. - At least one fan module does not work normally.
2	MST: stack master/slave indicator	-	Off: The switch is not a stack master.
		Green	- Steady on: The switch is a stack master or standalone switch. - Blinking: The switch is working in SVF mode. (Versions earlier than V100R005C10: Only the CE5810-24T4S-EI and CE5810-48T4S-EI support this indicator state. V100R005C10 and later versions: only the CE5810-24T4S-EI, CE5810-48T4S-EI, and CE5850-48T4S2Q-EI support this indicator state.)

No.	Indicator/Button	Color	Description
		Amber	Steady on: A master election error or another type of error has occurred in the stack. (The CE5855-48T4S2Q-EI and CE5855-24T4S2Q-EI do not support this indicator state.)
3	STAT: STAT mode indicator	-	Off: The STAT mode is not selected.
		Green	Steady on: The STAT mode (default mode) is selected, and service port indicators show the link connection states and link activity on ports.
4	SPEED: SPEED mode indicator	-	Off: The SPEED mode is not selected.
		Green	Steady on: The SPEED mode is selected, and service port indicators show the speed of each port.
5	STACK: STACK mode indicator	-	Off: The STACK mode is not selected.
		Green	Steady on: The STACK mode is selected, and service port indicators show the stack member ID or leaf ID of the local switch. NOTE In V100R002C00 and later versions, if the indicator mode on any member switch of a stack or SVF system is changed to STACK by pressing the MODE button, all the other member switches in the stack or SVF system change the stack mode to STACK. In this case, service port indicators on the member switches show stack member IDs or leaf IDs of these switches.
6	MODE/ID: mode switch button and ID indicator NOTE The mode switch button on the rear panel is integrated with the ID indicator. There is only an ID indicator and no mode switch button on the front panel.	Mode switch button: -	- When you press the MODE button once, the SPEED indicator turns green and service port indicators show the speed of each port. - When you press the MODE button a second time, the STACK indicator turns green and service port indicators show the stack member ID of the local switch. - When you press the button a third time, the STAT indicator turns green (default mode) and service port indicators show the link connection states and link activity on ports. If you do not press the MODE button within 45 seconds, only the STAT indicator blinks green and the SPEED and STACK indicators are off.
		ID indicator: -	Off: The ID indicator is not used (default state).

No.	Indicator/Button	Color	Description
		ID indicator or: blue	Steady on: The indicator identifies the device for maintenance. The ID indicator can be turned on or off remotely to help field engineers find the device to maintain.
7	Service port indicator (GE electrical port) NOTE The indicator on the left indicates the port at the top, and the indicator on the right indicates the port at the bottom.		The meaning of the service port indicators varies according to the current mode. For details, see Table 2-8 .
8	Service port indicator (10GE optical port) NOTE Each 10GE optical port has two single-color indicators. The one on the left is the ACT indicator (yellow), and the one on the right is the LINK indicator (green). Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top.		
9	Service port indicator (40GE optical port) NOTE Arrowheads show the positions of ports. A down arrowhead indicates a port at the bottom, and an up arrowhead indicates a port at the top.		
10	40GE Breakout indicators 1/2/3/4 (10GE ports converted from a 40GE port)	-	Off: 40GE ports are not split into four 10GE ports.

No.	Indicator/Button	Color	Description
	NOTE Indicators 1, 2, 3, 4 turn on in cyclic order, with each indicator keeping on for 5s.	Green	Steady on: At least one 40GE port has been split into four 10GE ports. When one or more 40GE ports are configured as four 10GE ports, these indicators identify the sequence numbers of the 10GE ports. A 40GE port indicator (9 in Figure 2-29) shows the status of a 10GE port converted from the 40GE port: ● When Breakout indicator 1 is on, each 40GE port indicator shows the status of the first 10GE port converted from the corresponding 40GE port. ● When Breakout indicator 2 is on, each 40GE port indicator shows the status of the second 10GE port converted from the corresponding 40GE port. ● When Breakout indicator 3 is on, each 40GE port indicator shows the status of the third 10GE port converted from the corresponding 40GE port. ● When Breakout indicator 4 is on, each 40GE port indicator shows the status of the fourth 10GE port converted from the corresponding 40GE port. The following is an example: The first 40GE port shown in Figure 2-29 is split into four 10GE ports, and the second 40GE port is not split. ● When Breakout indicator 1 is on, the indicator of 40GE port 1 shows the status of the first 10GE port converted from 40GE port 1, and the indicator of 40GE port 2 still shows the status of 40GE port 2. ● When Breakout indicator 2 is on, the indicator of 40GE port 1 shows the status of the second 10GE port converted from 40GE port 1, and the indicator of 40GE port 2 still shows the status of 40GE port 2.
11	ACT: USB-based deployment indicator	-	Off: USB-based deployment is disabled (default state).
		Green	● Steady on: A USB-based deployment has been completed. ● Blinking: The system is reading data from a USB flash drive.
		Red	Steady on: USB-based deployment has failed.

No.	Indicator/Button	Color	Description
12	L/A: ETH management port indicator	-	Off: No link is established on the port.
		Green	● Steady on: A link is established on the port. ● Blinking: The port is sending or receiving data.

Table 2-8 Service port indicators in various modes

Display Mode	Port	Color	Description
STAT	GE electrical port, 10GE electrical port, and 40GE optical port	-	Off: The port is not connected or has been shut down.
		Green	● Steady on: A link is established on the port. ● Blinking: The port is sending or receiving data.
	10GE optical port	Green	● Off: The port is not connected or has been shut down. ● Steady on: A link is established on the port.
		Amber	● Off: The port is not sending or receiving data. ● Blinking: The port is sending or receiving data.
SPEED	GE electrical port	-	Off: The port is not connected or has been shut down.
		Green	● Steady on: The port speed is 10/100 Mbit/s. ● Blinking: The port speed is 1000 Mbit/s.
	10GE electrical port	-	Off: The port is not connected or has been shut down.
		Green	● Steady on: The port speed is 100/1000 Mbit/s. ● Blinking: The port speed is 10 Gbit/s.
	10GE optical port	-	Off: The port is not connected or has been shut down.
		Green	● Steady on: The port speed is 1000 Mbit/s. ● Blinking: The port speed is 10 Gbit/s.
	40GE optical port	-	Off: The port is not connected or has been shut down.
		-	Off: The port is not connected or has been shut down.

Display Mode	Port	Color	Description
		Green	● Steady on: The 40GE port has been split into four 10GE ports. ● Blinking: The port is working as a 40GE port. NOTE The 40GE optical ports of the CE5850-48T4S2Q-EI cannot be split into four 10GE ports, so this indicator can only be off or blinking.
STACK	Green NOTE This row describes the states and meanings of port indicators on a switch working in stack mode.		● Off: Port indicators do not show the stack member ID of the switch. ● Steady on: If the indicator of a port is steady on, the port number is the stack member ID of the switch. NOTE In STACK mode, a 10GE optical port has only its LINK indicator on (green).
	Green NOTE This row describes the states and meanings of port indicators on a switch working in super virtual fabric (SVF) mode.		● Off: Port indicators do not show the leaf ID of the switch. ● Steady on: If the indicator of a port is steady on, the port number indicates the leaf ID of the switch. NOTE The leaf ID range supported by a switch depends on the number of downlink ports on the switch: ● On the CE5810-24T4S-EI and CE6810-24S2Q-LI, downlink ports 1 to 24 indicate leaf IDs 101 to 124. If the leaf ID of the switch is larger than 124, port indicators retain the original states before the switch changes to the SVF state and do not show the leaf ID. ● On the CE6810-48S4Q-EI, CE6810-48S4Q-LI, CE6810-48S-LI, CE6810-32T16S4Q-LI, CE6850-48T4Q-EI, CE5810-48T4S-EI, and CE5850-48T4S2Q-EI, downlink ports 1 to 48 indicate leaf IDs 101 to 148. If the leaf ID of the switch is larger than 148, port indicators retain the original states before the switch changes to the SVF state and do not show the leaf ID.

2.5 Ports

10/100/1000BASE-T Ethernet Electrical Port

A 10/100/1000BASE-T Ethernet electrical port receives and sends services at a speed of 10 Mbit/s, 100 Mbit/s, or 1000 Mbit/s. A 10/100/1000BASE-T Ethernet electrical port uses a

Category 5 or higher category cable. [Table 2-9](#) describes the attributes of a 10/100/1000BASE-T Ethernet electrical port.

Table 2-9 Attributes of a 10/100/1000BASE-T Ethernet electrical port

Attribute	Description
Connector	RJ45
Standards compliance	IEEE802.3ab
Applicable cable	Straight-through cable and crossover cable
Working mode	Supported rate: 10/100/1000 Mbit/s auto-sensing Full-duplex
Maximum transmission distance	100 m

10GBASE-T Ethernet Electrical Port

A 10GBASE-T Ethernet electrical port receives and sends services at a speed of 100 Mbit/s, 1000 Mbit/s, or 10 Gbit/s. The port can work in 100/1000M mode through auto-sensing. [Table 2-10](#) describes the attributes of a 10GBASE-T Ethernet electrical port. Category 6A shielded twisted pairs are recommended for the 10GBASE-T Ethernet electrical port.

Table 2-10 Attributes of a 10GBASE-T Ethernet electrical port

Attribute	Description
Connector	RJ45
Standards compliance	IEEE802.3an and IEEE802.3az
Applicable cable	Straight-through cable and crossover cable
Working mode	Supported rate: 100/1000 Mbit/s and 10 Gbit/s auto-sensing Full-duplex
Maximum transmission distance	100 m

10GE SFP+ Ethernet Optical Port

A 10GE SFP+ Ethernet optical port can automatically work in GE mode when it has a GE optical module installed. A 10GE SFP+ Ethernet optical port can receive and send services when the

network speed is 1 Gbit/s or 10 Gbit/s. [Table 2-11](#) describes the attributes of a 10GE SFP+ Ethernet optical port.

Table 2-11 Attributes of a 10GE SFP+ Ethernet optical port

Attribute	Description
Connector	LC
Optical port	Determined by modules or cables in use
Standards compliance	IEEE802.3ae
Working mode	Supported rate: 1000 Mbit/s, 10 Gbit/s auto-sensing Full-duplex

 NOTE

10GE SFP+ Ethernet optical ports of the CE6850U-48S6Q-HI and CE6850U-24S2Q-HI can use 2GE/4GE/8GE FC optical modules.

A 5 m SFP+ high-speed cable cannot be used to connect 10GE optical ports between the CE5850EI (running a version prior to V100R005C10) and CE5855EI switches. To connect the 10GE optical ports of the two switches, use any of the following methods:

- Use a 1 m, 3 m, 7 m, or 10 m SFP+ high-speed cable.
- Use an active optical cable (AOC) or optical modules and optical fibers.
- Upgrade the system software of the CE5850EI switch to V100R005C10 or a later version.

40GE QSFP+ Ethernet Optical Port

A 40GE QSFP+ Ethernet optical port receives and sends services at a speed of 40 Gbit/s. If a 40GE port is split into four 10GE ports, it must use 1-in-4-out QSFP+ optical modules and fibers or 1-in-4-out QSFP+ cables. [Table 2-12](#) describes the attributes of a 40GE QSFP+ Ethernet optical port.

Table 2-12 Attributes of a 40GE QSFP+ Ethernet optical port

Attribute	Description
Connector	LC/MPO
Optical port	Determined by modules or cables in use
Standards compliance	IEEE802.3ba
Working mode	Full-duplex

 NOTE

A 5 m 1-to-4 QSFP+ high-speed cable cannot be used to connect a 40GE optical port (split into four 10GE ports) and 10GE optical ports between the CE5850EI (running a version prior to V100R005C10) and CE5855EI switches. To connect the 10GE and 40GE optical ports of the two switches, use any of the following methods:

- Use a 1 m or 3 m 1-to-4 QSFP+ high-speed cable.
- Use optical modules and optical fibers.
- Upgrade the system software of the CE5850EI switch to V100R005C10 or a later version.

Console Port

The console port is connected to a console for onsite configuration. The port must use a **console cable**. [Table 2-13](#) describes the attributes of the console port.

 NOTE

- The console port and Mini USB port share one internal serial port. You can use the console port or Mini USB port as the serial port according to your needs. When the Mini USB is activated, the console port cannot be used.
- When both the console port and Mini-USB port have a cable connected, the Mini-USB port is used.

Table 2-13 Attributes of the console port

Attribute	Description
Connector	RJ45
Standards compliance	RS232
Working mode	Duplex Universal Asynchronous Receiver/Transmitter (UART)
Baud rate	9600 bit/s - 115200 bit/s Default value: 9600 bit/s

Mini USB Port

The Mini USB port can connect to a configuration terminal for onsite configuration of the system, but the configuration terminal must have a USB serial port driver installed. The Mini USB port is used as the serial port once a link is established on the port.

ETH Management Port (RJ45)

The ETH management port (RJ45) of a switch is connected to the network port of a configuration terminal or network management workstation to set up the onsite or remote configuration environment. The ETH management port (RJ45) uses a Category 5 or higher category cable. [Table 2-14](#) describes the attributes of the ETH management port (RJ45).

Table 2-14 Attributes of the ETH management port (RJ45)

Attribute	Description
Connector	RJ45
Standards compliance	IEEE802.3ab
Working mode	Supported rate: 10/100/1000 Mbit/s auto-sensing Full-duplex
Maximum transmission distance	100 m

ETH Management Port (Combo)

The ETH management port (combo) consists of an electrical port and an optical port. You can connect the electrical or optical port to a configuration terminal or network management workstation to set up the onsite or remote configuration environment. The electrical and optical ports are multiplexed, and only one of them can work at a time.

NOTE

The combo port automatically selects the working mode as follows:

- If the optical port has no optical module installed and the electrical port has no network cable connected, the port type depends on which port is connected first. If the electrical port is connected by a network cable first, the electrical port is used for data switching. If the optical port has an optical module installed first, the optical port is used for data switching.
- If the electrical port has a network cable connected and is in Up state, the electrical port is still used for data switching when the optical port has an optical module installed.
- If the optical port has an optical module installed and is in Up state, the optical port is still used for data switching when the electrical port has a network cable connected.
- If the optical port has an optical module and fiber installed and the electrical port has a network cable connected, the optical port is used for data switching after the switch restarts.

The combo electrical port uses a Category 5 or higher category network cable. [Table 2-15](#) describes the attributes of the combo electrical port.

Table 2-15 Attributes of the combo electrical port

Attribute	Description
Connector	RJ45
Standards compliance	IEEE802.3ab
Working mode	10/100/1000 Mbit/s auto-sensing Full-duplex

Attribute	Description
Maximum transmission distance	100 m

The combo optical port uses a 100M or GE optical module and matching fibers. A 100M optical module can be used only after the switch starts successfully. If a 10GE optical module is installed, the interface can go Up, but the system displays an alarm message, indicating that the interface does not support the optical module. If a GE copper module is installed and the remote interface has a GE copper module installed, the local interface can go Up but does not support rate configuration.

Table 2-16 describes the attributes of the combo optical port.

Table 2-16 Attributes of the combo optical port

Attribute	Description
Connector	LC
Standards compliance	IEEE802.3z
Working mode	100/1000 Mbit/s Full-duplex

The CE6850HI and CE6850U-HI switches have two ETH management ports (combo). Pay attention to the following when using the two management ports:

- The two ports cannot be used together, and you must choose one of them to use.
- Before start of a CE6850HI or CE6850U-HI switch, you can select interface 1 or interface 2 in the BIOS menu. Interface 1 is the default choice. For details, see Modify parameters.
- After registration of the switch succeeds:
 - If both the management ports have a cable connected and are in Up state, port 1 acts as the primary management port and port 2 becomes the backup automatically. The management interface number displayed on the command line interface is MEth0/0/0, regardless of which port is used.
 - If cables are connected to the two ETH management ports after successful registration of the switch, the port that is connected first is used as the primary management port.
 - If port 1 fails, the system switches management traffic to port 2 automatically. When port 1 recovers, management traffic cannot be switched back to port 1, unless port 2 fails or the switch restarts. You can observe indicators on the ETH management ports to determine which port is used currently. (The Link indicator of the ETH management port in use is steady green. If data is being transmitted on this port, its ACT indicator is blinking yellow. The indicators of the backup port are off.)

USB Port

A USB flash drive can be connected to the USB port for log backup, system software backup and uploading, or USB-based deployment.

2.6 Specifications

Table 2-17 lists the specifications of CE7800&6800&5800 series switches.

Table 2-17 Specifications

Item	Description
Physical specifications	● Dimensions (W x D x H): - CE7850-32Q-EI: 442.0 mm x 607.0 mm x 43.6 mm - CE6810-48S4Q-EI: 442.0 mm x 600.0 mm x 43.6 mm - CE6810-48S4Q-LI: 442.0 mm x 600.0 mm x 43.6 mm - CE6810-48S-LI: 442.0 mm x 600.0 mm x 43.6 mm - CE6810-32T16S4Q-LI: 442.0 mm x 420.0 mm x 43.6 mm - CE6810-24S2Q-LI: 442.0 mm x 600.0 mm x 43.6 mm - CE6850-48S4Q-EI: 442.0 mm x 600.0 mm x 43.6 mm - CE6850-48T4Q-EI: 442.0 mm x 600.0 mm x 43.6 mm - CE6850-48S6Q-HI: 442.0 mm x 600.0 mm x 43.6 mm - CE6850U-48S6Q-HI: 442.0 mm x 600.0 mm x 43.6 mm - CE6850-48T6Q-HI: 442.0 mm x 600.0 mm x 43.6 mm - CE6850U-24S2Q-HI: 442.0 mm x 600.0 mm x 43.6 mm - CE6851-48S6Q-HI: 442.0 mm x 420.0 mm x 43.6 mm - CE5810-24T4S-EI: 442.0 mm x 420.0 mm x 43.6 mm - CE5810-48T4S-EI: 442.0 mm x 420.0 mm x 43.6 mm - CE5850-48T4S2Q-EI: 442.0 mm x 420.0 mm x 43.6 mm - CE5850-48T4S2Q-HI: 442.0 mm x 420.0 mm x 43.6 mm - CE5855-48T4S2Q-EI: 442.0 mm x 420.0 mm x 43.6 mm - CE5855-24T4S2Q-EI: 442.0 mm x 420.0 mm x 43.6 mm

Item		Description
		● Weight (with two power modules and two fan modules, calculated based on the heaviest model if multiple models are supported): - CE7850-32Q-EI: 11.2 kg - CE6810-48S4Q-EI: 10.4 kg - CE6810-48S4Q-LI: 10.4 kg - CE6810-48S-LI: 10.2 kg - CE6810-32T16S4Q-LI: 8.5 kg - CE6810-24S2Q-LI: 10.1 kg - CE6850-48S4Q-EI: 11.05 kg - CE6850-48T4Q-EI: 11.35 kg - CE6850-48S6Q-HI: 11.6 kg - CE6850U-48S6Q-HI: 12.6 kg - CE6850-48T6Q-HI: 12.6 kg - CE6850U-24S2Q-HI: 12.3 kg - CE6851-48S6Q-HI: 8.7 kg - CE5810-24T4S-EI: 8.0 kg - CE5810-48T4S-EI: 8.2 kg - CE5850-48T4S2Q-EI: 8.85 kg - CE5850-48T4S2Q-HI: 8.8 kg - CE5855-48T4S2Q-EI: 8.4 kg - CE5855-24T4S2Q-EI: 8.1 kg
Environment parameters	Temperature	● Operating temperature: 0°C to 40°C (0 m to 1800 m) NOTE When the altitude is between 1800 m and 5000 m, the highest operating temperature reduces 1°C every time the altitude increases 220 m. ● Storage temperature: -40°C to +70°C
	Relative humidity	5% RH to 95% RH, noncondensing
	Altitude	< 5000 m

Item		Description
	Noise (sound pressure, 27°C)	● CE7850-32Q-EI: - Back-to-front airflow: < 55 dBA - Front-to-back airflow: < 54 dBA ● CE6810-48S4Q-EI: - Back-to-front airflow: < 51 dBA - Front-to-back airflow: < 48 dBA ● CE6810-48S4Q-LI: - Back-to-front airflow: < 51 dBA - Front-to-back airflow: < 48 dBA ● CE6810-48S-LI: - Back-to-front airflow: < 51 dBA - Front-to-back airflow: < 48 dBA ● CE6810-32T16S4Q-LI: - Back-to-front airflow: < 51 dBA - Front-to-back airflow: < 51 dBA ● CE6810-24S2Q-LI: - Back-to-front airflow: < 51 dBA - Front-to-back airflow: < 48 dBA ● CE6850-48S4Q-EI: - Back-to-front airflow: < 45 dBA - Front-to-back airflow: < 56 dBA ● CE6850-48T4Q-EI: - Back-to-front airflow: < 56 dBA - Front-to-back airflow: < 56 dBA ● CE6850-48S6Q-HI: - Back-to-front airflow: < 52 dBA - Front-to-back airflow: < 52 dBA ● CE6850U-48S6Q-HI: - Back-to-front airflow: < 52 dBA - Front-to-back airflow: < 52 dBA ● CE6850-48T6Q-HI: - Back-to-front airflow: < 53 dBA - Front-to-back airflow: < 53 dBA ● CE6850U-24S2Q-HI: - Back-to-front airflow: < 52 dBA - Front-to-back airflow: < 52 dBA ● CE6851-48S6Q-HI:

Item		Description
		- Back-to-front airflow: < 56 dBA - Front-to-back airflow: < 58 dBA ● CE5810-24T4S-EI: - Back-to-front airflow: < 43 dBA - Front-to-back airflow: < 47 dBA ● CE5810-48T4S-EI: - Back-to-front airflow: < 43 dBA - Front-to-back airflow: < 47 dBA ● CE5850-48T4S2Q-EI: - Back-to-front airflow: < 45 dBA - Front-to-back airflow: < 45 dBA ● CE5850-48T4S2Q-HI: - Back-to-front airflow: < 45 dBA - Front-to-back airflow: < 51 dBA ● CE5855-48T4S2Q-EI: - Back-to-front airflow: < 48 dBA - Front-to-back airflow: < 55 dBA ● CE5855-24T4S2Q-EI: - Back-to-front airflow: < 48 dBA - Front-to-back airflow: < 51 dBA
Power specifications	Power source type	AC/DC/high-voltage DC
	AC power input	● Rated input voltage range: 100 V AC to 240 V AC, 50/60 Hz ● Maximum input voltage range: - CE6850-48S6Q-HI, CE6850-48T6Q-HI, CE6850U-48S6Q-HI, and CE6850U-24S2Q-HI: 90 V AC to 290 V AC, 47 Hz to 63 Hz - CE5855-48T4S2Q-EI and CE5855-24T4S2Q-EI: 90 V AC to 264 V AC, 47 Hz to 63 Hz - Other models: 90 V AC to 290 V AC, 45 Hz to 65 Hz
	DC power input	The CE5800 series, CE6810 series, CE6850-48S4Q-EI, and CE6851-48S6Q-HI support DC power inputs: ● Rated voltage range: -48 V DC to -60 V DC ● Maximum voltage range: -38.4 V DC to -72 V DC

Item		Description
	High-voltage DC power input	The CE6850-48S6Q-HI, CE6850-48T6Q-HI, CE6850U-48S6Q-HI, and CE6850U-24S2Q-HI support high-voltage DC power inputs: ● Rated voltage range of 240 V high-voltage DC power input: 240 V DC ● Maximum voltage range of 240 V high-voltage DC power input: 188 V DC to 290 V DC ● Rated voltage range of 380 V high-voltage DC power input: 240 V DC to 380 V DC ● Maximum voltage range of 380 V high-voltage DC power input: 188 V DC to 400 V DC
	Maximum input current	● 150 W AC power (PAC-150WA): 2.5 A (90 V AC) ● 150 W AC power (ES0W2PSA0150): 3 A (90 V AC) ● 350 W AC power (PAC-350WA series): 5 A (90 V AC) ● 350 W DC power (PDC-350WA series): 11 A (-38.4 V DC) ● 600 W AC power (PAC-600WA series): 9 A (90 V AC) ● 600 W AC&240 V DC power module (PAC-600WB series): 9 A (90 V AC)/4 A (240 V DC) ● 600 W high-voltage DC power module (PHD-600WA series): 4 A (240 V DC)

Item		Description
Chassis power consumption	Maximum power consumption	● CE7850-32Q-EI: 431 W ● CE6810-48S4Q-EI: 238 W ● CE6810-48S4Q-LI: 238 W ● CE6810-48S-LI: 178 W ● CE6810-32T16S4Q-LI: 288 W ● CE6810-24S2Q-LI: 171 W ● CE6850-48S4Q-EI: 272 W ● CE6850-48T4Q-EI: 380 W ● CE6850-48S6Q-HI: 272 W ● CE6850U-48S6Q-HI: 339 W ● CE6850-48T6Q-HI: 379 W ● CE6850U-24S2Q-HI: 282 W ● CE6851-48S6Q-HI: 245 W ● CE5810-24T4S-EI: 68 W ● CE5810-48T4S-EI: 92 W ● CE5850-48T4S2Q-EI: 133 W ● CE5850-48T4S2Q-HI: 131 W ● CE5855-48T4S2Q-EI: 103 W ● CE5855-24T4S2Q-EI: 75 W

Item		Description
	Typical power consumption	● CE7850-32Q-EI: 297 W (100% traffic load, QSFP+ cables on 32 ports, double power modules) ● CE6810-48S4Q-EI: 101 W (100% traffic load, SFP + cables on 48 ports and QSFP+ cables on 4 ports, double power modules) ● CE6810-48S4Q-LI: 101 W (100% traffic load, SFP + cables on 48 ports and QSFP+ cables on 4 ports, double power modules) ● CE6810-48S-LI: 89 W (100% traffic load, SFP+ cables on 48 ports, double power modules) ● CE6810-32T16S4Q-LI: 204 W (100% traffic load, 3 m Ethernet cables on 32 ports, SFP+ high-speed copper cables on 16 ports, and QSFP+ high-speed copper cables on 4 ports, double power modules) ● CE6810-24S2Q-LI: 88 W (100% traffic load, SFP+ cables on 24 ports and QSFP+ cables on 2 ports, double power modules) ● CE6850-48S4Q-EI: 180 W (100% traffic load, SFP + cables on 48 ports and QSFP+ cables on 4 ports, double power modules) ● CE6850-48T4Q-EI: 305 W (100% traffic load, 3 m network cables on 48 ports and QSFP+ cables on 4 ports, double power modules) ● CE6850-48S6Q-HI: 166 W (100% traffic load, SFP + cables on 48 ports and QSFP+ cables on 6 ports, double power modules) ● CE6850U-48S6Q-HI: 235 W (100% traffic load, SFP+ cables on 48 ports and QSFP+ cables on 6 ports, double power modules) ● CE6850-48T6Q-HI: 252 W (100% traffic load, 3 m network cables on 48 ports and QSFP+ cables on 6 ports, double power modules) ● CE6850U-24S2Q-HI: 183 W (100% traffic load, SFP+ cables on 24 ports and QSFP+ cables on 2 ports, double power modules) ● CE6851-48S6Q-HI: 145 W (100% traffic load, SFP + cables on 48 ports and QSFP+ cables on 6 ports, double power modules) ● CE5810-24T4S-EI: 58 W (100% traffic load, 3 m network cables on 24 ports, SFP+ cables on 4 ports, double power modules) ● CE5810-48T4S-EI: 80 W (100% traffic load, 3 m network cables on 48 ports and SFP+ cables on 4 ports, double power modules)

Item		Description
		● CE5850-48T4S2Q-EI: 103 W (100% traffic load, 3 m network cables on 48 ports and SFP+ cables on 4 ports and QSFP+ cables on 2 ports, double power modules) ● CE5850-48T4S2Q-HI: 109 W (100% traffic load, 3 m network cables on 48 ports and SFP+ cables on 4 ports and QSFP+ cables on 2 ports, double power modules) ● CE5855-48T4S2Q-EI: 76 W (100% traffic load, 3 m network cables on 48 ports and SFP+ cables on 4 ports and QSFP+ cables on 2 ports, double power modules) ● CE5855-24T4S2Q-EI: 48 W (100% traffic load, 3 m network cables on 24 ports and SFP+ cables on 4 ports and QSFP+ cables on 2 ports, double power modules)
Chassis heat dissipation	Maximum heat dissipation	● CE7850-32Q-EI: 1470 BTU/hr ● CE6810-48S4Q-EI: 812 BTU/hr ● CE6810-48S4Q-LI: 812 BTU/hr ● CE6810-48S-LI: 607 BTU/hr ● CE6810-32T16S4Q-LI: 983 BTU/hr ● CE6810-24S2Q-LI: 583 BTU/hr ● CE6850-48S4Q-EI: 928 BTU/hr ● CE6850-48T4Q-EI: 1297 BTU/hr ● CE6850-48S6Q-HI: 928 BTU/hr ● CE6850U-48S6Q-HI: 1157 BTU/hr ● CE6850-48T6Q-HI: 1293 BTU/hr ● CE6850U-24S2Q-HI: 962 BTU/hr ● CE6851-48S6Q-HI: 836 BTU/hr ● CE5810-24T4S-EI: 232 BTU/hr ● CE5810-48T4S-EI: 314 BTU/hr ● CE5850-48T4S2Q-EI: 454 BTU/hr ● CE5850-48T4S2Q-HI: 447 BTU/hr ● CE5855-48T4S2Q-EI: 351 BTU/hr ● CE5855-24T4S2Q-EI: 256 BTU/hr

Item		Description
	Typical heat dissipation	● CE7850-32Q-EI: 1013 BTU/hr (100% traffic load, QSFP+ cables on 32 ports, double power modules) ● CE6810-48S4Q-EI: 344 BTU/hr (100% traffic load, SFP+ cables on 48 ports and QSFP+ cables on 4 ports, double power modules) ● CE6810-48S4Q-LI: 344 BTU/hr (100% traffic load, SFP+ cables on 48 ports and QSFP+ cables on 4 ports, double power modules) ● CE6810-48S-LI: 302 BTU/hr (100% traffic load, SFP+ cables on 48 ports, double power modules) ● CE6810-32T16S4Q-LI: 696 BTU/hr (100% traffic load, 3 m Ethernet cables on 32 ports, SFP+ high-speed copper cables on 16 ports, and QSFP+ high-speed copper cables on 4 ports, double power modules) ● CE6810-24S2Q-LI: 300 BTU/hr (100% traffic load, SFP+ cables on 24 ports and QSFP+ cables on 2 ports, double power modules) ● CE6850-48S4Q-EI: 614 BTU/hr (100% traffic load, SFP+ cables on 48 ports and QSFP+ cables on 4 ports, double power modules) ● CE6850-48T4Q-EI: 1041 BTU/hr (100% traffic load, 3 m network cables on 48 ports and QSFP+ cables on 4 ports, double power modules) ● CE6850-48S6Q-HI: 566 BTU/hr (100% traffic load, SFP+ cables on 48 ports and QSFP+ cables on 6 ports, double power modules) ● CE6850U-48S6Q-HI: 802 BTU/hr (100% traffic load, SFP+ cables on 48 ports and QSFP+ cables on 6 ports, double power modules) ● CE6850-48T6Q-HI: 860 BTU/hr (100% traffic load, 3 m network cables on 48 ports and QSFP+ cables on 6 ports, double power modules) ● CE6850U-24S2Q-HI: 624 BTU/hr (100% traffic load, SFP+ cables on 24 ports and QSFP+ cables on 2 ports, double power modules) ● CE6851-48S6Q-HI: 495 BTU/hr (100% traffic load, SFP+ cables on 48 ports and QSFP+ cables on 6 ports, double power modules) ● CE5810-24T4S-EI: 198 BTU/hr (100% traffic load, 3 m network cables on 24 ports, SFP+ cables on 4 ports, double power modules) ● CE5810-48T4S-EI: 273 BTU/hr (100% traffic load, 3 m network cables on 48 ports and SFP+ cables on 4 ports, double power modules)

Item		Description
		● CE5850-48T4S2Q-EI: 351 BTU/hr (100% traffic load, 3 m network cables on 48 ports and SFP+ cables on 4 ports and QSFP+ cables on 2 ports, double power modules) ● CE5850-48T4S2Q-HI: 372 BTU/hr (100% traffic load, 3 m network cables on 48 ports and SFP+ cables on 4 ports and QSFP+ cables on 2 ports, double power modules) ● CE5855-48T4S2Q-EI: 259 BTU/hr (100% traffic load, 3 m network cables on 48 ports and SFP+ cables on 4 ports and QSFP+ cables on 2 ports, double power modules) ● CE5855-24T4S2Q-EI: 164 BTU/hr (100% traffic load, 3 m network cables on 24 ports and SFP+ cables on 4 ports and QSFP+ cables on 2 ports, double power modules)
Surge protection		Ethernet electrical ports: 2 kV in common mode Power module: ● AC: – CE6850-48S6Q-HI, CE6850-48T6Q-HI, CE6850U-48S6Q-HI, and CE6850U-24S2Q-HI: 4 kV in common mode and 2.5 kV in differential mode – Other models: 6 kV in common mode and 6 kV in differential mode ● DC: 4 kV in common mode and 2 kV in differential mode
Heat dissipation	Heat dissipation mode	Air cooling
	Airflow	Front-to-back or back-to-front, which is determined by features of fan modules and power modules.
Reliability	Power module backup	1+1 backup
	Fan module backup	The CE5850EI and CE5810EI support 1+1 backup of fan modules NOTE A CE6800, CE7850, CE5855EI, or CE5850HI chassis uses two fan modules, with each fan module containing two fans. The four fans on the chassis work in 3+1 backup mode.
	Hot swap	All the power modules and fan modules support hot swap.

Item		Description
Technical specifications	Processor	● CE7850-32Q-EI: 1.5 GHz, quad-core ● CE6810-48S4Q-EI: 1.5 GHz, quad-core ● CE6810-48S4Q-LI: 1.2 GHz, quad-core ● CE6810-48S-LI: 1.2 GHz, quad-core ● CE6810-32T16S4Q-LI: 1.2 GHz, quad-core ● CE6810-24S2Q-LI: 1.2 GHz, quad-core ● CE6850-48S4Q-EI: 1.5 GHz, quad-core ● CE6850-48T4Q-EI: 1.5 GHz, quad-core ● CE6850-48S6Q-HI: 1.5 GHz, quad-core ● CE6850U-48S6Q-HI: 1.5 GHz, quad-core ● CE6850-48T6Q-HI: 1.2 GHz, quad-core ● CE6850U-24S2Q-HI: 1.5 GHz, quad-core ● CE6851-48S6Q-HI: 1.2 GHz, quad-core ● CE5810-24T4S-EI: 1.2 GHz, dual-core ● CE5810-48T4S-EI: 1.2 GHz, dual-core ● CE5850-48T4S2Q-EI: 1.2 GHz, quad-core ● CE5850-48T4S2Q-HI: 1.2 GHz, dual-core ● CE5855-48T4S2Q-EI: 1 GHz, dual-core ● CE5855-24T4S2Q-EI: 1 GHz, dual-core

Item		Description
	DRAM Memory	● CE7850-32Q-EI: 4 GB ● CE6810-48S4Q-EI: 2 GB ● CE6810-48S4Q-LI: 2 GB ● CE6810-48S-LI: 2 GB ● CE6810-32T16S4Q-LI: 2 GB ● CE6810-24S2Q-LI: 2 GB ● CE6850-48S4Q-EI: 2 GB ● CE6850-48T4Q-EI: 2 GB ● CE6850-48S6Q-HI: 4 GB ● CE6850U-48S6Q-HI: 4 GB ● CE6850-48T6Q-HI: 4 GB ● CE6850U-24S2Q-HI: 4 GB ● CE6851-48S6Q-HI: 2 GB ● CE5810-24T4S-EI: 2 GB ● CE5810-48T4S-EI: 2 GB ● CE5850-48T4S2Q-EI: 2 GB ● CE5850-48T4S2Q-HI: 2 GB ● CE5855-48T4S2Q-EI: 2 GB ● CE5855-24T4S2Q-EI: 2 GB

Item		Description
	NOR Flash	● CE7850-32Q-EI: 16 MB ● CE6810-48S4Q-EI: 16 MB ● CE6810-48S4Q-LI: 16 MB ● CE6810-48S-LI: 16 MB ● CE6810-32T16S4Q-LI: 16 MB ● CE6810-24S2Q-LI: 16 MB ● CE6850-48S4Q-EI: 8 MB ● CE6850-48T4Q-EI: 8 MB ● CE6850-48S6Q-HI: 16 MB ● CE6850U-48S6Q-HI: 16 MB ● CE6850-48T6Q-HI: 16 MB ● CE6850U-24S2Q-HI: 16 MB ● CE6851-48S6Q-HI: 16 MB ● CE5810-24T4S-EI: 16 MB ● CE5810-48T4S-EI: 16 MB ● CE5850-48T4S2Q-EI: 8 MB ● CE5850-48T4S2Q-HI: 16 MB ● CE5855-48T4S2Q-EI: 16 MB ● CE5855-24T4S2Q-EI: 16 MB

Item		Description
	NAND Flash	● CE7850-32Q-EI: 1 GB ● CE6810-48S4Q-EI: 1 GB ● CE6810-48S4Q-LI: 512 MB ● CE6810-48S-LI: 512 MB ● CE6810-32T16S4Q-LI: 512 MB ● CE6810-24S2Q-LI: 512 MB ● CE6850-48S4Q-EI: 1 GB ● CE6850-48T4Q-EI: 1 GB ● CE6850-48S6Q-HI: 1 GB ● CE6850U-48S6Q-HI: 1 GB ● CE6850-48T6Q-HI: 1 GB ● CE6850U-24S2Q-HI: 1 GB ● CE6851-48S6Q-HI: 1 GB ● CE5810-24T4S-EI: 512 MB ● CE5810-48T4S-EI: 512 MB ● CE5850-48T4S2Q-EI: 1 GB ● CE5850-48T4S2Q-HI: 1 GB ● CE5855-48T4S2Q-EI: 512 MB ● CE5855-24T4S2Q-EI: 512 MB

Item		Description
Stack	Service port supporting the stack function	● CE7850-32Q-EI: 40GE optical ports ● CE6810-48S4Q-EI: 10GE optical ports and 40GE optical ports ● CE6810-48S4Q-LI: 10GE optical ports and 40GE optical ports ● CE6810-48S-LI: 10GE optical ports ● CE6810-32T16S4Q-LI: 10GE electrical ports, 10GE optical ports, and 40GE optical ports ● CE6810-24S2Q-LI: 10GE optical ports and 40GE optical ports ● CE6850-48S4Q-EI: 10GE optical ports and 40GE optical ports ● CE6850-48T4Q-EI: 10GE electrical ports (V100R002 and later versions) and 40GE optical ports ● CE6850-48S6Q-HI: 10GE optical ports and 40GE optical ports ● CE6850U-48S6Q-HI: 10GE optical ports and 40GE optical ports ● CE6850-48T6Q-HI: 10GE electrical ports and 40GE optical ports ● CE6850U-24S2Q-HI: 10GE optical ports and 40GE optical ports ● CE6851-48S6Q-HI: 10GE optical ports and 40GE optical ports ● CE5810-24T4S-EI: 10GE optical ports ● CE5810-48T4S-EI: 10GE optical ports ● CE5850-48T4S2Q-EI: 10GE and 40GE optical ports ● CE5850-48T4S2Q-HI: 10GE and 40GE optical ports ● CE5855-48T4S2Q-EI: 10GE optical ports and 40GE optical ports ● CE5855-24T4S2Q-EI: 10GE optical ports and 40GE optical ports NOTE Downlink GE electrical ports cannot be used as stack ports.

Item	Description
Safety standards compliance	● EN 60950-1:2006+A11:2009+A1:2010+A12:2011 ● EN 60825-1:2007 ● EN 60825-2:2010 ● UL 60950-1:2007 2rd Edition ● CSA C22.2 No.650:2007 2rd Edition ● IEC 60950-1:2005+A1:2009 ● AS/NZS 60950-1:2011 ● GB4943:2011
EMC standards compliance	● FCC 47CFR Part15 CLASS A ● ETSI EN 300 386 V1.6.1:2012 ● ICES-003:2012 CLASS A ● CISPR 22:2008 CLASS A ● CISPR 24:2010 ● EN 55022:2010 CLASS A ● EN 55024:2010 ● AS/NZS CISPR 22:2009 CLASS A ● IEC 61000-3-2:2005+A1:2008+A2:2009/EN 61000-3-2:2006+A1:2009+A2:2009 ● IEC 61000-3-3:2008/EN 61000-3-3:2008 ● CNS 13438:2006 CLASS A ● VCCI V-4:2012 CLASS A ● VCCI V-3:2012 CLASS A ● EC Council Directive 2004/108/EC ● GB9254
Safety and environmental standards compliance	● 2002/95/EC, 2011/65/EU ● 2002/96/EC, 2012/19/EU ● EC NO.1907/2006 ● ETSI EN 300 019-1-1 V2.1.4 ● ETSI EN 300 019-1-2 V2.1.4 ● ETSI EN 300 019-1-3 V2.3.2 ● ETSI EN 300753 V1.2.1

2.7 Ordering Information

Ordering information is subject to change without notice in the case of product upgrades. Ordering information provided in this manual is for reference only. To obtain latest ordering information, contact Huawei switch distributors or local Huawei representative office.

Table 2-18 provides the ordering information.

Table 2-18 Ordering information

Part Number	Part Model	Part Description
02355272	CE5850-48T4S2Q-EI	CE5850-48T4S2Q-EI Switch (48-Port GE RJ45, 4-Port 10GE SFP+, 2-Port 40GE QSFP+, Without Fan and Power Module)
02350EXX	CE5850-48T4S2Q-EI-F	CE5850-48T4S2Q-EI Switch (48-Port GE RJ45, 4-Port 10G SFP+, 2-Port 40G QSFP+, 2*FAN Box, Port-side Exhaust, Without Power Module)
02350EXY	CE5850-48T4S2Q-EI-B	CE5850-48T4S2Q-EI Switch (48-Port GE RJ45, 4-Port 10G SFP+, 2-Port 40G QSFP+, 2*FAN Box, Port-side Intake, Without Power Module)
02359104	CE5850-EI-B00	CE5850-48T4S2Q-EI Switch (2*150W AC Power Module, 2*FAN Box, Port side Exhaust)
02350FCK	CE5850-EI-B-B00	CE5850-48T4S2Q-EI Switch (2*150W Power Module, 2*FAN Box, Port-side Intake)
02356878	CE5810-48T4S-EI	CE5810-48T4S-EI Switch (48-Port GE RJ45, 4-Port 10GE SFP+, Without Fan and Power Module)
02350EYW	CE5810-48T4S-EI-F	CE5810-48T4S-EI Switch (48-Port GE RJ45, 4-Port 10G SFP+, 2*FAN Box, Port-side Exhaust, Without Power Module)
02350EYX	CE5810-48T4S-EI-B	CE5810-48T4S-EI Switch (48-Port GE RJ45, 4-Port 10G SFP+, 2*FAN Box, Port-side Intake, Without Power Module)
02359081	CE5810-EI-B00	CE5810-48T4S-EI Switch (2*150W AC Power Module, 2*FAN Box, Port side Exhaust)
02350EYP	CE5810-EI-B-B00	CE5810-48T4S-EI Switch (2*150W Power Module, 2*FAN Box, Port-side Intake)
02350BGQ	CE5810-EI-B10	CE5810-48T4S-EI Bundle 10(CE5810-48T4S-EI mainframe, 8*SFP-10G-USR, Without Fan and Power Module)
02356879	CE5810-24T4S-EI	CE5810-24T4S-EI Switch (24-Port GE RJ45, 4-Port 10GE SFP+, Without Fan and Power Module)
02350EYF	CE5810-24T4S-EI-F	CE5810-24T4S-EI Switch (24-Port GE RJ45, 4-Port 10G SFP+, 2*FAN Box, Port-side Exhaust, Without Power Module)

Part Number	Part Model	Part Description
02350EYH	CE5810-24T4S-EI-B	CE5810-24T4S-EI Switch (24-Port GE RJ45, 4-Port 10G SFP+, 2*FAN Box, Port-side Intake, Without Power Module)
02359082	CE5810-EI-B01	CE5810-24T4S-EI Switch (2*150W AC Power Module, 2*FAN Box, Port side Exhaust)
02350EYN	CE5810-EI-B-B01	CE5810-24T4S-EI Switch (2*150W Power Module, 2*FAN Box, Port-side Intake)
02350BGP	CE5810-EI-B11	CE5810-24T4S-EI Bundle 11 (CE5810-24T4S-EI mainframe, 4*SFP-10G-USR, Without Fan and Power Module)
02358051	CE5850-48T4S2Q-HI	CE5850-48T4S2Q-HI Switch (48-Port GE RJ45, 4-Port 10GE SFP+, 2-Port 40GE QSFP+, Without Fan and Power Module)
02350EYC	CE5850-48T4S2Q-HI-F	CE5850-48T4S2Q-HI Switch (48-Port GE RJ45, 4-Port 10G SFP+, 2-Port 40G QSFP+, 2*FAN Box, Port-side Exhaust, Without Power Module)
02350EYD	CE5850-48T4S2Q-HI-B	CE5850-48T4S2Q-HI Switch (48-Port GE RJ45, 4-Port 10G SFP+, 2-Port 40G QSFP+, 2*FAN Box, Port-side Intake, Without Power Module)
02359246	CE5850-HI-B00	CE5850-48T4S2Q-HI Switch (2*150W AC Power Module, 2*FAN Box, Port side Exhaust)
02350FCL	CE5850-HI-B-B00	CE5850-48T4S2Q-HI Switch (2*150W Power Module, 2*FAN Box, Port-side Intake)
02350GTR	CE5855-48T4S2Q-EI-F	CE5855-48T4S2Q-EI Switch (48-Port GE RJ45, 4-Port 10G SFP+, 2-Port 40G QSFP+, 2*FAN Box, Port-side Exhaust, Without Power Module)
02350GTT	CE5855-48T4S2Q-EI-B	CE5855-48T4S2Q-EI Switch (48-Port GE RJ45, 4-Port 10G SFP+, 2-Port 40G QSFP+, 2*FAN Box, Port-side Intake, Without Power Module)
02350GTU	CE5855-EI-F-B00	CE5855-48T4S2Q-EI Switch (48-Port GE RJ45, 4-Port 10G SFP+, 2-Port 40G QSFP+, 2*AC Power Module, 2*FAN Box, Port-side Exhaust)
02350GTW	CE5855-EI-B-B00	CE5855-48T4S2Q-EI Switch (48-Port GE RJ45, 4-Port 10G SFP+, 2-Port 40G QSFP+, 2*AC Power Module, 2*FAN Box, Port-side Intake)
02350GTX	CE5855-24T4S2Q-EI-F	CE5855-24T4S2Q-EI Switch (24-Port GE RJ45, 4-Port 10G SFP+, 2-Port 40G QSFP+, 2*FAN Box, Port-side Exhaust, Without Power Module)

Part Number	Part Model	Part Description
02350GTY	CE5855-24T4S2Q-EI-B	CE5855-24T4S2Q-EI Switch (24-Port GE RJ45, 4-Port 10G SFP+, 2-Port 40G QSFP+, 2*FAN Box, Port-side Intake, Without Power Module)
02350GUA	CE5855-EI-F-B01	CE5855-24T4S2Q-EI Switch (24-Port GE RJ45, 4-Port 10G SFP+, 2-Port 40G QSFP+, 2*AC Power Module, 2*FAN Box, Port-side Exhaust)
02350GUB	CE5855-EI-B-B01	CE5855-24T4S2Q-EI Switch (24-Port GE RJ45, 4-Port 10G SFP+, 2-Port 40G QSFP+, 2*AC Power Module, 2*FAN Box, Port-side Intake)
02350JAR	CE6851-HI-F-B0A	CE6851-48S6Q-HI Switch (48-Port 10G SFP+, 6-Port 40GE QSFP+, 2*AC Power Module, 2*FAN Box, Port-side Exhaust)
02350JAS	CE6851-HI-B-B0A	CE6851-48S6Q-HI Switch (48-Port 10G SFP+, 6-Port 40GE QSFP+, 2*AC Power Module, 2*FAN Box, Port-side Intake)
02350JAP	CE6851-48S6Q-HI-F	CE6851-48S6Q-HI Switch (48-Port 10G SFP+, 6-Port 40GE QSFP+, 2*FAN Box, Port-side Exhaust, Without Power Module)
02350JAQ	CE6851-48S6Q-HI-B	CE6851-48S6Q-HI Switch (48-Port 10G SFP+, 6-Port 40GE QSFP+, 2*FAN Box, Port-side Intake, Without Power Module)
02350EWK	CE6850-HI-F-B00	CE6850-48T6Q-HI Switch (48-Port 10GE RJ45, 6-Port 40GE QSFP+, 2*AC Power Module, 2*FAN Box, Port-side Exhaust)
02350EWL	CE6850-HI-B-B00	CE6850-48T6Q-HI Switch (48-Port 10GE RJ45, 6-Port 40GE QSFP+, 2*AC Power Module, 2*FAN Box, Port-side Intake)
02350EWH	CE6850-48T6Q-HI-F	CE6850-48T6Q-HI Switch (48-Port 10GE RJ45, 6-Port 40GE QSFP+, 2*FAN Box, Port-side Exhaust, Without Power Module)
02350EWJ	CE6850-48T6Q-HI-B	CE6850-48T6Q-HI Switch (48-Port 10GE RJ45, 6-Port 40GE QSFP+, 2*FAN Box, Port-side Intake, Without Power Module)
02359314	CE6850-HI-B00	CE6850-48S6Q-HI Switch (2*600W AC Power Module, 2*FAN Box, Port Side Exhaust)
02350EHE	CE6850-HI-B-B0A	CE6850-48S6Q-HI Switch (2*600W AC Power Module, 2*FAN Box, Port-side Intake)
02350EHC	CE6850-48S6Q-HI-F	CE6850-48S6Q-HI Switch (48-Port 10G SFP+, 6-Port 40GE QSFP+, 2*FAN Box, Port-side Exhaust, Without Power Module)

Part Number	Part Model	Part Description
02350EHD	CE6850-48S6Q-HI-B	CE6850-48S6Q-HI Switch (48-Port 10G SFP+, 6-Port 40GE QSFP+, 2*FAN Box, Port-side Intake, Without Power Module)
02359312	CE6850U-HI-F-B0A	CE6850U-48S6Q-HI Switch (48-Port 10GE SFP+, support 2/4/8G FC, 6-Port 40GE QSFP+, 2*AC Power Module, 2*FAN Box, Port-side Exhaust)
02350EHH	CE6850U-HI-B-B0A	CE6850U-48S6Q-HI Switch (48-Port 10GE SFP+, support 2/4/8G FC, 6-Port 40GE QSFP+, 2*AC Power Module, 2*FAN Box, Port-side Intake)
02350EHF	CE6850U-48S6Q-HI-F	CE6850U-48S6Q-HI Switch (48-Port 10GE SFP+, support 2/4/8G FC, 6-Port 40GE QSFP+, 2*FAN Box, Port-side Exhaust, Without Power Module)
02350EHG	CE6850U-48S6Q-HI-B	CE6850U-48S6Q-HI Switch (48-Port 10GE SFP+, support 2/4/8G FC, 6-Port 40GE QSFP+, 2*FAN Box, Port-side Intake, Without Power Module)
02350GTP	CE6850U-HI-F-B0B	CE6850U-24S2Q-HI Switch (24-Port 10GE SFP+, support 2/4/8G FC, 2-Port 40GE QSFP+, 2*AC Power Module, 2*FAN Box, Port-side Exhaust)
02350GTQ	CE6850U-HI-B-B0B	CE6850U-24S2Q-HI Switch (24-Port 10GE SFP+, support 2/4/8G FC, 2-Port 40GE QSFP+, 2*AC Power Module, 2*FAN Box, Port-side Intake)
02350GTM	CE6850U-24S2Q-HI-F	CE6850U-24S2Q-HI Switch (24-Port 10GE SFP+, support 2/4/8G FC, 2-Port 40GE QSFP+, 2*FAN Box, Port-side Exhaust, Without Power Module)
02350GTN	CE6850U-24S2Q-HI-B	CE6850U-24S2Q-HI Switch (24-Port 10GE SFP+, support 2/4/8G FC, 2-Port 40GE QSFP+, 2*FAN Box, Port-side Intake, Without Power Module)
02359083	CE6850-EI-B00	CE6850-48S4Q-EI Switch (2*350W AC Power Module, 2*FAN Box, Port side Exhaust)
02350EXQ	CE6850-EI-B-B0A	CE6850-48S4Q-EI Switch (2*350W AC Power Module, 2*FAN Box, Port-side Intake)
02350EXD	CE6850-48S4Q-EI-F	CE6850-48S4Q-EI Switch (48-Port 10G SFP+, 4-Port 40G QSFP+, 2*FAN Box, Port-side Exhaust, Without Power Module)
02350EXE	CE6850-48S4Q-EI-B	CE6850-48S4Q-EI Switch (48-Port 10G SFP+, 4-Port 40G QSFP+, 2*FAN Box, Port-side Intake, Without Power Module)
02355264	CE6850-48S4Q-EI	CE6850-48S4Q-EI Switch (48-Port 10GE SFP+, 4-Port 40GE QSFP+, Without Fan and Power Module)

Part Number	Part Model	Part Description
02359084	CE6850-EI-B01	CE6850-48T4Q-EI Switch (2*600W AC Power Module, 2*FAN Box, Port side Exhaust)
02350EXT	CE6850-EI-B-B00	CE6850-48T4Q-EI Switch (2*600W AC Power Module, 2*FAN Box, Port-side Intake)
02350EXR	CE6850-48T4Q-EI-F	CE6850-48T4Q-EI Switch (48-Port 10G RJ45, 4-Port 40G QSFP+, 2*FAN Box, Port-side Exhaust, Without Power Module)
02350EXS	CE6850-48T4Q-EI-B	CE6850-48T4Q-EI Switch (48-Port 10G RJ45, 4-Port 40G QSFP+, 2*FAN Box, Port-side Intake, Without Power Module)
02355265	CE6850-48T4Q-EI	CE6850-48T4Q-EI Switch (48-Port 10GE RJ45, 4-Port 40GE QSFP+, Without Fan and Power Module)
02359247	CE6810-EI-B00	CE6810-48S4Q-EI Switch (2*600W AC Power Module, 2*FAN Box, Port side Exhaust)
02350EXW	CE6810-EI-B-B0A	CE6810-48S4Q-EI Switch (2*600W AC Power Module, 2*FAN Box, Port-side Intake)
02350EXU	CE6810-48S4Q-EI-F	CE6810-48S4Q-EI Switch (48-Port 10G SFP+, 4-Port 40G QSFP+, 2*FAN Box, Port-side Exhaust, Without Power Module)
02350EXV	CE6810-48S4Q-EI-B	CE6810-48S4Q-EI Switch (48-Port 10G SFP+, 4-Port 40G QSFP+, 2*FAN Box, Port-side Intake, Without Power Module)
02358856	CE6810-48S4Q-EI	CE6810-48S4Q-EI Switch (48-Port 10GE SFP+, 4-Port 40GE QSFP+, Without Fan and Power Module)
02350AQB	CE6810-LI-B00	CE6810-48S4Q-LI Switch (2*600W AC Power Module, 2*FAN Box, Port side Exhaust)
02350EGX	CE6810-LI-B-B0A	CE6810-48S4Q-LI Switch (2*600W AC Power Module, 2*FAN Box, Port-side Intake)
02350EGV	CE6810-48S4Q-LI-F	CE6810-48S4Q-LI Switch (48-Port 10G SFP+, 4-Port 40GE QSFP+, 2*FAN Box, Port-side Exhaust, Without Power Module)
02350EGW	CE6810-48S4Q-LI-B	CE6810-48S4Q-LI Switch (48-Port 10G SFP+, 4-Port 40GE QSFP+, 2*FAN Box, Port-side Intake, Without Power Module)
02350APY	CE6810-48S4Q-LI	CE6810-48S4Q-LI Switch (48-Port 10GE SFP+, 4-Port 40GE QSFP+, Without Fan and Power Module)
02350AQC	CE6810-LI-B01	CE6810-48S-LI Switch (2*600W AC Power Module, 2*FAN Box, Port side Exhaust)

Part Number	Part Model	Part Description
02350EHB	CE6810-LI-B-B0B	CE6810-48S-LI Switch (2*600W AC Power Module, 2*FAN Box, Port-side Intake)
02350EGY	CE6810-48S-LI-F	CE6810-48S-LI Switch (48-Port 10G SFP+, 2*FAN Box, Port-side Exhaust, Without Power Module)
02350EHA	CE6810-48S-LI-B	CE6810-48S-LI Switch (48-Port 10G SFP+, 2*FAN Box, Port-side Intake, Without Power Module)
02350AQA	CE6810-48S-LI	CE6810-48S-LI Switch (48-Port 10GE SFP+, Without Fan and Power Module)
02350GUE	CE6810-LI-F-B0C	CE6810-24S2Q-LI Switch (24-Port 10G SFP+, 2-Port 40GE QSFP+, 2*AC Power Module, 2*FAN Box, Port-side Exhaust)
02350GUF	CE6810-LI-B-B0C	CE6810-24S2Q-LI Switch (24-Port 10G SFP+, 2-Port 40GE QSFP+, 2*AC Power Module, 2*FAN Box, Port-side Intake)
02350GUC	CE6810-24S2Q-LI-F	CE6810-24S2Q-LI Switch (24-Port 10G SFP+, 2-Port 40GE QSFP+, 2*FAN Box, Port-side Exhaust, Without Power Module)
02350GUD	CE6810-24S2Q-LI-B	CE6810-24S2Q-LI Switch (24-Port 10G SFP+, 2-Port 40GE QSFP+, 2*FAN Box, Port-side Intake, Without Power Module)
02350EWD	CE6810-LI-F-B00	CE6810-32T16S4Q-LI Switch (32-Port 10G RJ45, 16-Port 10G SFP+, 4-Port 40G QSFP+, 2*AC Power Module, 2*FAN Box, Port-side Exhaust)
02350EWE	CE6810-LI-B-B00	CE6810-32T16S4Q-LI Switch (32-Port 10G RJ45, 16-Port 10G SFP+, 4-Port 40G QSFP+, 2*AC Power Module, 2*FAN Box, Port-side Intake)
02350EWB	CE6810-32T16S4Q-LI-F	CE6810-32T16S4Q-LI Switch (32-Port 10G RJ45, 16-Port 10G SFP+, 4-Port 40G QSFP+, 2*FAN Box, Port-side Exhaust, Without Power Module)
02350EWC	CE6810-32T16S4Q-LI-B	CE6810-32T16S4Q-LI Switch (32-Port 10G RJ45, 16-Port 10G SFP+, 4-Port 40G QSFP+, 2*FAN Box, Port-side Intake, Without Power Module)
02350EVU	CE6810-LI-F-B1A	CE6810-48S4Q-LI Bundle (With 4*40GE QSFP+ Optical Module, 2*FAN Box, Port-side Exhaust, Without Power Module)
02350EVV	CE6810-LI-B-B1A	CE6810-48S4Q-LI Bundle (With 4*40GE QSFP+ Optical Module, 2*FAN Box, Port-side Intake, Without Power Module)

Part Number	Part Model	Part Description
02350EVW	CE6810-LI-F-B1B	CE6810-48S-LI Bundle (With 2*40GE QSFP +&8*10GE SFP+ Optical Module, 2*FAN Box, Port-side Exhaust, Without Power Module)
02350EVX	CE6810-LI-B-B1B	CE6810-48S-LI Bundle (With 2*40GE QSFP +&8*10GE SFP+ Optical Module, 2*FAN Box, Port-side Intake, Without Power Module)
02350CXP	CE6810-LI-B10	CE6810-48S4Q-LI Bundle 10(CE6810-48S4Q-LI mainframe, 4*QSFP-40G-iSR4, Without Fan and Power Module)
02350DBW	CE6810-LI-B11	CE6810-48S-LI Bundle 11(CE6810-48S-LI mainframe, 2*QSFP-40G-iSR4 , 8*SFP-10G-USR, Without Fan and Power Module)
02358859	CE7850-32Q-EI	CE7850-32Q-EI Switch (32-Port 40GE QSFP+, Without Fan and Power Module)
02359250	CE7850-EI-B00	CE7850-32Q-EI Switch (2*600W AC Power Module, 2*FAN Box, Port side Exhaust)
02350EYY	CE7850-32Q-EI-F	CE7850-32Q-EI Switch (32-Port 40G QSFP+, 2*FAN Box, Port-side Exhaust, Without Power Module)
02350FAB	CE7850-32Q-EI-B	CE7850-32Q-EI Switch (32-Port 40G QSFP+, 2*FAN Box, Port-side Intake, Without Power Module)
02350EYQ	CE7850-EI-B-B0A	CE7850-32Q-EI Switch (2*600W AC Power Module, 2*FAN Box, Port-side Intake)

3 Power Module

About This Chapter



NOTICE

- Power modules in a chassis must have the same power and same heat dissipation method.
 - A switch must use the power modules it supports.
 - When two power modules work in 1+1 backup mode, you can hot swap one of them.
 - When only one power module is installed in a chassis, install a filler panel on the empty power supply slot.
 - Before powering off a switch, turn off all its power modules.
-

[3.1 150 W AC Power Module \(PAC-150WA\)](#)

[3.2 150 W AC Power Module \(ES0W2PSA0150\)](#)

[3.3 350 W AC Power Module](#)

[3.4 350 W DC Power Module](#)

[3.5 600 W AC Power Module](#)

[3.6 600 W AC&240 V DC Power Module](#)

[3.7 600 W High-Voltage DC Power Module](#)

3.1 150 W AC Power Module (PAC-150WA)

Version Mapping

Table 3-1 describes the mapping between switch models and the PAC-150WA.

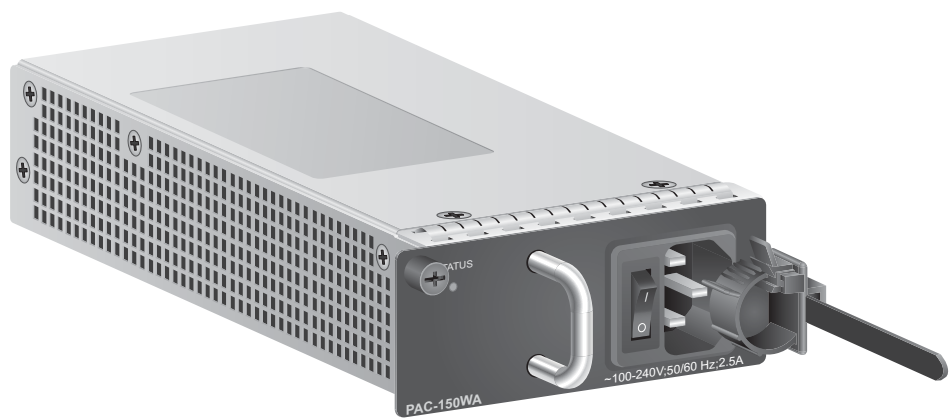
Table 3-1 Mapping between switch models and the PAC-150WA

Switch Model	PAC-150WA
CE5810-24T4S-EI CE5810-48T4S-EI	Supported NOTE This power module is supported in V100R002C00 version and later versions.
CE5850-48T4S2Q-EI	Supported NOTE This power module is supported in V100R001C00 version and later versions.
CE5850-48T4S2Q-HI	Supported NOTE This power module is supported in V100R003C00 version and later versions.
CE6850-48S4Q-EI CE6850-48T4Q-EI CE6850-48S6Q-HI CE6810-48S4Q-EI CE6810-48S4Q-LI CE6810-48S-LI CE7850-32Q-EI CE6850U-48S6Q-HI CE6850-48T6Q-HI CE6850U-24S2Q-HI CE6851-48S6Q-HI CE6810-32T16S4Q-LI CE6810-24S2Q-LI CE5855-48T4S2Q-EI CE5855-24T4S2Q-EI	Not supported

Appearance

Figure 3-1 shows the appearance of the PAC-150WA.

Figure 3-1 Appearance of the PAC-150WA



Function

Table 3-2 describes the functions of the PAC-150WA.

Table 3-2 Functions of the PAC-150WA

Function		Description
Input protection	Input undervoltage protection	In this protection state, the power module stops supplying power. When the input voltage restores to the normal range, the power module automatically resumes power supply.
	Input overcurrent protection	In this protection state, the power module stops supplying power and cannot automatically resume power supply when the input current restores to the normal range.
Output protection	Output overvoltage protection	In this protection state, the power module supplies power intermittently. When the output voltage restores to the normal range, the power module automatically resumes power supply.
	Output overcurrent protection	In this protection state, the power module supplies power intermittently. When the output current is limited within a range, the power module automatically resumes power supply.
	Output short-circuit protection	In this protection state, the power module supplies power intermittently. When the short circuit is removed, the power module automatically resumes power supply.
Overtemperature protection		When the temperature of the power module exceeds a specified threshold, the power module stops supplying power. When the temperature falls into the normal range, the power module automatically resumes power supply.
Heat dissipation		Natural heat dissipation
Hot swap		Supported

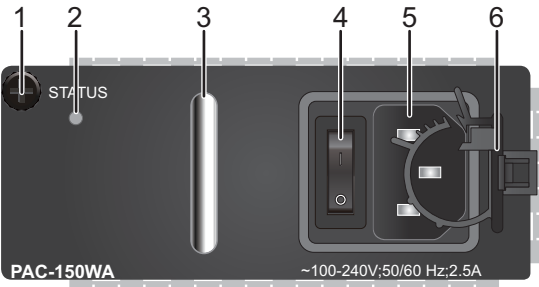
 NOTE

When a power module enters overtemperature protection state, take measures to lower the temperature of the power module. The power module can automatically resume power supply when the temperature falls within the normal range.

Panel

Figure 3-2 shows the panel of the PAC-150WA.

Figure 3-2 Panel of the PAC-150WA



1. Captive screw	2. Indicator	3. Handle	4. Power switch
5. AC power socket	6. AC terminal locking latch	-	-

Table 3-3 describes the indicator on the PAC-150WA panel.

Table 3-3 Description of the indicator on the PAC-150WA panel

Indicator	Color	Description
STATUS: power indicator	Green	● Off: The power input is abnormal (for example, no input, overvoltage, or undervoltage) or the power output is abnormal (for example, overvoltage, overcurrent, short-circuit, or overtemperature). ● Steady on: The power module is working normally.

Specifications

Table 3-4 lists the technical specifications of the PAC-150WA.

Table 3-4 Technical specifications of the PAC-150WA

Item	PAC-150WA
Dimensions (W x D x H)	100 mm x 205 mm x 40 mm
Weight	1 kg
Rated input voltage	100 V AC-240 V AC, 50/60 Hz
Maximum input voltage	90 V AC-290 V AC, 45 Hz-65 Hz
Maximum input current	2.5 A
Maximum output current	12.5 A
Rated output voltage	12 V

Item	PAC-150WA
Rated output power	150 W
Part Number	02130969

3.2 150 W AC Power Module (ES0W2PSA0150)

Version Mapping

Table 3-5 describes the mapping between switch models and the ES0W2PSA0150.

Table 3-5 Mapping between switch models and the ES0W2PSA0150

Switch Model	ES0W2PSA0150
CE5810-24T4S-EI CE5810-48T4S-EI CE5850-48T4S2Q-EI CE5850-48T4S2Q-HI CE6810-48S4Q-EI CE6810-48S4Q-LI CE6810-48S-LI CE6810-32T16S4Q-LI CE6810-24S2Q-LI CE6850-48S4Q-EI CE6850-48T4Q-EI CE6850-48S6Q-HI CE6850-48T6Q-HI CE6851-48S6Q-HI CE6850U-48S6Q-HI CE6850U-24S2Q-HI CE7850-32Q-EI	Not supported
CE5855-48T4S2Q-EI CE5855-24T4S2Q-EI	Supported NOTE This power module is supported in V100R005C10 and later versions.

 NOTE

The ES0W2PSA0150 power module can only be used in the CE5855EI and cannot be used in any other models.

Appearance

Figure 3-3 shows the appearance of the ES0W2PSA0150.

Figure 3-3 Appearance of the ES0W2PSA0150



Function

Table 3-6 describes the functions of the ES0W2PSA0150.

Table 3-6 Functions of the ES0W2PSA0150

Function		Description
Input protection	Input overvoltage protection and undervoltage protection	In either of the two protection states, the power module stops supplying power. When the input voltage restores to the normal range, the power module automatically resumes power supply.
	Input overcurrent protection	In this protection state, the power module stops supplying power and cannot automatically resume power supply when the input current restores to the normal range.
Output protection	Output overvoltage protection	In this protection state, the power module stops supplying power intermittently. When the output voltage restores to the normal range, the power module automatically resumes power supply.
	Output overcurrent protection	In this protection state, the power module supplies power intermittently. When the output current is limited within a range, the power module automatically resumes power supply.
	Output short-circuit protection	In this protection state, the power module supplies power intermittently. When the short circuit is removed, the power module automatically resumes power supply.

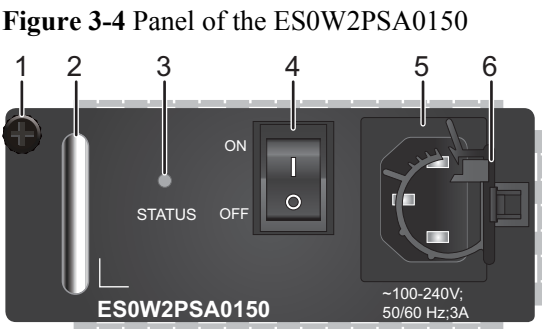
Function	Description
Overtemperature protection	When the temperature of the power module exceeds a specified threshold, the power module stops supplying power. When the temperature falls into the normal range, the power module automatically resumes power supply.
Heat dissipation	Natural heat dissipation
Hot swap	Supported

 **NOTE**

When a power module enters overtemperature protection state, take measures to lower the temperature of the power module. The power module can automatically resume power supply when the temperature falls within the normal range.

Panel

Figure 3-4 shows the panel of the ES0W2PSA0150.



1. Captive screw	2. Handle	3. Indicator	4. Power switch
5. AC power socket	6. AC terminal locking latch	-	-

Table 3-7 describes the indicators on the ES0W2PSA0150 panel.

Table 3-7 Description of the indicator on the ES0W2PSA0150 panel

Indicator	Color	Description
STATUS: running status indicator	Green	● Off: The power input is abnormal (for example, no input, overvoltage, or undervoltage) or the power output is abnormal (for example, undervoltage or overtemperature). ● Steady on: The power module is working normally. ● Blinking: The power output is abnormal (for example, output overvoltage, overcurrent, or short circuit).

Specifications

Table 3-8 lists the technical specifications of the ES0W2PSA0150.

Table 3-8 Technical specifications of the ES0W2PSA0150

Item	ES0W2PSA0150
Dimensions (W x D x H)	100 mm x 205 mm x 40 mm
Weight	0.8 kg
Rated input voltage range	100 V AC to 240 V AC, 50/60 Hz
Maximum input voltage	90 V AC to 264 V AC, 47 Hz to 63 Hz
Maximum input current	3 A
Maximum output current	12.5 A
Rated output voltage	12 V
Rated output power	150 W
Part Number	02310JFA

3.3 350 W AC Power Module

Version Mapping

350 W AC power modules include PAC-350WA-B (B: back-to-front airflow, air exhaust on front panel) and PAC-350WA-F (F: front-to-back airflow, air intake on front panel).

Table 3-9 describes the mapping between switch models and 350 W AC power modules.

Table 3-9 Mapping between switch models and 350 W AC power modules

Switch Model	PAC-350WA-B	PAC-350WA-F
CE5810-24T4S-EI CE5810-48T4S-EI CE5850-48T4S2Q-EI CE5850-48T4S2Q-HI CE6810-48S4Q-EI CE6810-48S4Q-LI CE6810-48S-LI CE6850-48S6Q-HI CE7850-32Q-EI CE6850U-48S6Q-HI CE6850-48T6Q-HI CE6850U-24S2Q-HI CE6851-48S6Q-HI CE5855-48T4S2Q-EI CE5855-24T4S2Q-EI CE6810-32T16S4Q-LI CE6810-24S2Q-LI	Not supported	Not supported
CE6850-48S4Q-EI	Supported NOTE This power module is supported in V100R001C00 version and later versions.	Supported NOTE This power module is supported in V100R001C00 version and later versions.
CE6850-48T4Q-EI	Supported NOTE This power module is supported in V100R001C00 and later versions. 600 W AC power modules are recommended for the CE6850-48T4Q-EI of V100R002C00 or a later version.	Supported NOTE This power module is supported in V100R001C00 and later versions. 600 W AC power modules are recommended for the CE6850-48T4Q-EI of V100R002C00 or a later version.

Appearance

Figure 3-5 shows the appearance of a PAC-350WA-B power module, and **Figure 3-6** shows the appearance of a PAC-350WA-F power module.

Figure 3-5 Appearance of a PAC-350WA-B power module



Figure 3-6 Appearance of a PAC-350WA-F power module



Function

PAC-350WA-B and PAC-350WA-F power modules use different airflow designs but have the same functions. **Table 3-10** describes the functions of a 350 W AC power module.

Table 3-10 Functions of a 350 W AC power module

Function		Description
Input protection	Input undervoltage protection	In this protection state, the power module stops supplying power. When the input voltage restores to the normal range, the power module automatically resumes power supply.
	Input overcurrent protection	In this protection state, the power module stops supplying power and cannot automatically resume power supply when the input current restores to the normal range.
Output protection	Output overvoltage protection	In this protection state, the power module supplies power intermittently. When the output voltage restores to the normal range, the power module automatically resumes power supply.
	Output overcurrent protection	In this protection state, the power module supplies power intermittently. When the output current is limited within a range, the power module automatically resumes power supply.
	Output short-circuit protection	In this protection state, the power module supplies power intermittently. When the short circuit is removed, the power module automatically resumes power supply.
Overtemperature protection		When the temperature of the power module exceeds a specified threshold, the power module stops supplying power. When the temperature falls into the normal range, the power module automatically resumes power supply.
Heat dissipation		● PAC-350WA-B: back-to-front airflow ● PAC-350WA-F: front-to-back airflow
Hot swap		Supported

 NOTE

When a power module enters overtemperature protection state, take measures to lower the temperature of the power module. The power module can automatically resume power supply when the temperature falls within the normal range.

Panel

Figure 3-7 and **Figure 3-8** show the panel of a 350 W AC power module.

Figure 3-7 Panel of a PAC-350WA-B AC power module

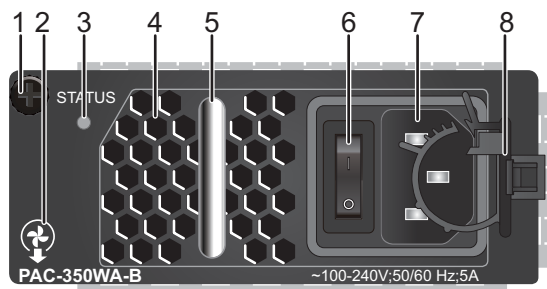
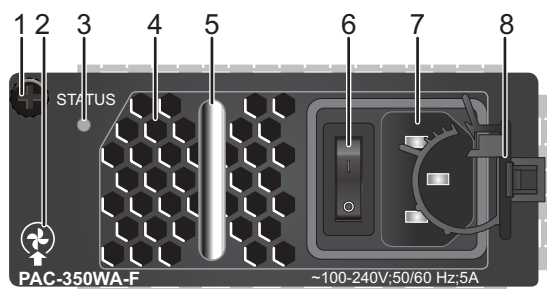


Figure 3-8 Panel of a PAC-350WA-F AC power module



1. Captive screw	2. Airflow flag  ● : back-to-front airflow  ● : front-to-back airflow	3. Indicator	4. Fan air vent
5. Handle	6. Power switch	7. AC power socket	8. AC terminal locking latch

Table 3-11 describes the indicator on the 350 W AC power module panel.

Table 3-11 Description of the indicator on the 350 W AC power module panel

Indicator	Color	Description
STATUS: power indicator	Green	● Off: The power input is abnormal (for example, no input, overvoltage, or undervoltage) or the power output is abnormal (for example, overvoltage, overcurrent, short-circuit, or overtemperature). ● Steady on: The power module is working normally.

Specifications

Table 3-12 lists the technical specifications of 350 W AC power modules.

Table 3-12 Technical specifications of 350 W AC power modules

Item	PAC-350WA-B	PAC-350WA-F
Dimensions (width x depth x height)	100 mm x 205 mm x 40 mm	
Weight	0.92 kg	
Rated input voltage	100 V AC-240 V AC, 50/60 Hz	
Maximum input voltage	90 V AC-290 V AC, 45 Hz-65 Hz	
Maximum input current	5 A	
Maximum output current	29.17 A	
Rated output voltage	12 V	
Rated output power	350 W	
Part Number	02130971	02130970

3.4 350 W DC Power Module

Version Mapping

350 W DC power modules include PDC-350WA-B (B: back-to-front airflow, air exhaust on front panel) and PDC-350WA-F (F: front-to-back airflow, air intake on front panel).

Table 3-13 describes the mapping between switch models and 350 W DC power modules.

Table 3-13 Mapping between switch models and 350 W DC power modules

Switch Model	PDC-350WA-B	PDC-350WA-F
CE5810-24T4S-EI CE5810-48T4S-EI CE5850-48T4S2Q-EI CE6850-48S4Q-EI	Supported NOTE This power module is supported in V100R002C00 version and later versions.	Supported NOTE This power module is supported in V100R002C00 version and later versions.

Switch Model	PDC-350WA-B	PDC-350WA-F
CE6850-48T4Q-EI CE6850-48S6Q-HI CE6850U-48S6Q-HI CE6850-48T6Q-HI CE6850U-24S2Q-HI CE7850-32Q-EI	Not supported	Not supported
CE5850-48T4S2Q-HI CE6810-48S4Q-EI	Supported NOTE This power module is supported in V100R003C00 version and later versions.	Supported NOTE This power module is supported in V100R003C00 version and later versions.
CE6810-48S4Q-LI CE6810-48S-LI	Supported NOTE This fan module is supported in V100R003C10 version and later versions.	Supported NOTE This fan module is supported in V100R003C10 version and later versions.
CE5855-48T4S2Q-EI CE5855-24T4S2Q-EI CE6810-32T16S4Q-LI CE6810-24S2Q-LI CE6851-48S6Q-HI	Supported NOTE This fan module is supported in V100R005C10 version and later versions.	Supported NOTE This fan module is supported in V100R005C10 version and later versions.

Appearance

Figure 3-9 shows the appearance of a PDC-350WA-B power module, and **Figure 3-10** shows the appearance of a PDC-350WA-F power module.

Figure 3-9 Appearance of a PDC-350WA-B power module



Figure 3-10 Appearance of a PDC-350WA-F power module



Function

PDC-350WA-B and PDC-350WA-F power modules use different airflow designs but have the same functions. [Table 3-14](#) describes the functions of a 350 W DC power module.

Table 3-14 Functions of a 350 W DC power module

Function		Description
Input protection	Input undervoltage protection	In this protection state, the power module stops supplying power. When the input voltage restores to the normal range, the power module automatically resumes power supply.
	Input overcurrent protection	In this protection state, the power module stops supplying power and cannot automatically resume power supply when the input current restores to the normal range.
Output protection	Output overvoltage protection	In this protection state, the power module supplies power intermittently. When the output voltage restores to the normal range, the power module automatically resumes power supply.
	Output overcurrent protection	In this protection state, the power module supplies power intermittently. When the output current is limited within a range, the power module automatically resumes power supply.
	Output short-circuit protection	In this protection state, the power module supplies power intermittently. When the short circuit is removed, the power module automatically resumes power supply.
Overtemperature protection		When the temperature of the power module exceeds a specified threshold, the power module stops supplying power. When the temperature falls into the normal range, the power module automatically resumes power supply.
Heat dissipation		● PDC-350WA-B: back-to-front airflow ● PDC-350WA-F: front-to-back airflow
Hot swap		Supported

 NOTE

When a power module enters overtemperature protection state, take measures to lower the temperature of the power module. The power module can automatically resume power supply when the temperature falls within the normal range.

Panel

[Figure 3-11](#) and [Figure 3-12](#) show the panel of a 350 W DC power module.

Figure 3-11 Panel of a PDC-350WA-B DC power module

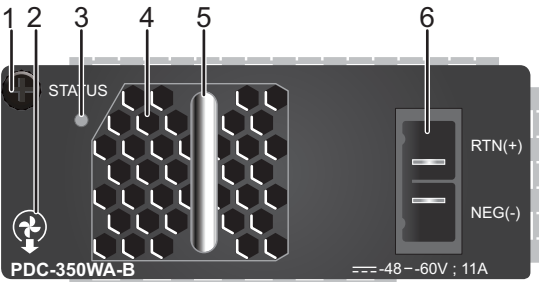
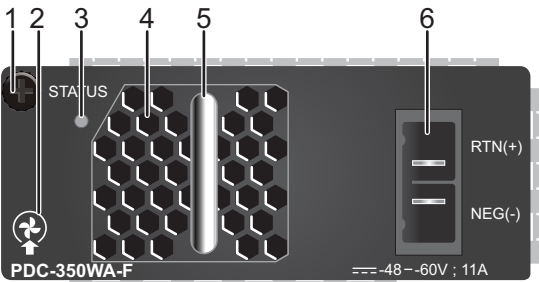


Figure 3-12 Panel of a PDC-350WA-F DC power module



1. Captive screw	2. Airflow flag  : back-to-front airflow  : front-to-back airflow	3. Indicator	4. Fan air vent
5. Handle	6. DC power socket	-	-

Table 3-15 describes the indicator on the 350 W DC power module panel.

Table 3-15 Description of the indicator on the 350 W DC power module panel

Indicator	Color	Description
STATUS: power indicator	Green	- Off: The input power is abnormal (no DC input power, DC input overvoltage, or DC input undervoltage) or the output power is abnormal (output overvoltage, overcurrent, short-circuit, or overtemperature). - Steady on: The power module is working properly.

Specifications

Table 3-16 lists the technical specifications of 350 W DC power modules.

Table 3-16 Technical specifications of 350 W DC power modules

Item	PDC-350WA-B	PDC-350WA-F
Dimensions (width x depth x height)	100 mm x 205 mm x 40 mm	
Weight	0.72 kg	
Rated input voltage	-48 V DC to -60 V DC	
Maximum input voltage	-38.4 V DC to -72 V DC	
Maximum input current	11 A	
Maximum output current	29.17 A	
Rated output voltage	12 V	
Rated output power	350 W	
Part Number	02310PQN	02310PQP

3.5 600 W AC Power Module

Version Mapping

600 W AC power modules include PAC-600WA-B (B: back-to-front airflow, air exhaust on front panel) and PAC-600WA-F (F: front-to-back airflow, air intake on front panel).

Table 3-17 describes the mapping between switch models and 600 W AC power modules.

Table 3-17 Mapping between switch models and 600 W AC power modules

Switch Model	PAC-600WA-B	PAC-600WA-F
CE5850-48T4S2Q-EI CE6850-48S4Q-EI CE6850-48S6Q-HI CE5810-24T4S-EI CE5810-48T4S-EI CE5850-48T4S2Q-HI CE6850U-48S6Q-HI CE6850-48T6Q-HI CE6850U-24S2Q-HI CE5855-48T4S2Q-EI CE5855-24T4S2Q-EI	Not supported	Not supported
CE6850-48T4Q-EI	Supported NOTE This power module is supported in V100R002C00 version and later versions.	Supported NOTE This power module is supported in V100R002C00 version and later versions.
CE6810-48S4Q-EI CE7850-32Q-EI	Supported NOTE This power module is supported in V100R003C00 version and later versions.	Supported NOTE This power module is supported in V100R003C00 version and later versions.
CE6810-48S4Q-LI CE6810-48S-LI	Supported NOTE This fan module is supported in V100R003C10 version and later versions.	Supported NOTE This fan module is supported in V100R003C10 version and later versions.
CE6810-32T16S4Q-LI CE6810-24S2Q-LI CE6851-48S6Q-HI	Supported NOTE This fan module is supported in V100R005C10 version and later versions.	Supported NOTE This fan module is supported in V100R005C10 version and later versions.

Appearance

Figure 3-13 shows the appearance of a PAC-600WA-B power module, and **Figure 3-14** shows the appearance of a PAC-600WA-F power module.

Figure 3-13 Appearance of a PAC-600WA-B power module



Figure 3-14 Appearance of a PAC-600WA-F power module



Function

PAC-600WA-B and PAC-600WA-F power modules use different airflow designs but have the same functions. [Table 3-18](#) describes the functions of a 600 W AC power module.

Table 3-18 Functions of a 600 W AC power module

Function		Description
Input protection	Input undervoltage protection	In this protection state, the power module stops supplying power. When the input voltage restores to the normal range, the power module automatically resumes power supply.
	Input overcurrent protection	In this protection state, the power module stops supplying power and cannot automatically resume power supply when the input current restores to the normal range.
Output protection	Output overvoltage protection	In this protection state, the power module supplies power intermittently. When the output voltage restores to the normal range, the power module automatically resumes power supply.
	Output overcurrent protection	In this protection state, the power module supplies power intermittently. When the output current is limited within a range, the power module automatically resumes power supply.
	Output short-circuit protection	In this protection state, the power module supplies power intermittently. When the short circuit is removed, the power module automatically resumes power supply.
Overtemperature protection		When the temperature of the power module exceeds a specified threshold, the power module stops supplying power. When the temperature falls into the normal range, the power module automatically resumes power supply.
Heat dissipation		● PAC-600WA-B: back-to-front airflow ● PAC-600WA-F: front-to-back airflow
Hot swap		Supported

 NOTE

When a power module enters overtemperature protection state, take measures to lower the temperature of the power module. The power module can automatically resume power supply when the temperature falls within the normal range.

Panel

Figure 3-15 and **Figure 3-16** show the panel of a 600 W AC power module.

Figure 3-15 Panel of a PAC-600WA-B AC power module

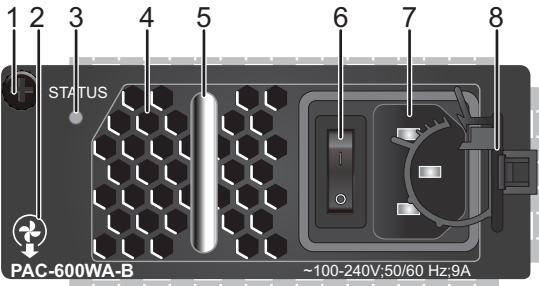
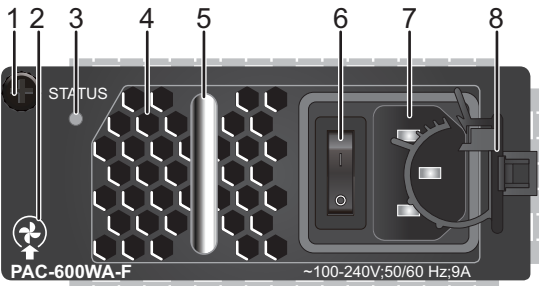


Figure 3-16 Panel of a PAC-600WA-F AC power module



1. Captive screw	2. Ventilation channel flag : back-to-front airflow : front-to-back airflow	3. Indicator	4. Fan air vent
5. Handle	6. Power switch	7. AC power socket	8. AC terminal locking latch

Table 3-19 describes the indicator on the 600 W AC power module panel.

Table 3-19 Description of the indicator on the 600 W AC power module panel

Indicator	Color	Description
STATUS: power indicator	Green	● Off: The input power is abnormal (no AC input power, AC input overvoltage, or AC input undervoltage) or the output power is abnormal (output overvoltage, overcurrent, short-circuit, or overtemperature). ● Steady on: The power module is working properly.

Specifications

Table 3-20 lists the technical specifications of 600 W AC power modules.

Table 3-20 Technical specifications of 600 W AC power modules

Item	PAC-600WA-B	PAC-600WA-F
Dimensions (width x depth x height)	100 mm x 205 mm x 40 mm	
Weight	1 kg	
Rated input voltage	100 V AC-240 V AC, 50/60 Hz	
Maximum input voltage	90 V AC-290 V AC, 45 Hz-65 Hz	
Maximum input current	9 A	
Maximum output current	50 A	
Rated output voltage	12 V	
Rated output power	600 W	
Part Number	02310PMH	02310PMJ

3.6 600 W AC&240 V DC Power Module

Version Mapping

600 W AC&240 V DC power modules can receive AC inputs or 240 V high-voltage inputs. They are classified into two types depending on the airflow designs: PAC-600WB-B (back-to-front airflow, air exhaust on front panel) and PAC-600WB-F (F: front-to-back airflow, air intake on front panel).

Table 3-21 describes the mapping between switch models and 600 W AC&240 V DC power modules.

Table 3-21 Mapping between switch models and 600 W AC&240 V DC power modules

Switch Model	PAC-600WB-B	PAC-600WB-F
CE7850-32Q-EI CE6810-48S4Q-EI CE6810-48S4Q-LI CE6810-48S-LI CE6850-48S4Q-EI CE6850-48T4Q-EI CE5810-24T4S-EI CE5810-48T4S-EI CE5850-48T4S2Q-EI CE5850-48T4S2Q-HI CE6810-32T16S4Q-LI CE6810-24S2Q-LI CE6851-48S6Q-HI CE5855-48T4S2Q-EI CE5855-24T4S2Q-EI	Not supported	Not supported
CE6850-48S6Q-HI	Supported NOTE This power module is supported in V100R005C00 and later versions.	Supported NOTE This power module is supported in V100R005C00 and later versions.
CE6850U-48S6Q-HI CE6850-48T6Q-HI CE6850U-24S2Q-HI	Supported NOTE This power module is supported in V100R005C10 and later versions.	Supported NOTE This power module is supported in V100R005C10 and later versions.

Appearance

Figure 3-17 shows the appearance of a PAC-600WB-B power module, and **Figure 3-18** shows the appearance of a PAC-600WB-F power module.

Figure 3-17 Appearance of a PAC-600WB-B power module



Figure 3-18 Appearance of a PAC-600WB-F power module



Function

PAC-600WB-B and PAC-600WB-F power modules use different airflow designs but have the same functions. Table 3-22 describes the functions of them.

Table 3-22 Functions of a 600 W AC&240 V DC power module

Function		Description
Input protection	Input overvoltage protection and undervoltage protection	In either of the two protection states, the power module stops supplying power. When the input voltage restores to the normal range, the power module automatically resumes power supply.

Function		Description
	Input overcurrent protection	In this protection state, the power module stops supplying power and cannot automatically resume power supply when the input current restores to the normal range.
Output protection	Output overvoltage protection	In this protection state, the power module stops supplying power intermittently. When the system recovers from output overvoltage, the power module automatically resumes power supply.
	Output overcurrent protection	In this protection state, the power module supplies power intermittently. When the output current is limited within a range, the power module automatically resumes power supply.
	Output short-circuit protection	In this protection state, the power module supplies power intermittently. When the short circuit is removed, the power module automatically resumes power supply.
Overtemperature protection		When the temperature of the power module exceeds a specified threshold, the power module stops supplying power. When the temperature falls into the normal range, the power module automatically resumes power supply.
Heat dissipation		● PAC-600WB-B: back-to-front airflow ● PAC-600WB-F: front-to-back airflow
Hot swap		Supported

NOTE

When a power module enters overtemperature protection state, take measures to lower the ambient temperature. The power module can automatically resume power supply when the temperature falls within the normal range.

Panel

Figure 3-19 and **Figure 3-20** show the panels of 600 W AC&240 V DC power modules.

Figure 3-19 Panel of a PAC-600WB-B power module

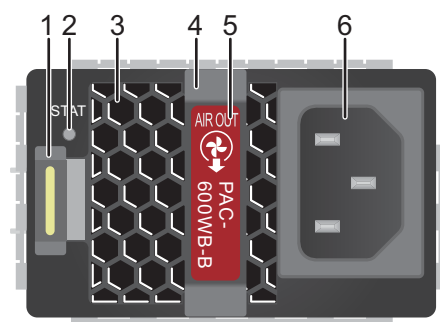
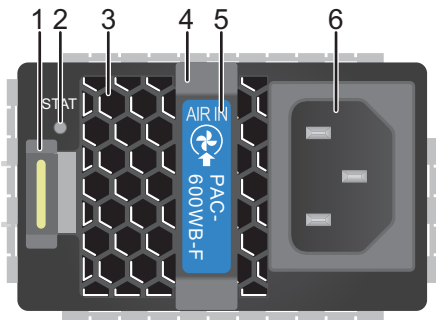


Figure 3-20 Panel of a PAC-600WB-F power module



1. Lock	2. Indicator	3. Fan air vent	4. Handle NOTE Each 600 W AC&240 V DC power module is delivered with a velcro strap on the handle. This velcro strap is used to bundle the power cable to the handle.
5. Airflow flag AIR OUT ● : back-to-front airflow AIR IN ● : front-to-back airflow	6. Power socket	-	-

Table 3-23 describes the indicator on a 600 W AC&240 V DC power module panel.

Table 3-23 Description of the indicator on a 600 W AC&240 V DC power module panel

Indicator	Color	Description
STATUS: running status indicator	-	Off: The power module receives no power input.
	Green	● Steady on: The power module is working normally. ● Blinking: The power module is in loading or standby state, or the power cable has been connected but the power module is not installed in the switch.
	Red	Steady on: ● Fans of the power module fail. ● The power module is in overtemperature protection state. ● The power input is abnormal (input undervoltage or input overvoltage). ● The power output is abnormal (output overcurrent, output short-circuit, or output overvoltage).

Specifications

Table 3-24 describes the technical specifications of 600 W AC&240 V DC power modules.

Table 3-24 Technical specifications of 600 W AC&240 V DC power modules

Item	PAC-600WB-B	PAC-600WB-F
Dimensions (W x D x H)	66.0 mm x 340.0 mm x 39.6 mm	
Weight	1.5 kg	
Rated AC input voltage range	100 V AC to 240 V AC, 50/60 Hz	
Maximum AC input voltage range	90 V AC to 290 V AC, 47 Hz to 63 Hz	
Rated voltage range of 240 V high-voltage DC power input	240 V DC	
Maximum voltage range of 240 V high-voltage DC power input	188 V DC to 290 V DC	
Maximum input current	● 9 A (90 V AC) ● 4 A (240 V DC)	
Maximum output current	50 A	
Rated output voltage	12 V	
Rated output power	600 W	

Item	PAC-600WB-B	PAC-600WB-F
Part Number	02310YQN	02310YQP

3.7 600 W High-Voltage DC Power Module

Version Mapping

600 W high-voltage DC power modules are classified into two types depending on the airflow designs: PHD-600WA-B (back-to-front airflow, air exhaust on front panel) and PHD-600WA-F (F: front-to-back airflow, air intake on front panel).

Table 3-25 describes the mapping between switch models and 600 W high-voltage DC power modules.

Table 3-25 Mapping between switch models and 600 W high-voltage DC power modules

Switch Model	PHD-600WA-B	PHD-600WA-F
CE7850-32Q-EI CE6810-48S4Q-EI CE6810-48S4Q-LI CE6810-48S-LI CE6850-48S4Q-EI CE6850-48T4Q-EI CE5810-24T4S-EI CE5810-48T4S-EI CE5850-48T4S2Q-EI CE5850-48T4S2Q-HI CE6810-32T16S4Q-LI CE6810-24S2Q-LI CE6851-48S6Q-HI CE5855-48T4S2Q-EI CE5855-24T4S2Q-EI	Not supported	Not supported
CE6850-48S6Q-HI	Supported NOTE This power module is supported in V100R005C00 and later versions.	Supported NOTE This power module is supported in V100R005C00 and later versions.

Switch Model	PHD-600WA-B	PHD-600WA-F
CE6850U-48S6Q-HI CE6850-48T6Q-HI CE6850U-24S2Q-HI	Supported NOTE This power module is supported in V100R005C10 and later versions.	Supported NOTE This power module is supported in V100R005C10 and later versions.

Appearance

Figure 3-21 shows the appearance of a PHD-600WA-B power module, and Figure 3-22 shows the appearance of a PHD-600WA-F power module.

Figure 3-21 Appearance of a PHD-600WA-B power module



Figure 3-22 Appearance of a PHD-600WA-F power module



Function

PHD-600WA-B and PHD-600WA-F power modules use different airflow designs but have the same functions. [Table 3-26](#) describes the functions of them.

Table 3-26 Functions of 600 W high-voltage DC power modules

Function		Description
Input protection	Input overvoltage protection and undervoltage protection	In either of the two protection states, the power module stops supplying power. When the input voltage restores to the normal range, the power module automatically resumes power supply.
	Input overcurrent protection	In this protection state, the power module stops supplying power and cannot automatically resume power supply when the input current restores to the normal range.
Output protection	Output overvoltage protection	In this protection state, the power module stops supplying power intermittently. When the system recovers from output overvoltage, the power module automatically resumes power supply.
	Output overcurrent protection	In this protection state, the power module supplies power intermittently. When the output current is limited within a range, the power module automatically resumes power supply.
	Output short-circuit protection	In this protection state, the power module supplies power intermittently. When the short circuit is removed, the power module automatically resumes power supply.
Overtemperature protection		When the temperature of the power module exceeds a specified threshold, the power module stops supplying power. When the temperature falls into the normal range, the power module automatically resumes power supply.
Heat dissipation		● PHD-600WA-B: back-to-front airflow ● PHD-600WA-F: front-to-back airflow
Hot swap		Supported

NOTE

When a power module enters overtemperature protection state, take measures to lower the ambient temperature. The power module can automatically resume power supply when the temperature falls within the normal range.

Panel

[Figure 3-23](#) and [Figure 3-24](#) show the panels of 600 W high-voltage DC power modules.

Figure 3-23 Panel of a PHD-600WA-B power module

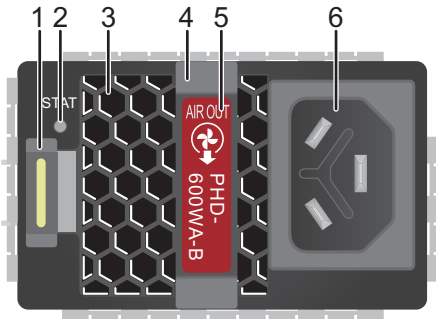
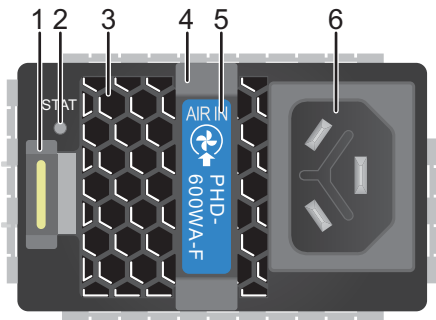


Figure 3-24 Panel of a PHD-600WA-F power module



1. Lock	2. Indicator	3. Fan air vent	4. Handle NOTE Each 600 W high-voltage DC power module is delivered with a velcro strap on the handle. This velcro strap is used to bundle the power cable to the handle.
5. Airflow flag AIR OUT  • : back-to-front airflow AIR IN  • : front-to-back airflow	6. Power socket	-	-

Table 3-27 describes the indicator on a 600 W high-voltage DC power module panel.

Table 3-27 Description of the indicator on the 600 W high-voltage DC power module panel

Indicator	Color	Description
STATUS: running status indicator	-	Off: The power module receives no power input.
	Green	● Steady on: The power module is working normally. ● Blinking: The power module is in loading or standby state, or the power cable has been connected but the power module is not installed in the switch.
	Red	Steady on: ● Fans of the power module fail. ● The power module is in overtemperature protection state. ● The power input is abnormal (input undervoltage or input overvoltage). ● The power output is abnormal (output overcurrent, output short-circuit, or output overvoltage).

Specifications

Table 3-28 describes the technical specifications of 600 W high-voltage DC power modules.

Table 3-28 Technical specifications of 600 W high-voltage DC power modules

Item	PHD-600WA-B	PHD-600WA-F
Dimensions (W x D x H)	66.0 mm x 340.0 mm x 39.6 mm	
Weight	1.5 kg	
Rated voltage range of 380 V high-voltage DC power input	240 V DC to 380 V DC	
Maximum voltage range of 380 V high-voltage DC power input	188 V DC to 400 V DC	
Maximum input current	4 A	
Maximum output current	50 A	
Rated output voltage	12 V	
Rated output power	600 W	
Part Number	02310YQQ	02310YQR

4 Fan Module

About This Chapter



NOTICE

- A chassis must use fan modules with the same airflow direction.
 - A chassis must use fan modules of the same series.
 - A chassis can work properly only when two fan modules are running. If one of fan modules is removed from the chassis, reinstall it into the chassis within 3 minutes.
-

[4.1 FAN-40EA Series Fan Modules](#)

[4.2 FAN-40SB Series Fan Modules](#)

[4.3 FAN-40HA Series Fan Modules](#)

[4.4 FAN-040A Series Fan Modules](#)

[4.5 FAN-060A Series Fan Modules](#)

4.1 FAN-40EA Series Fan Modules

Version Mapping

The FAN-40EA series fan modules are classified into two types depending on the airflow designs: FAN-40EA-B (B: back-to-front airflow, air exhaust on front panel) and FAN-40EA-F (F: front-to-back airflow, air intake on front panel).

Table 4-1 describes the mapping between switch models and FAN-40EA fan modules.

Table 4-1 Mapping between switch models and FAN-40EA fan modules

Switch Model	FAN-40EA-B	FAN-40EA-F
CE5810-24T4S-EI CE5810-48T4S-EI CE6850-48S6Q-HI CE7850-32Q-EI CE6850U-48S6Q-HI CE6850-48T6Q-HI CE6850U-24S2Q-HI CE5855-48T4S2Q-EI CE5855-24T4S2Q-EI	Not supported	Not supported
CE5850-48T4S2Q-EI CE6850-48S4Q-EI CE6850-48T4Q-EI	Supported NOTE This fan module is supported in V100R001C00 version and later versions.	Supported NOTE This fan module is supported in V100R001C00 version and later versions.
CE5850-48T4S2Q-HI CE6810-48S4Q-EI	Supported NOTE This fan module is supported in V100R003C00 version and later versions.	Supported NOTE This fan module is supported in V100R003C00 version and later versions.
CE6810-48S4Q-LI CE6810-48S-LI	Supported NOTE This fan module is supported in V100R003C10 version and later versions.	Supported NOTE This fan module is supported in V100R003C10 version and later versions.
CE6810-32T16S4Q-LI CE6810-24S2Q-LI CE6851-48S6Q-HI	Supported NOTE This fan module is supported in V100R005C10 version and later versions.	Supported NOTE This fan module is supported in V100R005C10 version and later versions.

Appearance

Figure 4-1 shows the appearance of a FAN-40EA-B fan module, and **Figure 4-2** shows the appearance of a FAN-40EA-F fan module.

NOTE

A FAN-40EA fan module consists of two fans.

Figure 4-1 Appearance of a FAN-40EA-B fan module



Figure 4-2 Appearance of a FAN-40EA-F fan module



Function

A FAN-40EA fan module consists of a fan tray, two fans, and a fan control unit. FAN-40EA-B and FAN-40EA-F fan modules use different airflow designs but have the same functions. **Table 4-2** describes the functions of them.

Table 4-2 Functions of a FAN-40EA fan module

Function	Description
Automatic fan speed adjustment	After the fan module communicates normally with the switch, the switch controls the speed of fans based on temperature of the chassis.
Hot swap	Supported
Heat dissipation	- FAN-40EA-B: back-to-front airflow - FAN-40EA-F: front-to-back airflow

Panel

Figure 4-3 and Figure 4-4 show the panels of the FAN-40EA series fan modules.

Figure 4-3 Panel of a FAN-40EA-B fan module

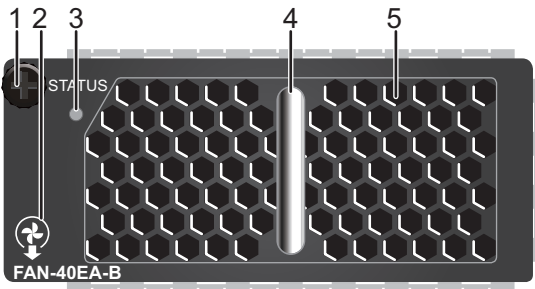
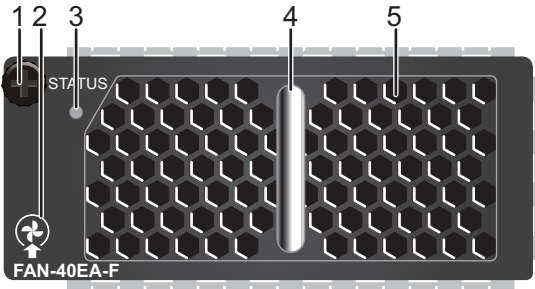


Figure 4-4 Panel of a FAN-40EA-F fan module



1. Captive screw	2. Airflow flag : back-to-front airflow : front-to-back airflow	3. Indicator
------------------	--	--------------

4. Handle	5. Fan air vent	-
-----------	-----------------	---

Table 4-3 describes the indicator on the panels of the FAN-40EA fan modules.

Table 4-3 Indicator on panels of the FAN-40EA fan modules

Indicator	Color	Description
STATUS: fan indicator	-	Off: The fan module is not running.
	Green	● Slow blinking: The fan module is working properly and communicating normally with the system. ● Fast blinking: The fan module is working properly but has not established communication with the system.
	Red	● Steady on: The fan module has a hardware fault and must be replaced. ● Blinking: An alarm has been generated, and you need to handle it accordingly. Common causes of this alarm include errors of dual in-line package (DIP) switches, short circuits, fan blades blocked, and fault of the fan module.

Specifications

Table 4-4 describes the technical specifications of the FAN-40EA series fan modules.

Table 4-4 Technical specifications of the FAN-40EA fan modules

Item	FAN-40EA-B	FAN-40EA-F
Dimensions (W x D x H)	94.5 mm x 183.1 mm x 39.8 mm	
Number of fans	2	
Weight	0.325 kg	
Maximum power consumption	12.71 W	
Maximum fan speed	18500±10% revolutions per minute (RPM)	
Maximum wind rate	46 cubic feet per minute (CFM)	
Part Number	02355338	02355421

4.2 FAN-40SB Series Fan Modules

Version Mapping

The FAN-40S series fan modules are classified into two types depending on the airflow designs: FAN-40SB-B (B: back-to-front airflow, air exhaust on front panel) and FAN-40SB-F (F: front-to-back airflow, air intake on front panel).

Table 4-5 lists the mapping between switch models and FAN-40SB fan modules.

Table 4-5 Mapping between switch models and FAN-40SB fan modules

Switch Model	FAN-40SB-B	FAN-40SB-F
CE5810-24T4S-EI CE5810-48T4S-EI	Supported NOTE This fan module is supported in V100R002C00 version and later versions.	Supported NOTE This fan module is supported in V100R002C00 version and later versions.
CE5850-48T4S2Q-EI CE5850-48T4S2Q-HI CE6850-48S4Q-EI CE6850-48T4Q-EI CE6810-48S4Q-EI CE6850-48S6Q-HI CE6851-48S6Q-HI CE6810-48S4Q-LI CE6810-48S-LI CE7850-32Q-EI CE6850U-48S6Q-HI CE6850-48T6Q-HI CE6850U-24S2Q-HI CE6810-32T16S4Q-LI CE6810-24S2Q-LI CE5855-48T4S2Q-EI CE5855-24T4S2Q-EI	Not supported	Not supported

Appearance

Figure 4-5 shows the appearance of a FAN-40SB-B fan module, and **Figure 4-6** shows the appearance of a FAN-40SB-F fan module.

 NOTE

A FAN-40SB fan module has only one fan.

Figure 4-5 Appearance of a FAN-40SB-B fan module

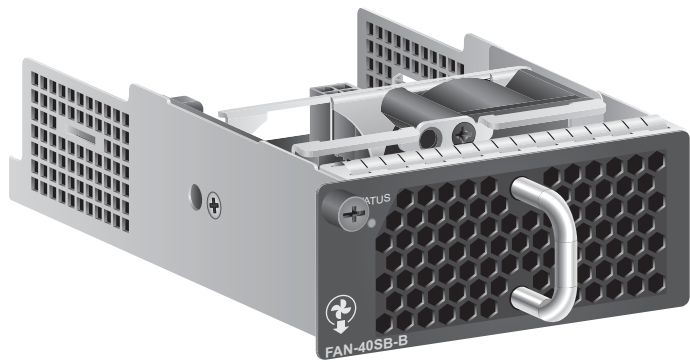
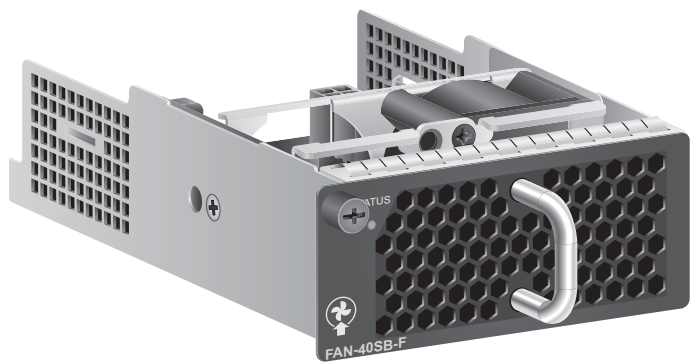


Figure 4-6 Appearance of a FAN-40SB-F fan module



Function

A FAN-40SB fan module consists of a fan tray, a fan, and a fan control unit. FAN-40SB-B and FAN-40SB-F fan modules use different airflow designs but have the same functions. Table 4-6 describes the functions of them.

Table 4-6 Functions of a FAN-40SB fan module

Function	Description
Automatic fan speed adjustment	After the fan module communicates normally with the switch, the switch controls the speed of fans based on temperature of the chassis.
Hot swap	Supported
Heat dissipation	- FAN-40SB-B: back-to-front airflow - FAN-40SB-F: front-to-back airflow

Panel

Figure 4-7 and Figure 4-8 show the panels of the FAN-40SB fan modules.

Figure 4-7 Panel of a FAN-40SB-B fan module

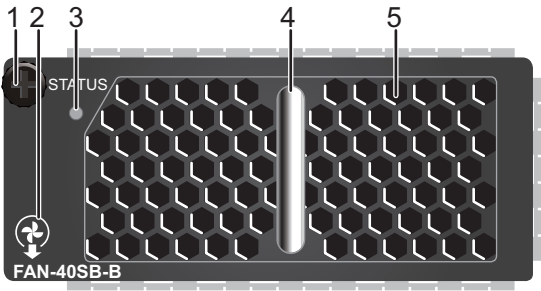
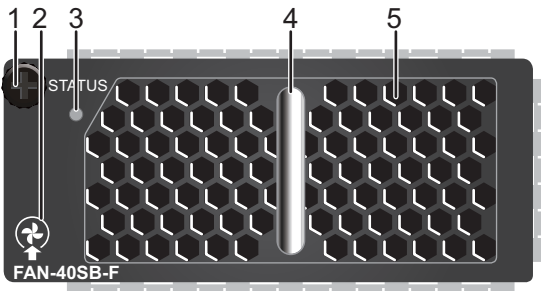


Figure 4-8 Panel of a FAN-40SB-F fan module



1. Captive screw	2. Airflow flag  • : back-to-front airflow  • : front-to-back airflow	3. Indicator
4. Handle	5. Fan air vent	-

Table 4-7 describes the indicator on the panels of the FAN-40SB fan modules.

Table 4-7 Indicator on panels of the FAN-40SB fan modules

Indicator	Color	Description
STATUS: fan indicator	-	Off: The fan module is not running.
	Green	● Slow blinking: The fan module is working properly and communicating normally with the system. ● Fast blinking: The fan module is working properly but has not established communication with the system.
	Red	● Steady on: The fan module has a hardware fault and must be replaced. ● Blinking: An alarm has been generated, and you need to handle it accordingly. Common causes of this alarm include errors of dual in-line package (DIP) switches, short circuits, fan blades blocked, and fault of the fan module.

Specifications

Table 4-8 describes the technical specifications of the FAN-40SB fan modules.

Table 4-8 Technical specifications of the FAN-40SB fan modules

Item	FAN-40SB-B	FAN-40SB-F
Dimensions (W x D x H)	94.5 mm x 183.1 mm x 39.8 mm	
Number of fans	1	
Weight	0.3 kg	
Maximum power consumption	4.3 W	
Maximum fan speed	16000±10% revolutions per minute (RPM) NOTE RPM: revolutions per minute.	
Maximum wind rate	20 cubic feet per minute (CFM)	
Part Number	02356152	02356153

4.3 FAN-40HA Series Fan Modules

Version Mapping

The FAN-40HA series fan modules are classified into two types depending on the airflow designs: FAN-40HA-B (B: back-to-front airflow, air exhaust on front panel) and FAN-40HA-F (F: front-to-back airflow, air intake on front panel).

Table 4-9 lists the mapping between switch models and FAN-40HA series fan modules.

Table 4-9 Mapping between switch models and FAN-40HA series fan modules

Switch Model	FAN-40HA-B	FAN-40HA-F
CE5810-24T4S-EI CE5810-48T4S-EI CE5850-48T4S2Q-EI CE5850-48T4S2Q-HI CE6850-48S4Q-EI CE6850-48T4Q-EI CE6850-48S6Q-HI CE6810-48S4Q-EI CE6810-48S4Q-LI CE6810-48S-LI CE6850U-48S6Q-HI CE6850-48T6Q-HI CE6850U-24S2Q-HI CE6851-48S6Q-HI CE6810-32T16S4Q-LI CE6810-24S2Q-LI CE5855-48T4S2Q-EI CE5855-24T4S2Q-EI	Not supported	Not supported
CE7850-32Q-EI	Supported NOTE This fan module is supported in V100R003C00 version and later versions.	Supported NOTE This fan module is supported in V100R003C00 version and later versions.

Appearance

Figure 4-9 shows the appearance of a FAN-40HA-B fan module, and **Figure 4-10** shows the appearance of a FAN-40HA-F fan module.

NOTE

A FAN-40HA fan module consists of two counter-rotating fans.

Figure 4-9 Appearance of a FAN-40HA-B fan module



Figure 4-10 Appearance of a FAN-40HA-F fan module



Function

A FAN-40HA fan module consists of a fan frame, two counter-rotating fans, and a fan control unit. FAN-40HA-B and FAN-40HA-F fan modules use different airflow designs but have the same functions. Table 4-10 describes the functions of them.

Table 4-10 Functions of a FAN-40HA fan module

Function	Description
Automatic fan speed adjustment	After the fan module communicates normally with the switch, the switch controls the speed of fans based on temperature of the chassis.
Hot swap	Supported
Heat dissipation	● FAN-40HA-B: back-to-front airflow ● FAN-40HA-F: front-to-back airflow

Panel

Figure 4-11 and Figure 4-12 show the panels of the FAN-40HA fan modules.

Figure 4-11 Panel of a FAN-40HA-B fan module

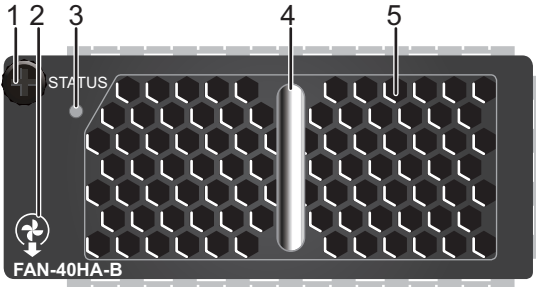
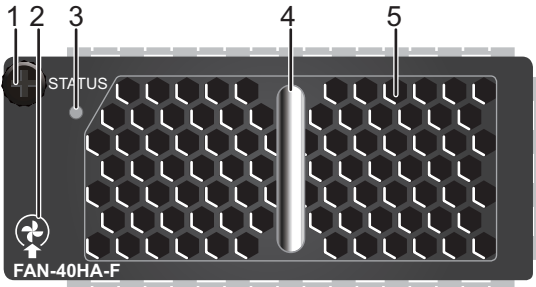


Figure 4-12 Panel of a FAN-40HA-F fan module



1. Captive screw	2. Airflow flag  • : back-to-front airflow  • : front-to-back airflow	3. Indicator
4. Handle	5. Fan air vent	-

Table 4-11 describes the indicator on the panels of the FAN-40HA fan modules.

Table 4-11 Indicator on panels of the FAN-40HA fan modules

Indicator	Color	Description
STATUS: fan indicator	-	Off: The fan module is not running.
	Green	● Slow blinking: The fan module is working properly and communicating normally with the system. ● Fast blinking: The fan module is working properly but has not established communication with the system.
	Red	● Steady on: The fan module has a hardware fault and must be replaced. ● Blinking: An alarm has been generated, and you need to handle it accordingly. Common causes of this alarm include errors of dual in-line package (DIP) switches, short circuits, fan blades blocked, and fault of the fan module.

Specifications

Table 4-12 describes the technical specifications of the FAN-40HA fan modules.

Table 4-12 Technical specifications of the FAN-40HA fan modules

Item	FAN-40HA-B	FAN-40HA-F
Dimensions (W x D x H)	94.5 mm x 183.1 mm x 39.8 mm	
Number of fans	2	
Weight	0.415 kg	
Maximum power consumption	40 W	
Maximum fan speed	19000±10% revolutions per minute (RPM)	
Maximum wind rate	64 cubic feet per minute (CFM)	
Part Number	02359097	02359096

4.4 FAN-040A Series Fan Modules

Version Mapping

The FAN-040A series fan modules are classified into two types depending on the airflow designs: FAN-040A-B (B: back-to-front airflow, air exhaust on front panel) and FAN-040A-F (F: front-to-back airflow, air intake on front panel).

Table 4-13 describes the mapping between switch models and FAN-040A fan modules.

Table 4-13 Mapping between switch models and FAN-040A fan modules

Switch Model	FAN-040A-B	FAN-040A-F
CE5810-24T4S-EI CE5810-48T4S-EI CE5850-48T4S2Q-EI CE5850-48T4S2Q-HI CE6810-48S4Q-LI CE6810-48S-LI CE6810-32T16S4Q-LI CE6810-24S2Q-LI CE6810-48S4Q-EI CE6850-48S4Q-EI CE6850-48T4Q-EI CE6850-48T6Q-HI CE6850-48S6Q-HI CE6851-48S6Q-HI CE6850U-48S6Q-HI CE6850U-24S2Q-HI CE7850-32Q-EI	Not supported	Not supported
CE5855-48T4S2Q-EI CE5855-24T4S2Q-EI	Supported NOTE This fan module is supported in V100R005C10 version and later versions.	Supported NOTE This fan module is supported in V100R005C10 version and later versions.

Appearance

Figure 4-13 shows the appearance of a FAN-040A-B fan module, and **Figure 4-14** shows the appearance of a FAN-040A-F fan module.

NOTE

A FAN-040A fan module consists of two fans.

Figure 4-13 Appearance of a FAN-040A-B fan module



Figure 4-14 Appearance of a FAN-040A-F fan module



Function

A FAN-040A fan module consists of a fan tray, two fans, and a fan control unit. FAN-040A-B and FAN-040A-F fan modules use different airflow designs but have the same functions. Table 4-14 describes the functions of them.

Table 4-14 Functions of a FAN-040A fan module

Function	Description
Automatic fan speed adjustment	After the fan module communicates normally with the switch, the switch controls the speed of fans based on temperature of the chassis.
Hot swap	Supported
Heat dissipation	● FAN-040A-B: back-to-front airflow ● FAN-040A-F: front-to-back airflow

Panel

Figure 4-15 and Figure 4-16 show the panels of the FAN-040A series fan modules.

Figure 4-15 Panel of a FAN-040A-B fan module

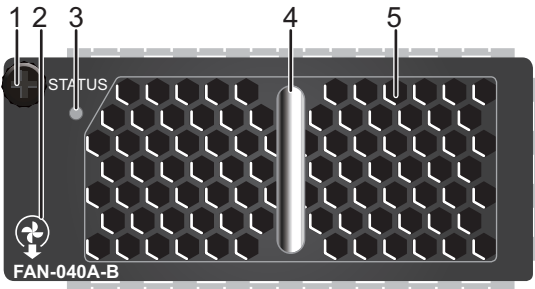
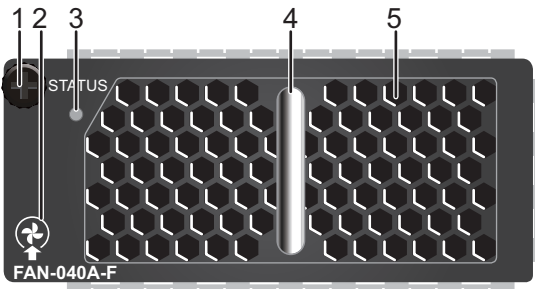


Figure 4-16 Panel of a FAN-040A-F fan module



1. Captive screw	2. Airflow flag : back-to-front airflow : front-to-back airflow	3. Indicator
4. Handle	5. Fan air vent	-

Table 4-15 describes the indicator on the panels of the FAN-040A fan modules.

Table 4-15 Indicator on panels of the FAN-040A fan modules

Indicator	Color	Description
STATUS: fan indicator	-	Off: The fan module is not running.
	Green	● Slow blinking: The fan module is working properly and communicating normally with the system. ● Fast blinking: The fan module is working properly but has not established communication with the system.
	Red	● Steady on: The fan module has a hardware fault and must be replaced. ● Blinking: An alarm has been generated, and you need to handle it accordingly. Common causes of this alarm include errors of dual in-line package (DIP) switches, short circuits, fan blades blocked, and fault of the fan module.

Specifications

Table 4-16 describes the technical specifications of the FAN-040A series fan modules.

Table 4-16 Technical specifications of the FAN-040A fan modules

Item	FAN-040A-B	FAN-040A-F
Dimensions (W x D x H)	94.5 mm x 183.1 mm x 39.8 mm	
Number of fans	2	
Weight	0.259 kg	
Maximum power consumption	12 W	
Maximum fan speed	16000±10% revolutions per minute (RPM)	
Maximum wind rate	40 cubic feet per minute (CFM)	
Part Number	02350JFA	02350JEY

4.5 FAN-060A Series Fan Modules

Version Mapping

The FAN-060A series fan modules are classified into two types depending on the airflow designs: FAN-060A-B (B: back-to-front airflow, air exhaust on front panel) and FAN-060A-F (F: front-to-back airflow, air intake on front panel).

Table 4-17 describes the mapping between device models and FAN-060A series fan modules.

Table 4-17 Mapping between switch models and FAN-060A series fan modules

Switch Model	FAN-060A-B	FAN-060A-F
CE7850-32Q-EI CE6810-48S4Q-EI CE6810-48S4Q-LI CE6810-48S-LI CE6850-48S4Q-EI CE6850-48T4Q-EI CE5810-24T4S-EI CE5810-48T4S-EI CE5850-48T4S2Q-EI CE5850-48T4S2Q-HI CE6810-32T16S4Q-LI CE6810-24S2Q-LI CE6851-48S6Q-HI CE5855-48T4S2Q-EI CE5855-24T4S2Q-EI	Not supported	Not supported
CE6850-48S6Q-HI	Supported NOTE The fan module is supported in V100R005C00 and later versions.	Supported NOTE The fan module is supported in V100R005C00 and later versions.
CE6850U-48S6Q-HI CE6850-48T6Q-HI CE6850U-24S2Q-HI	Supported NOTE The fan module is supported in V100R005C10 and later versions.	Supported NOTE The fan module is supported in V100R005C10 and later versions.

Appearance

Figure 4-17 shows the appearance of a FAN-060A-B fan module, and **Figure 4-18** shows the appearance of a FAN-060A-F fan module.

 NOTE

Each FAN-060A fan module has two counter-rotating fans.

Figure 4-17 Appearance of a FAN-060A-B fan module



Figure 4-18 Appearance of a FAN-060A-F fan module



Function

A FAN-060A fan module consists of a fan tray, two counter-rotating fans, and a fan monitoring unit. FAN-060A-B and FAN-060A-F fan modules use different airflow designs but have the same functions. Table 4-18 describes the functions of them.

Table 4-18 Functions of a FAN-060A fan module

Function	Description
Automatic fan speed adjustment	After the fan module communicates normally with the switch, the switch controls the speed of fans based on temperature of the chassis.
Hot swap	Supported
Heat dissipation	- FAN-060A-B: back-to-front airflow - FAN-060A-F: front-to-back airflow

Panel

Figure 4-19 and Figure 4-20 show the panels of FAN-060A series fan modules.

Figure 4-19 Panel of a FAN-060A-B fan module

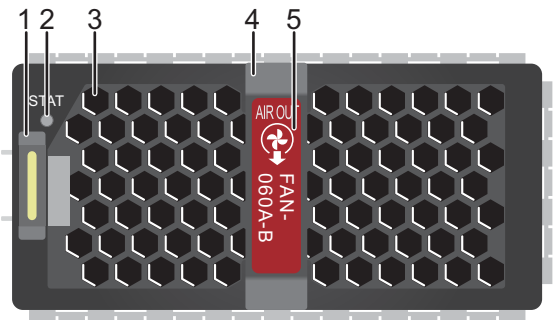
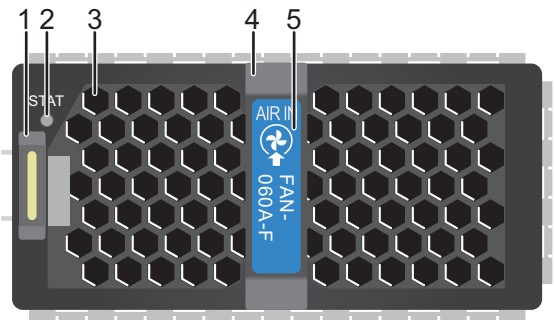


Figure 4-20 Panel of a FAN-060A-F fan module



1. Lock	2. Indicator	3. Fan air vent
4. Handle	5. Airflow flag AIR OUT AIR IN • : back-to-front airflow • : front-to-back airflow	-

Table 4-19 describes the indicator on a FAN-060A fan module.

Table 4-19 Indicator on a FAN-060A fan module

Indicator	Color	Description
STATUS: running status indicator	-	Off: The fan module is not running.
	Green	● Slow blinking: The fan module is working properly and communicating normally with the system. ● Fast blinking: The fan module is working properly but has not established communication with the system.
	Red	● Steady on: The fan module has a hardware fault and must be replaced. ● Blinking: An alarm has been generated, and you need to handle it accordingly. Common causes of this alarm include short circuits, fan blades blocked, and fault of the fan module.

Specifications

Table 4-20 describes the technical specifications of the FAN-060A series fan modules.

Table 4-20 Technical specifications of FAN-060A series fan modules

Item	FAN-060A-B	FAN-060A-F
Dimensions (W x D x H)	86.0 mm x 198.3 mm x 40.0 mm	
Number of fans	2	
Weight	0.478 kg	
Maximum power consumption	40 W	
Maximum fan speed	19000±10% revolutions per minute (RPM)	
Maximum wind rate	64 cubic feet per minute (CFM)	
Part Number	02359310	02359308

5 Cables

About This Chapter

- 5.1 AC Power Cable
- 5.2 DC Power Cable
- 5.3 380 V High-Voltage DC Power Cable
- 5.4 Ground Cable
- 5.5 Console Cable
- 5.6 Network Cable
- 5.7 Optical Fiber
- 5.8 AOC Cable
- 5.9 Copper Cable

5.1 AC Power Cable

Types of AC Power Cables

 NOTE

The AC power cables delivered must comply with the standards used in the delivery destination. This section uses the AC power cables complying with China's national standards as an example.

AC power cables are classified into two types: C13 straight female to PI straight male AC power cable and C13 straight female to C14 straight male AC power cable.

Appearance and Structure

Figure 5-1 shows the appearance of a C13 straight female to PI straight male AC power cable.

Figure 5-1 Appearance of a C13 straight female to PI straight male AC power cable



Figure 5-2 shows the appearance of a C13 straight female to C14 straight male AC power cable.

Figure 5-2 Appearance of a C13 straight female to C14 straight male AC power cable



Connection

An AC power cable is connected to the AC power module of the device:

- The C13 straight female connector is connected to the power socket of a power module.
- The PI straight male or C14 straight male connector is connected to a power source.

When a **600 W AC&240 V DC power module** uses 240 V high-voltage power input, it must be connected to the power supply device using a C13 straight female to C14 straight male AC power cable. This power cable is connected as follows:

- The C13 straight female connector is connected to the power socket of the **600 W AC&240 V DC power module**.
- The C14 straight male connector is connected to a high-voltage DC PDU. If a high-voltage DC power distribution box is used, make OT or cord end terminals for the cable. Cut the C14 straight male connector off and crimp OT or cord end terminals on the bare wires. Connect the blue wire to a positive terminal on the DC power distribution box, the brown wire to a negative terminal, and the yellow-green wire to a protection ground.

5.2 DC Power Cable

Appearance and Structure

Figure 5-3 shows the appearance of a DC power cable.

Figure 5-3 Appearance of a DC power cable

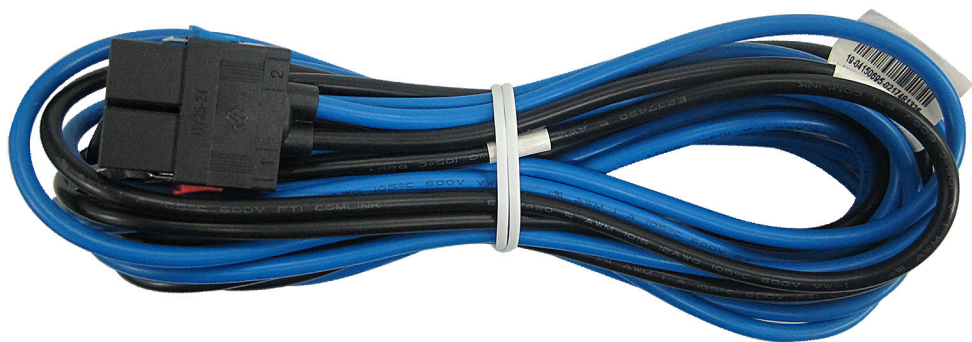
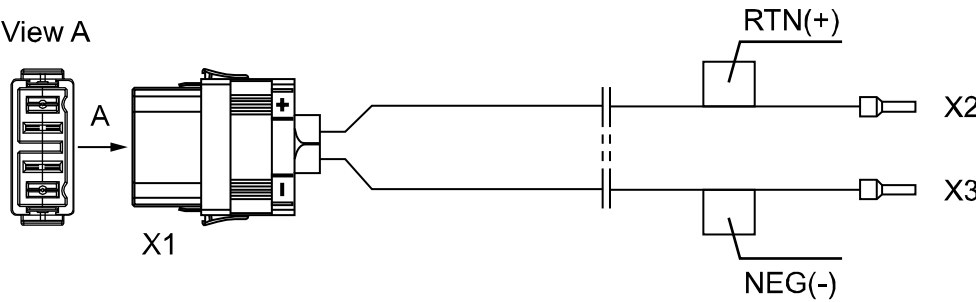


Figure 5-4 shows the structure of a DC power cable.

Figure 5-4 Structure of a DC power cable



Pin Assignments

Table 5-1 lists the pin assignments of a DC power cable.

Table 5-1 Pin assignments of a DC power cable

X1	X2	X3	Length	Conductor Cross-Sectional Area
2 female	Cord end terminal 4^2 grey	Cord end terminal 4^2 grey	3 m	3.332 mm ² (12AWG)

Connection

A DC power cable connects to the DC power module of the device:

- X1 connector connects to the input port on the DC power module.

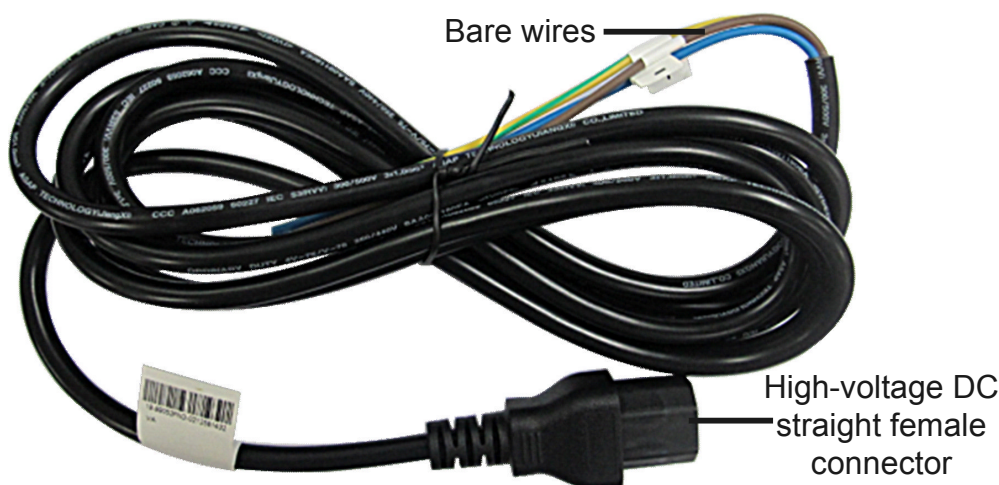
- X2/X3 cord end terminal connects to an external power module.

5.3 380 V High-Voltage DC Power Cable

Appearance and Structure

Figure 5-5 shows the appearance of a 380 V high-voltage DC power cable.

Figure 5-5 Appearance of a 380 V high-voltage DC power cable (high-voltage DC straight female connector to bare wires)



Connection

A 380 V high-voltage DC power cable has a high-voltage DC straight female connector at one end and bare wires at the other end, and is used to connect a **600 W high-voltage DC power module** to a power supply device:

- The high-voltage DC straight female connector is connected to the power socket of the **600 W high-voltage DC power module**.
- The bare wires are connected to a 380 V high-voltage DC power distribution frame or power distribution box. Crimp OT or cord end terminals on the bare wires, and then connect the blue wire to a negative terminal, the brown wire to a positive terminal, and the yellow-green wire to a protection ground.

5.4 Ground Cable

Appearance and Structure

NOTE

Different types of ground cables have similar appearance, except for the cross-sectional area, size of the cable lugs, and cable length. The following figure is for reference.

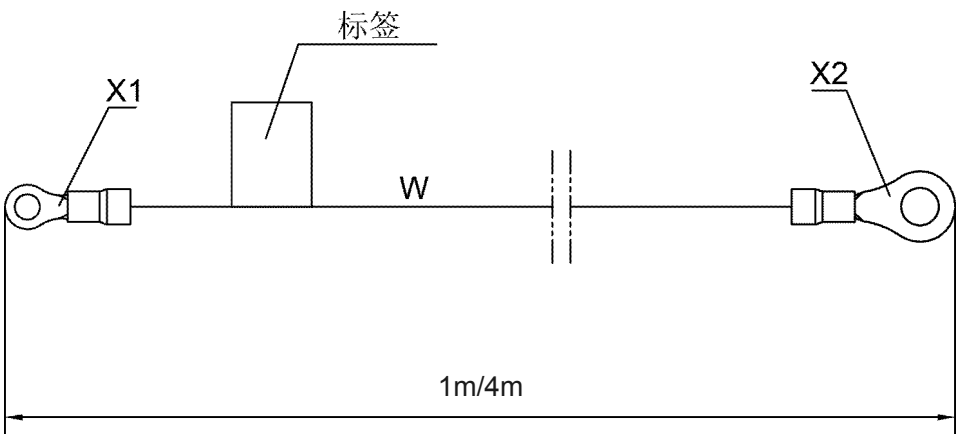
Figure 5-6 shows the appearance of a ground cable.

Figure 5-6 Appearance of a ground cable



Figure 5-7 shows the structure of a ground cable.

Figure 5-7 Structure of a ground cable



Pin Assignments

Table 5-2 lists the pin assignments of a ground cable.

Table 5-2 Pin assignments of a ground cable

X1	X2	Wire Color	Conductor Cross-Sectional Area	Length
OT6-4	OT6-6	Green-yellow	4 mm ²	1 m or 4 m NOTE The default ground cable delivered with a switch is 1 m long. You can also order a 4 m ground cable for a switch based on your installation environment.

Connection

A ground cable grounds a device to protect it from lightning and electromagnetic interference. A ground cable is connected to a chassis in the following way:

- The OT6-4 naked crimping connector connects to the ground point on the chassis.
- The OT6-6 naked crimping connector connects to the ground point on the cabinet.

5.5 Console Cable

Appearance and Structure

Figure 5-8 shows the appearance of a console cable.

Figure 5-8 Appearance of a console cable

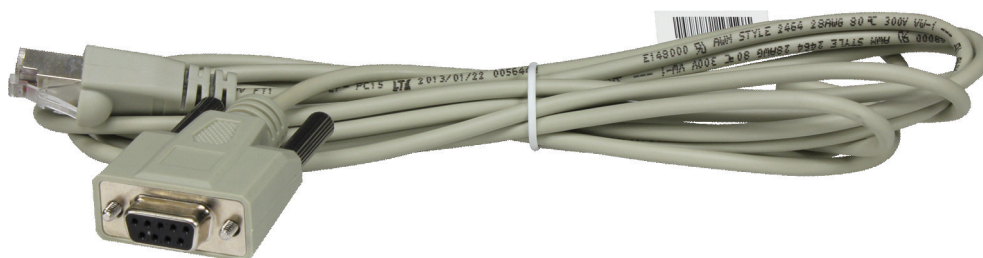
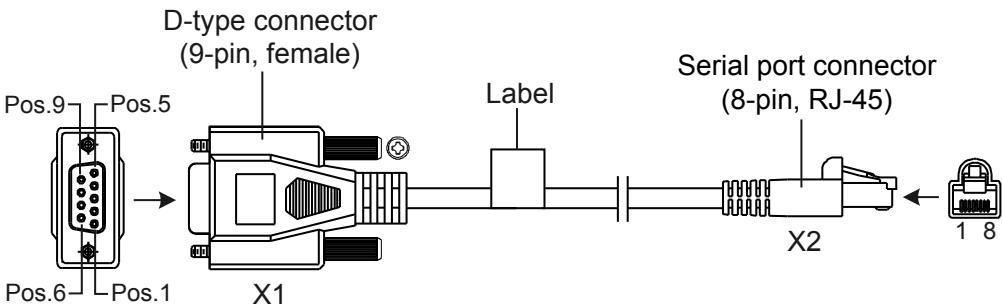


Figure 5-9 shows the structure of a console cable.

Figure 5-9 Structure of a console cable



Pin Assignments

Table 5-3 lists the pin assignments of console cable connectors.

Table 5-3 Pin assignments of console cable connectors

Connector	X1 (DB-9)	X2 (RJ45)
Pin assignment	2	3
	3	6
	5	5

Connection

A console cable connects the console port of a device to the serial port of an operation terminal, enabling users to commission or locally maintain the device.

A console cable connects a device and a console as follows:

- The 8-pin RJ45 connector is connected to the console port of the device.
- The DB-9 female connector is connected to a maintenance terminal, such as a computer.

5.6 Network Cable

Types of Network Cables

Network cables are classified into straight-through cables and crossover cables.

- Straight-through cable: The pin assignments of RJ45 connectors at both ends are shown in Table 5-4.
- Crossover cable: The pin assignments of RJ45 connectors at both ends are shown in Table 5-5.

Appearance and Structure

NOTE

- Straight-through cables and crossover cables are standard unshielded network cables that use RJ45 connectors.
- A straight-through cable and a crossover cable have the same appearance.

Figure 5-10 and **Figure 5-11** show the appearance of a network cable.

Figure 5-10 Appearance of a network cable (1)

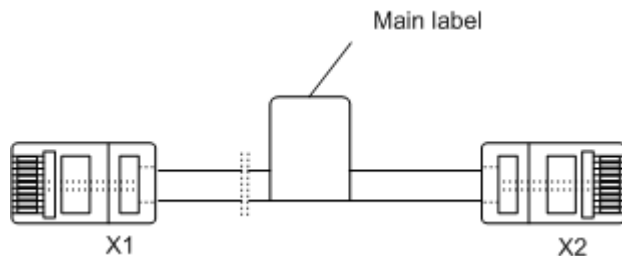


Figure 5-11 Appearance of a network cable (2)



Figure 5-12 shows the structure of a network cable.

Figure 5-12 Structure of a network cable



Pin Assignments

Table 5-4 lists the pin assignments of a straight-through cable.

Table 5-4 Pin assignments of a straight-through cable

X1 Pin	Wire Color	X2 Pin
1	White and orange	1
2	Orange	2
3	White and green	3
4	Blue	4
5	White and blue	5
6	Green	6
7	White and brown	7
8	Brown	8

Table 5-5 lists the pin assignments of a crossover cable.

Table 5-5 Pin assignments of a crossover cable

X1 Pin	Wire Color	X2 Pin
1	White and orange	3
2	Orange	6
3	White and green	1
4	Blue	4

X1 Pin	Wire Color	X2 Pin
5	White and blue	5
6	Green	2
7	White and brown	7
8	Brown	8

 NOTE

To achieve the best electrical transmission performance, ensure that the wires connected to pins 1 and 2 and to pins 3 and 6 are twisted pairs.

Connection

Network cables connect network devices to each other to enable the devices to communicate or to allow local maintenance and remote access.

- A straight-through cable connects a terminal (such as a PC or switch) to a network device.
- A crossover cable connects two terminals (such as PCs and switches).

Supported Cabling Types for 10GBASE-T

Table 5-6 describes the supported cabling types for a 10GBASE-T Ethernet electrical port.

Table 5-6 Supported cabling types for 10GBASE-T

Item	Category 7 STP	Category 6A STP	Category 6A UTP	Category 6 STP	Category 6 UTP
Cable Description	Category 7 shielded twisted pair (STP)	Category 6A shielded twisted pair	Category 6A unshielded twisted pair (UTP)	Category 6 shielded twisted pair	Category 6 unshielded twisted pair
Type	Class F	Class Ea	Class Ea	Class E	Class E
Maximum transmission distance	100 m	100 m	100 m	100 m	37 m - 55 m NOTE Category 6 unshielded cables may be limited by alien crosstalk beyond 37-meter channels.

Item	Category 7 STP	Category 6A STP	Category 6A UTP	Category 6 STP	Category 6 UTP
Cabling system bandwidth	600 MHz NOTE The cabling system exceeds the requirements for IEEE 10GBASE-T performance.	500 MHz NOTE The cabling system exceeds the requirements for IEEE 10GBASE-T performance.		250 MHz NOTE Category 6 is specified only to 250 MHz. The cabling system must be certified to 500 MHz according to TSB-155 to help ensure 10GBASE-T compliance.	

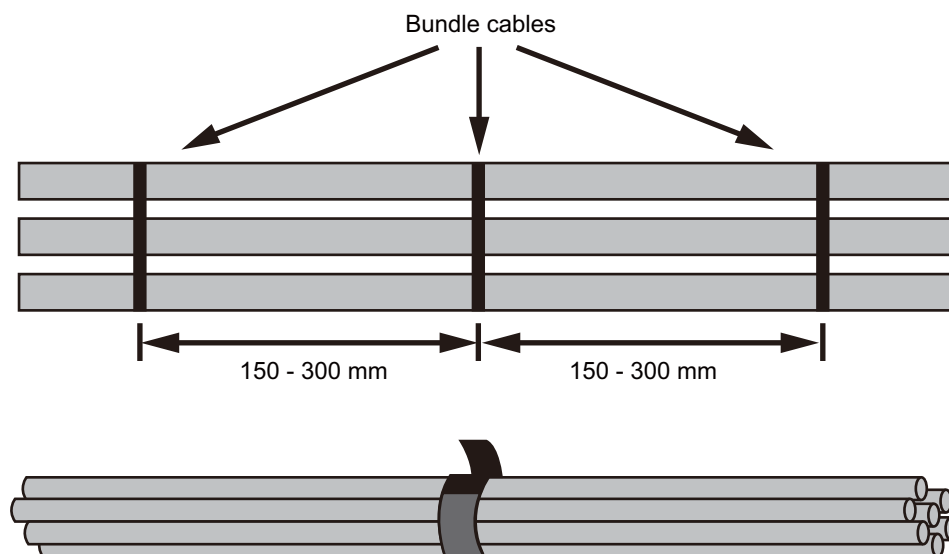
 NOTE

- In a new built equipment room, Category 6A shielded twisted pairs or Category 7 twisted pairs are recommended. These cables can avoid alien crosstalk while having no special installation requirements. In addition, they can be used with other types of cables.
- If Category 6 shielded or unshielded twisted pairs are used in an equipment room and the cabling systems can meet requirements of TSB-155, follow these rules route these cables:
 - Separate these cables with other types of cables. If they must be routed in the same cable trough with other types of cables, separated them from other cables using a metal plate.
 - Separate cables as much as possible at the outlet and keep the cables parallel with each other. Most alien crosstalk appears within 20 m away from the outlet. To reduce alien crosstalk, do not bundle cables in the first 5 m to 20 m.
 - If cables need to be bundled, bundle cables with cable ties placed every 150 mm to 300 mm. See [Table 5-7](#). Bundle cables loosely, as shown in [Figure 5-13](#).
 - You are advised to add no more than 12 cables in a bundle. A bundle cannot have more than 24 cables.
 - Replace Category 6 connectors with Category 6A connectors.

Table 5-7 Intervals between cable ties

Diameter of a Network Cable Bundle (mm)	Interval Between Cable Ties (mm)
< 10	150
10-30	200
> 30	300

Figure 5-13 Method to bundle cables



5.7 Optical Fiber

Types of Optical Fibers

Optical fibers are classified into two categories:

- Optical fibers for 10GE optical modules:
 - LC-LC fiber: Single-fiber push-on (LC) connectors at both ends
 - Single-mode fiber: yellow
 - Multi-mode fiber: orange
- Optical fibers for 40GE optical modules:
 - LC-LC single-mode fiber: Single-fiber push-on (LC) connectors at both ends
 - MPO-MPO fiber: multi-fiber push-on (MPO) connectors at both ends
 - MPO-4*DLC fiber: MPO connector at one end and four pairs of double LC (DLC) connectors at the other end
 - MPO-8*FC fiber: MPO connector at one end and eight FC connectors at the other end

Appearance and Structure

Figure 5-14 shows the appearance of an LC single-mode fiber.

Figure 5-14 Appearance of an LC single-mode fiber



Figure 5-15 shows the appearance of an LC multi-mode fiber.

Figure 5-15 Appearance of an LC multi-mode fiber

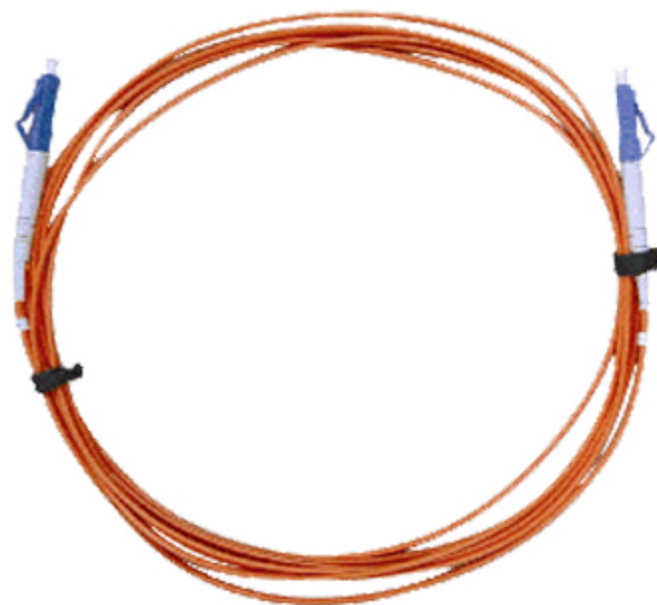


Figure 5-16 shows the appearance of an MPO-MPO fiber.

Figure 5-16 Appearance of an MPO-MPO fiber



Figure 5-17 shows the appearance of an MPO-4*DLC fiber.

Figure 5-17 Appearance of an MPO-4*DLC fiber



Figure 5-18 shows the appearance of an MPO-8*FC fiber.

Figure 5-18 Appearance of an MPO-8*FC fiber



Figure 5-19 shows the structure of an MPO-MPO fiber.

Figure 5-19 Structure of an MPO-MPO fiber

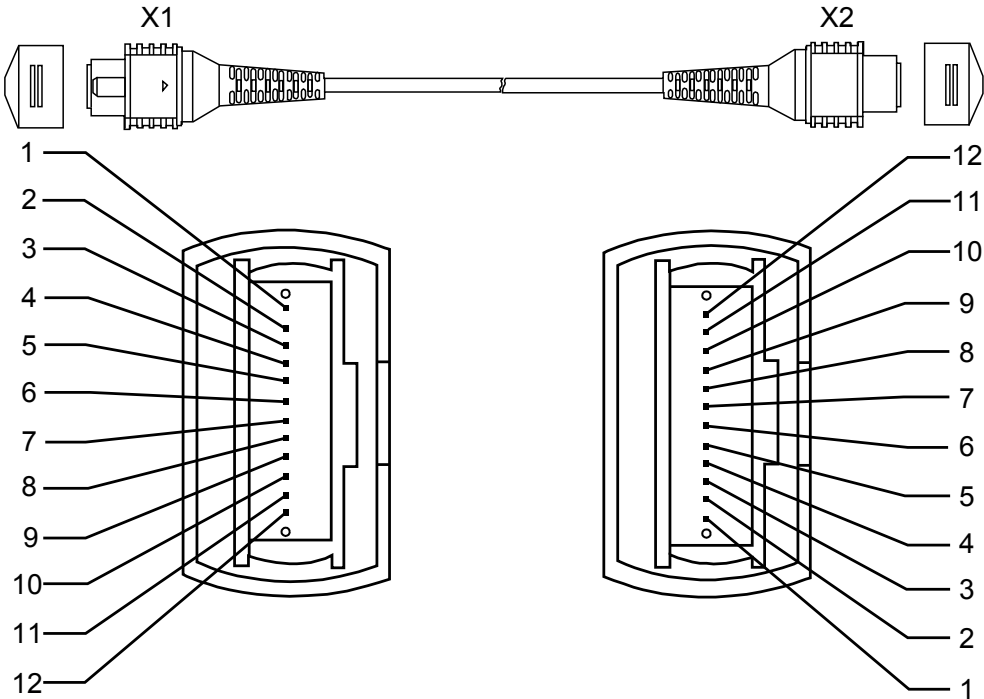


Figure 5-20 shows the structure of an MPO-4*DLC fiber.

Figure 5-20 Structure of an MPO-4*DLC fiber

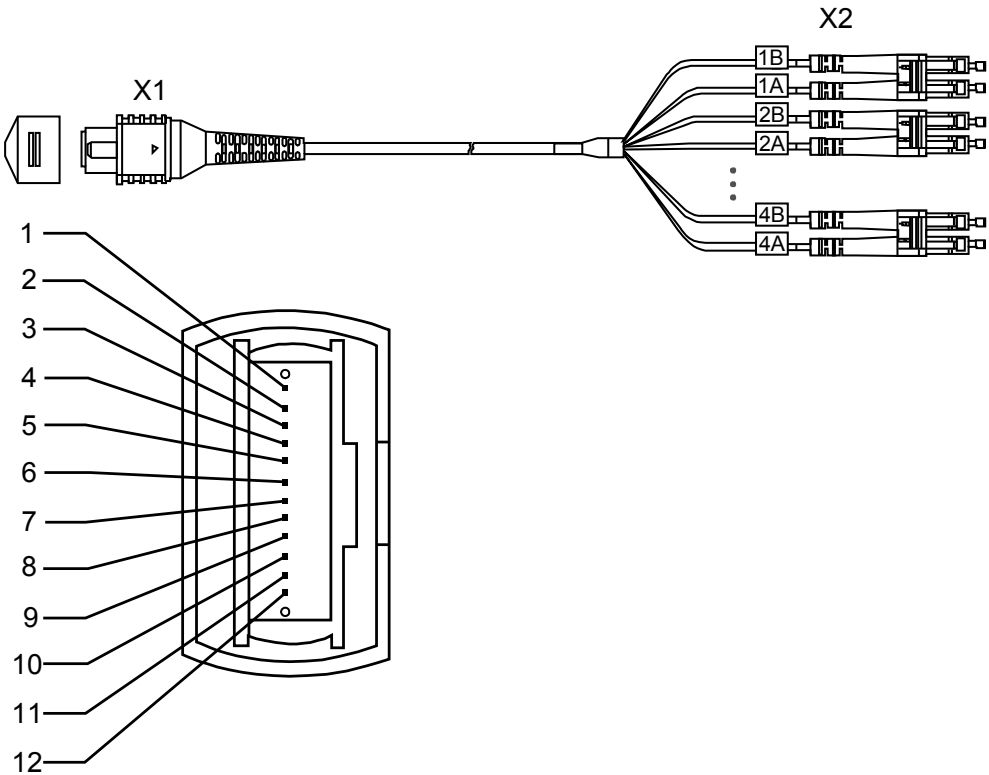
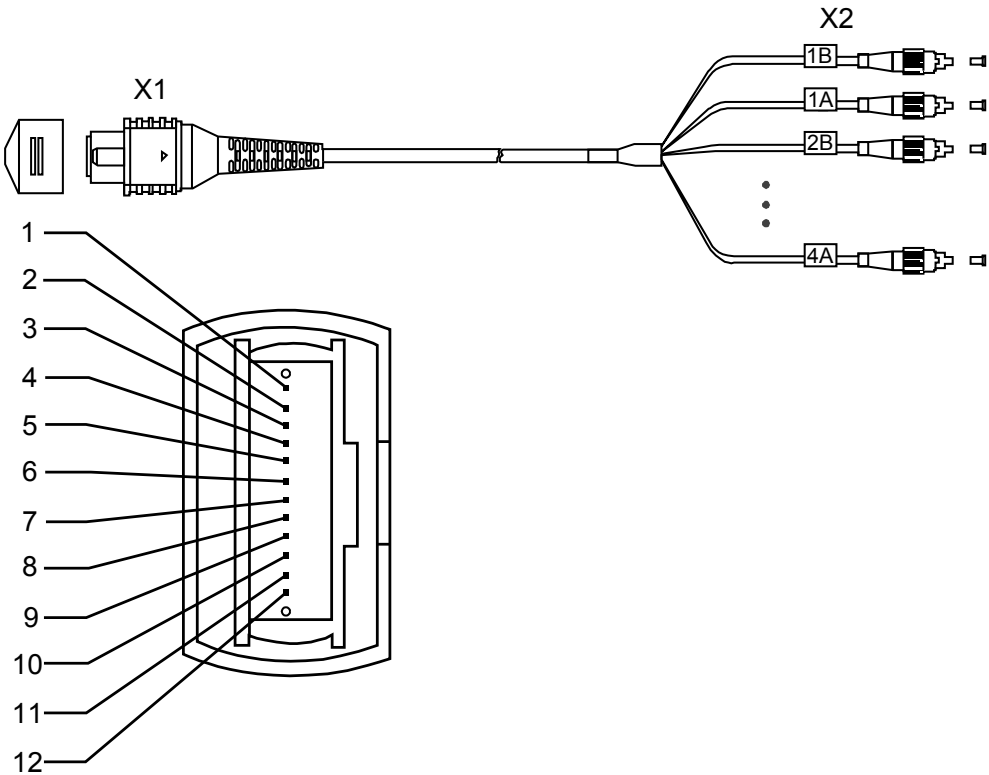


Figure 5-21 shows the structure of an MPO-8*FC fiber.

Figure 5-21 Structure of an MPO-8*FC fiber



Pin Assignments

Table 5-8 lists the pin assignments of an MPO-MPO fiber.

Table 5-8 Pin assignments of an MPO-MPO fiber

X1 Pin	X2 Pin
1	1
2	2
3	3
4	4
9	9
10	10
11	11
12	12

MPO-4*DLC and MPO-8*FC fibers have the same pin assignments, as shown in Table 5-9.

Table 5-9 Pin assignments of MPO-4*DLC and MPO-8*FC fibers

X1 Pin	X2 Pin
1	1B
2	2B
3	3B
4	4B
9	4A
10	3A
11	2A
12	1A

Connection

Table 5-10 describes usage scenarios of optical fibers and fiber connections in these scenarios.

 NOTE

Table 5-10 only describes connections of two ends of each fiber. In most cases, two devices are not directly connected by one fiber, and there are optical distribution frames (ODFs) between them.

Table 5-10 Fiber usage scenarios and connections

Fiber	Connection
LC-LC fiber	● Scenario 1: used for connection between two CE7800&6800&5800 series switches for stacking or service aggregation. ● Scenario 2: used for connection between 10GE or 40GE ports of CE7800&6800&5800 series switches and CE12800. Each end connects to a 10GE port using an SFP+ optical module. Each end connects to a 40GE port using an QSFP+ optical module.
MPO-MPO fiber	● Scenario 1: used for connection between two CE7800&6800&5800 series switches for stacking or service aggregation. ● Scenario 2: used for connection between 40GE ports of CE7800&6800&5800 series switches and CE12800. Each end connects to a 40GE port using a QSFP+ optical module.

Fiber	Connection
MPO-4*DLC fiber	When a 40GE port is configured as four 10GE ports, one end connects of this fiber connects to a 40GE port using a QSFP+ optical module, and the other end connects to four 10GE ports using SFP+ optical modules. Used for connection between a switch and an ODF. The MPO connector connects to the switch using a QSFP+ optical module, and the 4*DLC connectors connect to the ODF.
MPO-8*FC fiber	Used for connection between a switch and an ODF. The MPO connector connects to the switch using a QSFP+ optical module, and the eight FC connectors connect to the ODF.

5.8 AOC Cable

Types of AOC Cables

An active optical cable (AOC) is an active optical fiber with optical modules at both ends, and therefore is easy to use. [Figure 5-22](#), and [Figure 5-23](#) show different types of AOC cables.

Figure 5-22 Appearance of an SFP+ to SFP+ AOC cable



Figure 5-23 Appearance of a QSFP+ to QSFP+ AOC cable



[Table 5-11](#) lists the attributes of various AOC cables.

Table 5-11 Attributes of AOC cables

Model	Version Support	Length	Operating Wavelength	Connector Type	Operating Temperature
SFP-10G-AOC10M	Supported in V100R003C10 and later versions	10 m	850 nm	SFP+ connectors at both ends	0°C to 70°C
SFP-10G-AOC20M	Supported in V100R003C10 and later versions	20 m	850 nm	SFP+ connectors at both ends	
QSFP-H40G-AOC10M	Supported in V100R003C10 and later versions	10 m	850 nm	QSFP+ connectors at both ends	
SFP-10G-AOC3M	Supported in V100R005C00 and later versions	3 m	850 nm	SFP+ connectors at both ends	

Connection

Table 5-12 describes usage scenarios of AOC cables and cable connections in these scenarios.

Table 5-12 AOC cable usage scenarios and connections

Cable Type	Connection
SFP+ to SFP+ AOC cable	- Scenario 1: used for 10GE optical port connection or stacking between CE7800&6800&5800 series switches. - Scenarios 2: used for 10GE connection between CE12800 and CE7800&6800&5800 switches. Both ends connect to a 10GE optical port.
QSFP+ to QSFP+ AOC cable	- Scenario 1: used for 40GE optical port connection or stacking between CE7800&6800&5800 series switches. - Scenarios 2: used for 40GE connection between CE12800 and CE7800&6800&5800 switches. Both ends connect to a 40GE optical port.

5.9 Copper Cable

Types of Copper Cables

Table 5-13 shows the types of copper cables.

Table 5-13 Types of copper cables

Model	Length	Electrical attribute	Connector Type	Part Number
SFP-10G-CU1M	1 m	Passive	SFP+ to SFP+	02310MUN
SFP-10G-CU3M	3 m	Passive	SFP+ to SFP+	02310MUP
SFP-10G-CU5M	5 m	Passive	SFP+ to SFP+	02310QPR
SFP-10G-AC7M	7 m	Active	SFP+ to SFP+	02310QPS
SFP-10G-AC10M	10 m	Active	SFP+ to SFP+	02310MUQ
QSFP-40G-CU1M	1 m	Passive	QSFP+ to QSFP+	02310MUG
QSFP-40G-CU3M	3 m	Passive	QSFP+ to QSFP+	02310MUH
QSFP-40G-CU5M	5 m	Passive	QSFP+ to QSFP+	02310MUJ
QSFP-4SFP10G-CU1M	1 m	Passive	QSFP+ to 4*SFP+	02310MUK
QSFP-4SFP10G-CU3M	3 m	Passive	QSFP+ to 4*SFP+	02310MUL
QSFP-4SFP10G-CU5M	5 m	Passive	QSFP+ to 4*SFP+	02310MUM

Appearance and Structure

Figure 5-24 shows the appearance of an SFP+ to SFP+ copper cable.

Figure 5-24 Appearance of an SFP+ to SFP+ copper cable



Figure 5-25 shows the appearance of a QSFP+ to QSFP+ copper cable.

Figure 5-25 Appearance of a QSFP+ to QSFP+ copper cable



Figure 5-26 shows the appearance of a QSFP+ to 4*SFP+ copper cable.

Figure 5-26 Appearance of a QSFP+ to 4*SFP+ copper cable



Figure 5-27 shows the structure of an SFP+ to SFP+ copper cable.

Figure 5-27 Structure of an SFP+ to SFP+ copper cable

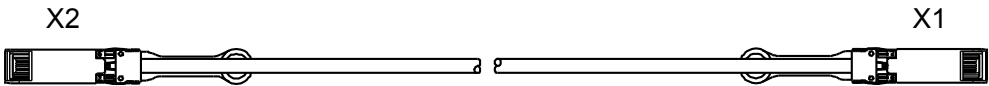


Figure 5-28 shows the structure of a QSFP+ to QSFP+ copper cable.

Figure 5-28 Structure of a QSFP+ to QSFP+ copper cable

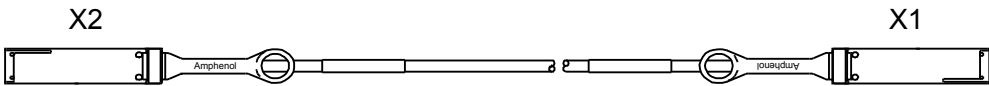
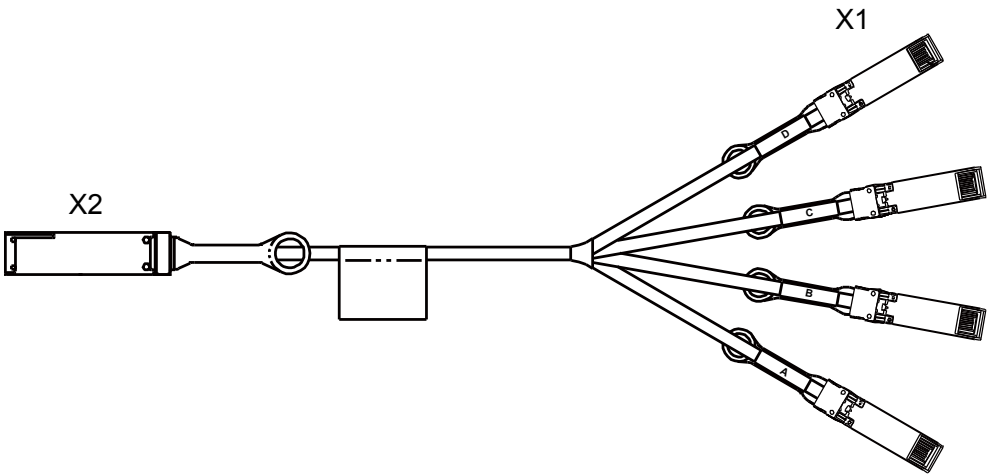


Figure 5-29 shows the structure of a QSFP+ to 4*SFP+ copper cable.

Figure 5-29 Structure of a QSFP+ to 4*SFP+ copper cable



Connection

Table 5-14 describes usage scenarios of copper cables and cable connections in these scenarios.

Table 5-14 Copper cable usage scenarios and connections

Copper Cable	Connection	Remarks
SFP+ to SFP+ copper cable	● Scenario 1: used for connection between 10GE optical ports of CE7800&6800&5800 series switches. ● Scenario 2: used to set up a stack of CE7800&6800&5800 series switches using 10GE optical ports. Each end connects to a 10GE optical port.	● Minimum clearance for cable routing: 60 mm ● Minimum bend radius: 35 mm
QSFP+ to QSFP+ copper cable	● Scenario 1: used for connection between 40GE optical ports of CE7800&6800&5800 series switches. ● Scenario 2: used to set up a stack of CE7800&6800&5800 series switches using 40GE optical ports. Each end connects to a 40GE optical port.	● Minimum clearance for cable routing: 75 mm ● Minimum bend radius: 50 mm
QSFP+ to 4*SFP+ copper cable	When a 40GE optical port is configured as four 10GE optical ports, one end of this cable connects to a 40GE optical port, and the other end connects to four 10GE optical ports.	QSFP+: ● Minimum clearance for cable routing: 75 mm ● Minimum bend radius: 50 mm SFP+: ● Minimum clearance for cable routing: 60 mm ● Minimum bend radius: 35 mm

6 Optical Module

About This Chapter

NOTE

- In this document, optical modules are classified based on encapsulation types, and optical modules of each encapsulation type are classified based on interface rates.
- The actual optical modules depend on the delivered ones. The appearance of optical modules in this document is for reference only.
- Use optical modules certified for Huawei switches. Non-certified optical modules cannot ensure transmission reliability and may affect service stability on the switch. Huawei is not responsible for any problem caused by non-certified optical modules and will not fix such problems.
- All the optical modules listed in the documentation are Huawei certified optical modules.

[6.1 Concepts](#)

[6.2 SFP/SFP+ Modules](#)

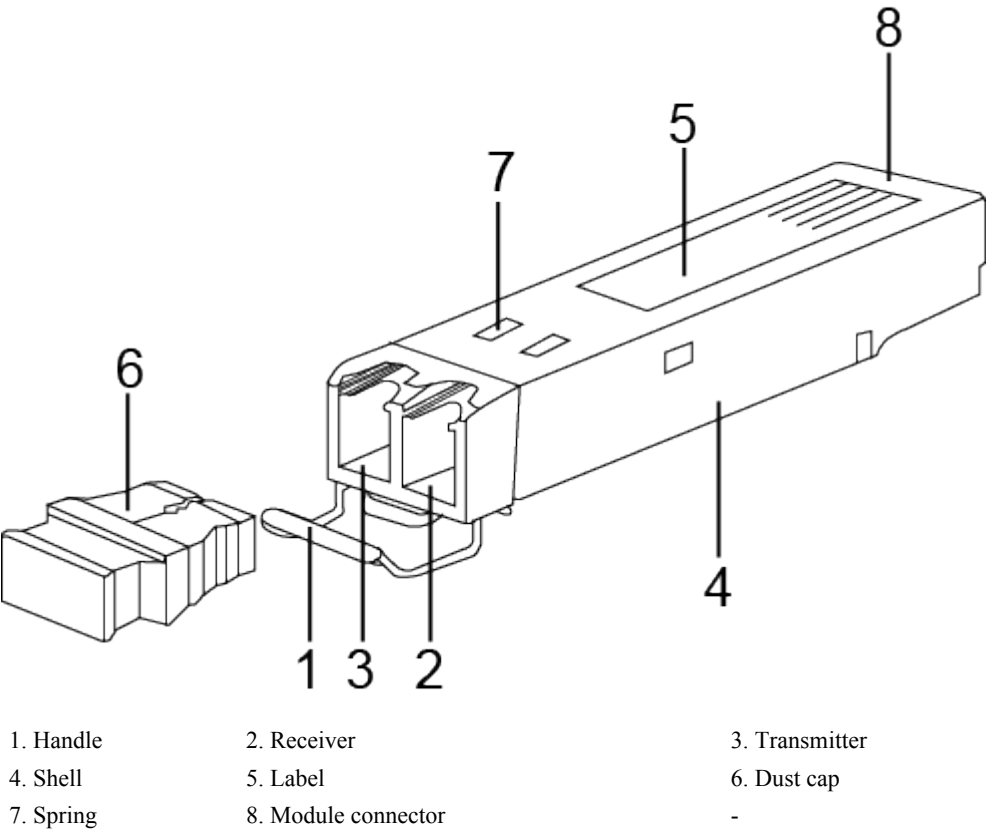
[6.3 QSFP+ Modules](#)

6.1 Concepts

Appearance and Structure

Figure 6-1 shows the appearance of an optical module.

Figure 6-1 Appearance of an optical module (SFP optical module as an example)



Terms

Transmission distance	Maximum distance over which optical signals can transmit. Optical signals sent from different types of sources can transmit over different distances due to negative effects of optical fibers, such as dispersion and attenuation.
Interface rate	Maximum rate of electrical signals that an optical device can transmit without bit errors. Various interface rates are defined in Ethernet standards, such as 125 Mbit/s, 1.25 Gbit/s, 10.3125 Gbit/s, and 41.25 Gbit/s.

Encapsulation type	Appearance type of an optical module. Encapsulation types of optical modules include SFP, eSFP, SFP+, XFP, and QSFP+. ● SFP: small form-factor pluggable. ● eSFP: enhanced small form-factor pluggable. An eSFP module is an SFP module that supports monitoring of voltage, temperature, bias current, transmit optical power, and receive optical power. Because all the SFP optical modules support these monitoring functions, eSFP is also called SFP. ● SFP+: small form-factor pluggable plus, SFP with a higher rate. SFP+ modules are more sensitive to electromagnetic interference (EMI) because they have a higher rate. To reduce EMI, SFP+ modules have more springs than SFP modules. ● XFP: 10GE optical module. X is the Roman numeral 10. ● QSFP+: Quad SFP+, four-channel SFP+.
Center wavelength	Wavelength measured at the midpoint of the half-amplitude line in the transmit spectrum.
Fiber mode	Mode of fibers defining based on core diameters and features of optical fibers. Optical fibers are classified into single-mode fibers and multi-mode fibers. Generally, multi-mode fibers have large core diameters and severe dispersion, so they transmit optical signals over short distances when working with multi-mode optical modules. Single-mode fibers have small dispersion and can transmit optical signals over long distances when working with single-mode optical modules.
Modal bandwidth	Bandwidth measured at a point with transmit power several dB lower than that of the point with the peak center wavelength. Modal bandwidth reflects spectrum characteristics of an optical module.
Fiber diameter	Diameter of the core of a fiber. According to international standards for optical fibers, the diameter of a multi-mode fiber is 62.5 um or 50 um, and the diameter of a single-mode fiber is 9 um.
Fiber class	Optical signals with different wavelengths have their best working windows in different optical fibers. To help efficiently adjust wavelengths or dispersion features of optical fibers and change their refractive indexes, the following classes are defined: multi-mode fiber (G.651), common single-mode fiber (G.652), shifted dispersion fiber (G.653), and non-zero shifted dispersion fiber (G.655). Multi-mode fiber (G.651) and common single-mode fiber (G.652) are commonly used fiber classes.
Connector type	Type of the interface on an optical module to accommodate a fiber. Commonly used connector types are LC (applicable to all the SFP, SFP+, and XFP modules) and MPO (applicable to some of QSFP+ modules).
Transmit optical power	Output optical power of an optical module when it is working properly.
Maximum receiver sensitivity	Minimum average input optical power that the receiver of an optical module can receive within a range of bit error rate (BER = 10^{-12}).

Overload optical power Maximum average input optical power that the receiver of an optical module can receive within a range of bit error rate ($BER = 10^{-12}$).

Extinction ratio Minimum ratio of the average optical power with signals transmitted against the average optical power without signals transmitted in complete modulation mode. The extinction ratio indicates the capability of an optical module to identify signal 0 and signal 1.

6.2 SFP/SFP+ Modules

Appearance

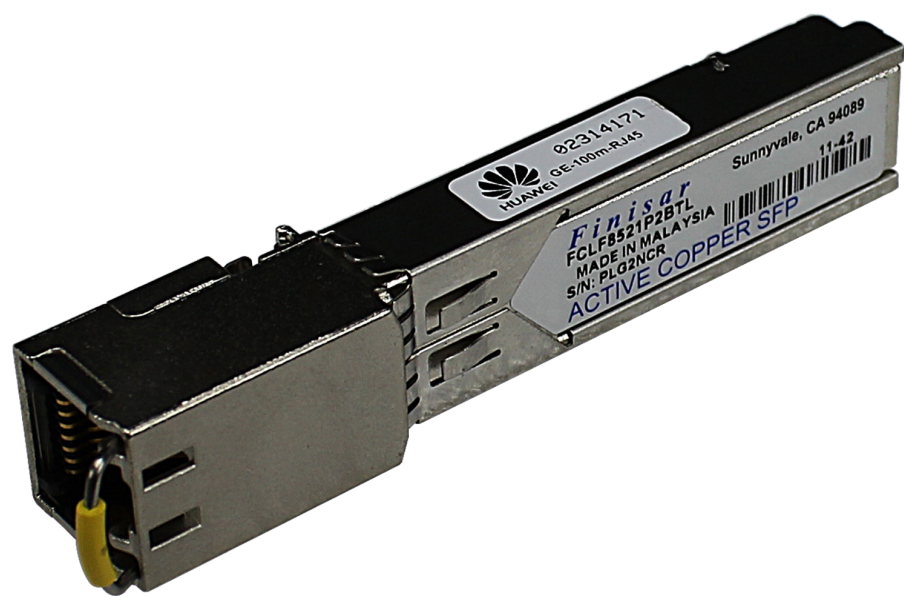
Figure 6-2 shows an SFP/SFP+ module.

Figure 6-2 Appearance of an SFP/SFP+ module



Figure 6-3 shows a GE copper module.

Figure 6-3 Appearance of a GE copper module



10GE Optical Module

Table 6-1 and Table 6-2 list the attributes of 10GE optical modules.

Table 6-1 Attributes of 10GE optical modules

Part Number	Model	Version Support	Transmission Distance	Standard	Fiber Mode, Modal Bandwidth (MHz*km), Core Diameter (μm), Connector Type	Operating Temperature
02318169	OMXD30000	Supported in V100R001C00 and later versions	≤ 26 m	10GBASE-SR	Multi-mode, 160, 62.5, LC	0°C to 70°C
			≤ 33 m		Multi-mode (OM1), 200, 62.5, LC	
			≤ 66 m		Multi-mode, 400, 50, LC	
			≤ 82 m		Multi-mode (OM2), 500, 50, LC	
			≤ 300 m		Multi-mode (OM3), 2000, 50, LC	
			≤ 400 m		Multi-mode (OM4), 4700, 50, LC	
02310CRM	OSXD22N00		≤ 220 m	10GBASE-LRM	Multi-mode, 160/500, 62.5, LC	

Part Number	Model	Version Support	Transmission Distance	Standard	Fiber Mode, Modal Bandwidth (MHz*km), Core Diameter (μm), Connector Type	Operating Temperature
			≤ 220 m		Multi-mode (OM1), 200/500, 62.5, LC	
			≤ 100 m		Multi-mode, 400/400, 50, LC	
			≤ 220 m		Multi-mode (OM2), 500/500, 50, LC	
			≤ 220 m		Multi-mode (OM3), 1500/500, 50, LC	
02310 QDJ	SFP-10G-LR		≤ 10 km	10GBASE-LR	Single-mode (G.652), -, 9, LC	
02310 CNF	OSX040N01		≤ 40 km	10GBASE-ER	Single-mode (G.652), -, 9, LC	
02310 SNN	SFP-10G-ZR		≤ 80 km	10GBASE-ZR	Single-mode (G.652), -, 9, LC	
02310J FE	LE2MXSC80FF0	Supported only in V100R001C00 and V100R002C00	≤ 80 km	10GBASE-ZR	Single-mode (G.652), -, 9, LC	
02310 MNW	SFP-10G-USR	Supported in V100R002C00 and later versions	≤ 30 m	10GBASE-USR	Multi-mode (OM2), 500, 50, LC	
			≤ 100 m		Multi-mode (OM3), 2000, 50, LC	
			≤ 150 m		Multi-mode (OM4), 4700, 50, LC	

Part Number	Model	Version Support	Transmission Distance	Standard	Fiber Mode, Modal Bandwidth (MHz*km), Core Diameter (μm), Connector Type	Operating Temperature
02311 BJB	SFP-10G-ER-SM1330-BIDI (Single-fiber bidirectional optical module)	Supported in V100R005C00 and later versions	≤ 40 km	10GBASE-BDER	Single-mode (G.652), -, 9, LC	
02311 BJC	SFP-10G-ER-SM1270-BIDI (Single-fiber bidirectional optical module)					
02311 BJJ	SFP-10G-iLR	Supported in V100R005C00 and later versions	≤ 1.4 km	10GBASE-iLR	Single-mode (G.652), -, 9, LC	

Table 6-2 Optical parameters of 10GE optical modules

Model	Operating Wavelength (nm)	Transmit Optical Power (dBm)	Receiver Sensitivity (dBm)	Overload Optical Power (dBm)	Extinction Ratio (dB)
OMXD30000	850	-7.3 to -1	≤ -11.1	-1	≥ 3
OSXD22N00	1310	-6.5 to +0.5	≤ -6.5	1.5	≥ 3.5
SFP-10G-LR	1310	-8.2 to +0.5	≤ -12.6	0.5	≥ 3.5
OSX040N01	1550	-4.7 to +4	≤ -14.1	-1	≥ 3
SFP-10G-ZR	1550	0 to 4	≤ -24	-7	≥ 9
LE2MXSC80FF0	1550	0 to 4	≤ -24	-7	≥ 9
SFP-10G-USR	850	-7.3 to -1	≤ -10.7	0.5	≥ 3
SFP-10G-ER-SM1330-BIDI (Single-fiber bidirectional optical module)	Tx1330/ Rx1270	0 to 5	≤ -18	-9	≥ 3.5
SFP-10G-ER-SM1270-BIDI (Single-fiber bidirectional optical module)	Tx1270/ Rx1330				
SFP-10G-iLR	1310	-8.2 to +0.5	≤ -14.4	0.5	≥ 3.5

10GE CWDM Optical Module

Table 6-3 lists the attributes of 10GE coarse wavelength division multiplexing (CWDM) optical modules.

Table 6-3 Attributes of 10GE CWDM optical modules

Operating Wavelength (nm), Model, Part Number	1471, SFP-10G-ZCW1471, 02310SSG	1491, SFP-10G-ZCW1491, 02310SSF	1511, SFP-10G-ZCW1511, 02310SSE	1531, SFP-10G-ZCW1531, 02310SSD
	1551, SFP-10G-ZCW1551, 02310SSC	1571, SFP-10G-ZCW1571, 02310SSB	1591, SFP-10G-ZCW1591, 02310SSA	1611, SFP-10G-ZCW1611, 02310SRY
Version Support	Supported in V100R003C00 and later versions			
Attributes and Parameters	Encapsulation type: SFP+			
	Transmission distance: ≤ 70 km			
	Fiber mode, connector type: single-mode, LC			
	Standard: CWDM			
	Operating temperature: 0°C to 70°C			
	● Transmit power: 0 dBm to 4 dBm ● Receiver sensitivity: ≤ -23 dBm ● Overload optical power: -7 dBm ● Extinction ratio: ≥ 8.2 dB			

GE Optical Module

Table 6-4 and **Table 6-5** list the attributes of GE optical modules.

Table 6-4 Attributes of GE optical modules

Part Number	Model	Version Support	Transmission Distance	Standard	Fiber Mode, Modal Bandwidth (MHz*km), Core Diameter (μ m), Connector Type	Operating Temperature
02315-204	eSFP-GE-	Supported in	≤ 220 m	1000BASE-SX	Multi-mode, 160, 62.5, LC	0°C to 70°C

Part Number	Model	Version Support	Transmission Distance	Standard	Fiber Mode, Modal Bandwidth (MHz*km), Core Diameter (μm), Connector Type	Operating Temperature
	SX-MM8500	V100R001C00 and later versions	≤ 275 m		Multi-mode (OM1), 200, 62.5, LC	
			≤ 500 m		Multi-mode, 400, 50, LC	
			≤ 550 m		Multi-mode (OM2), 500, 50, LC	
02315200	SFP-GE-LX-SM1310		≤ 550 m	1000BASE-LX10	Multi-mode (OM1), 200/500, 62.5, LC	
			≤ 550 m		Multi-mode, 400/400, 50, LC	
			≤ 550 m		Multi-mode (OM2), 500/500, 50, LC	
			≤ 10 km		Single-mode (G.652), -, 9, LC	
02317346	S-SFP-GE-LH40-SM1310		≤ 40 km	1000BASE-EX	Single-mode (G.652), -, 9, LC	
02317348	S-SFP-GE-LH80-SM1550		≤ 80 km	1000BASE-ZX	Single-mode (G.652), -, 9, LC	
02315206	eSFP-GE-ZX100-SM1550		≤ 100 km	-	Single-mode, -, -, LC	

Part Number	Model	Version Support	Transmission Distance	Standard	Fiber Mode, Modal Bandwidth (MHz*km), Core Diameter (μm), Connector Type	Operating Temperature
02315285	SFP-GE-LX-SM1310-BIDI (Single-fiber bidirectional optical module)		≤ 10 km	1000BASE-BX10-D 1000BASE-BX10-U	Single-mode (G.652), -, 9, LC	
02315286	SFP-GE-LX-SM1490-BIDI (Single-fiber bidirectional optical module)					
02310KVU	LE2MGSC40ED00 (Single-fiber bidirectional optical module)	Supported in V100R002C00 and later versions	≤ 40 km	-	Single-mode (G.652), -, -, LC	

Part Number	Model	Version Support	Transmission Distance	Standard	Fiber Mode, Modal Bandwidth (MHz*km), Core Diameter (μm), Connector Type	Operating Temperature
02310 KVV	LE2M GSC4 0DE0 (Single-fiber bidirectional optical module)					

 NOTE

Single-fiber bidirectional (BIDI) optical modules must be used in pairs. For example, SFP-GE-LX-SM1310-BIDI must be used with SFP-GE-LX-SM1490-BIDI.

Table 6-5 Optical parameters of GE optical modules

Model	Operating Wavelength (nm)	Transmit Optical Power (dBm)	Receiver Sensitivity (dBm)	Overload Optical Power (dBm)	Extinction Ratio (dB)
eSFP-GE-SX-MM850	850	-9.5 to -2.5	≤ -17	0	≥ 9
SFP-GE-LX-SM1310	1310	-9 to -3	≤ -20	-3	≥ 9
S-SFP-GE-LH40-SM1310	1310	-5 to 0	≤ -23	-3	≥ 9
S-SFP-GE-LH80-SM1550	1550	-2 to +5	≤ -23	-3	≥ 9
eSFP-GE-ZX100-SM1550	1550	0 to 5	≤ -30	-9	≥ 8

Model	Operating Wavelength (nm)	Transmit Optical Power (dBm)	Receiver Sensitivity (dBm)	Overload Optical Power (dBm)	Extinction Ratio (dB)
SFP-GE-LX-SM1310-BIDI (Single-fiber bidirectional optical module)	Tx1310/ Rx1490	-9 to -3	≤ -19.5	-3	≥ 6
SFP-GE-LX-SM1490-BIDI (Single-fiber bidirectional optical module)	Tx1490/ Rx1310				
LE2MGS C40ED0 (Single-fiber bidirectional optical module)	Tx1490/ Rx1310	-2 to +3	≤ -23	-3	≥ 9
LE2MGS C40DE0 (Single-fiber bidirectional optical module)	Tx1310/ Rx1490				

GE CWDM Optical Module

Table 6-6 lists the attributes of GE CWDM optical modules.

Table 6-6 Attributes of GE CWDM optical modules

Operating Wavelength (nm), Model, Part Number	1471, CWDM-SFPGE-1471, 02310LPN	1491, CWDM-SFPGE-1491, 02310LPK	1511, CWDM-SFPGE-1511, 02310LPH	1531, CWDM-SFPGE-1531, 02310LPL
	1551, CWDM-SFPGE-1551, 02310LPG	1571, CWDM-SFPGE-1571, 02310LNS	1591, CWDM-SFPGE-1591, 02310LNT	1611, CWDM-SFPGE-1611, 02310LPJ
Version Support	Supported in V100R002C00 and later versions			
Attributes and Parameters	Encapsulation type: eSFP			
	Transmission distance: ≤ 80 km			
	Fiber mode, connector type: single-mode, LC			
	Standard: CWDM			
	Operating temperature: 0°C to 70°C			
	● Transmit power: 0 dBm to 5 dBm ● Receiver sensitivity: ≤ -28 dBm ● Overload optical power: -9 dBm ● Extinction ratio: ≥ 8.5 dB			

GE Copper Module

Table 6-7 lists the attributes of GE copper modules.

Table 6-7 Attributes of GE copper modules

Part Number	Model	Version Support	Transmission Distance	Standard	Cable Type	Connector Type
02314171	SFP-1000BaseT	Supported in V100R001C00 and later versions	≤ 100 m	1000BASE-T (SFP-GE-T)	CAT5 UTP/STP	RJ45

100M Optical Module

Table 6-8 and **Table 6-9** list the attributes of 100M optical modules.

Table 6-8 Attributes of 100M optical modules

Part Number	Model	Version Support	Transmission Distance	Standard	Fiber Mode, Modal Bandwidth (MHz*km), Core Diameter (μm), Connector Type	Operating Temperature
02315233	SFP-FE-SX-MM1310	Supported in V100R005C00 and later versions	≤ 2 km	100BASE-FX	Multi-mode (OM1), 200, 62.5, LC	0°C to 70°C
					Multi-mode, 400, 50, LC	
					Multi-mode (OM2), 500, 50, LC	
02315205	eSFP-FE-LX-SM1310		≤ 15 km	100BASE-LX	Single-mode (G.652), -, 9, LC	
02317344	S-SFP-FE-LH40-SM1310		≤ 40 km	100BASE-EX	Single-mode (G.652), -, 9, LC	
02315203	SFP-FE-LX-SM1310-BIDI (Single-fiber bidirectional optical module)		≤ 15 km	100BASE-BX	Single-mode (G.652), -, 9, LC	

Part Number	Model	Version Support	Transmission Distance	Standard	Fiber Mode, Modal Bandwidth (MHz*km), Core Diameter (μm), Connector Type	Operating Temperature
02315202	SFP-FE-LX-SM1550-BIDI (Single-fiber bidirectional optical module)					

 NOTE

BIDI optical modules must be used in pairs. For example, SFP-FE-LX-SM1310-BIDI must be used with SFP-FE-LX-SM1550-BIDI.

Table 6-9 Optical parameters of 100M optical modules

Model	Operating Wavelength (nm)	Transmit Optical Power (dBm)	Receiver Sensitivity (dBm)	Overload Optical Power (dBm)	Extinction Ratio (dB)
SFP-FE-SX-MM1310	1310	-19 to -14	≤ -30	-14	≥ 10
eSFP-FE-LX-SM1310	1310	-15 to -8	≤ -31	-8	≥ 8.2
S-SFP-FE-LH40-SM1310	1310	-5 to 0	≤ -37	-10	≥ 10.5

Model	Operating Wavelength (nm)	Transmit Optical Power (dBm)	Receiver Sensitivity (dBm)	Overload Optical Power (dBm)	Extinction Ratio (dB)
SFP-FE-LX-SM1310-BIDI (Single-fiber bidirectional optical module)	Tx1310/ Rx1550	-15 to -8	≤ -32	-8	≥ 8.5
SFP-FE-LX-SM1550-BIDI (Single-fiber bidirectional optical module)	Tx1550/ Rx1310	-15 to -8	≤ -32	-8	≥ 8.5

FC Optical Module

[Table 6-10](#) and [Table 6-11](#) list the attributes and optical parameters of an FC optical module.

Table 6-10 Attributes of an FC optical module

Part Number	Model	Version Support	Rate	Transmission Distance	Fiber Mode, Modal Bandwidth (MHz*km), Core Diameter (μm), Connector Type	Operating Temperature
02311BJH	SFP-FC2G-SW-SR	Supported in V100R005C10 and later versions	2GE	≤ 0.3 km	Multi-mode (OM2), 500, 50, LC	-20°C to 85°C
				≤ 0.5 km	Multi-mode (OM3), 2000, 50, LC	

Part Number	Model	Version Support	Rate	Trans mission Distance	Fiber Mode, Modal Bandwidth (MHz*km), Core Diameter (μm), Connector Type	Operating Temperature
02311BJG	SFP-FC2G-LW-LR		2 Gbit/s	≤ 15 km	Single-mode (G.652), -, 9, LC	0°C to 70°C
02311BJF	SFP-FC4G-SW-SR		2GE/4GE	≤ 0.3 km	Multi-mode (OM3), 2000, 50, LC	-20°C to 85°C
02311BJE	SFP-FC4G-LW-LR		2GE/4GE	≤ 10 km	Single-mode (G.652), -, 9, LC	0°C to 70°C
02311BJL	SFP-FC8G-SW-SR		8GE	≤ 0.05 km	Multi-mode (OM2), 500, 50, LC	0°C to 70°C
				≤ 0.15 km	Multi-mode (OM3), 2000, 50, LC	
			4GE	≤ 0.15 km	Multi-mode (OM2), 500, 50, LC	
				≤ 0.38 km	Multi-mode (OM3), 2000, 50, LC	
			2GE	≤ 0.3 km	Multi-mode (OM2), 500, 50, LC	
				≤ 0.5 km	Multi-mode (OM3), 2000, 50, LC	
02311BJA	SFP-FC8G-LW-LR		2GE/4GE/8GE	≤ 10 km	Single-mode (G.652), -, 9, LC	0°C to 70°C

Table 6-11 Optical parameters of an FC optical module

Model	Operating Wavelength (nm)	Transmit Optical Power (dBm)	Receiver Sensitivity (dBm)	Overload Optical Power (dBm)	Extinction Ratio (dB)
SFP-FC2G-SW-SR	850	-9.5 to -2.5	≤ -17	0	≥ 9
SFP-FC2G-LW-LR	1310	-5 to 0	≤ -21	0	≥ 8.2
SFP-FC4G-SW-SR	850	-9 to -1.5	≤ -15	0	≥ 3
SFP-FC4G-LW-LR	1310	-8.4 to -1	≤ -18	0	≥ 9
SFP-FC8G-SW-SR	850	-8.2 to -1.3	≤ -11.2	0	≥ 9
SFP-FC8G-LW-LR	1310	-8.4 to 0.5	≤ -13.8	0.5	≥ 3.5

6.3 QSFP+ Modules

Appearance

Figure 6-4 shows the appearance of a QSFP+ module.

Figure 6-4 Appearance of a QSFP+ module



40GE Optical Module

Table 6-12 and Table 6-13 list the attributes of 40GE optical modules.

Table 6-12 Attributes of 40GE optical modules

Part Number	Model	Version Support	Transmission Distance	Standard	Fiber Mode, Modal Bandwidth (MHz*km), Core Diameter (μm), Connector Type	Operating Temperature
02310 MHR	QSFP-40G-iSR4	Supported in V100R001C00 and later versions	≤ 30 m	40GBase-SR4	Multi-mode (OM2), 500, 50, MPO	0°C to 70°C
			≤ 100 m	10GBase-USR (four lanes)	Multi-mode (OM3), 2000, 50, MPO	
			≤ 150 m		Multi-mode (OM4), 4700, 50, MPO	
02310 MHS	QSFP-40G-LR4		≤ 10 km	40GBase-LR4	Single-mode (G.652), -, 9, LC	
02310 RMB	QSFP-40G-eSR4	Supported in V100R003C00 and later	≤ 82 m	40GBase-eSR4	Multi-mode (OM2), 500, 50, MPO	
			≤ 300 m	10GBase-SR (four lanes)	Multi-mode (OM3), 2000, 50, MPO	

Part Number	Model	Version Support	Transmission Distance	Standard	Fiber Mode, Modal Bandwidth (MHz*km), Core Diameter (μm), Connector Type	Operating Temperature
		versions	≤ 400 m		Multi-mode (OM4), 4700, 50, MPO	
02311 BKT	QSFP-40G-ER4	Supported in V100R005C00 and later versions	≤ 40 km	40GBase-ER4	Single-mode (G.652), -, 9, LC	
02311 DRW	QSFP-40G-iSM4		≤ 1.4 km	40GBASE-iSM4	Single-mode (G.652), -, 9, MPO	
02311 DTR	QSFP-40G-eSM4		≤ 10 km	40GBASE-eSM4	Single-mode (G.652), -, 9, MPO	

Table 6-13 Optical parameters of 40GE optical modules

Model	Center Wavelength (nm)	Transmit Optical Power (dBm)	Receiver Sensitivity (dBm)	Overload Optical Power (dBm)	Extinction Ratio (dB)
QSFP-40G-iSR4	850	-7.6 to -1	≤ -9.5	2.4	≥ 3
QSFP-40G-LR4	1271, 1291, 1311, 1331	-7 to +2.3	≤ -11.5	3.3	≥ 3.5
QSFP-40G-eSR4	850	-7.6 to -1	≤ -5.4	2.4	≥ 3
QSFP-40G-ER4	1271, 1291, 1311, 1331	-2.7 to +4.5	≤ -19.5	-4.5	≥ 5.5
QSFP-40G-iSM4	1310	-8.2 to +0.5	≤ -11.5	0.5	≥ 3.5
QSFP-40G-eSM4	1310	-8.2 to +0.5	≤ -12.6	0.5	≥ 3.5